

Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs

Statistical formulation, theory, algorithms, and empirical results

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Exponential-family segment models, irregular designs, multi-sequence hierarchies, and latent groups.

A technical treatment of generalized hierarchical Bayesian segmentation for ordered data. Segment marginal likelihoods from conjugate exponential-family models are combined with exact dynamic programming over contiguous partitions, with extensions for irregular designs, shared and group-specific changepoints, latent groups, and explicitly approximate nonconjugate segment models.

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Scope of this volume

This volume is the technical reference for *Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs*. It is based on the editable manuscript, the implementation archive, and the populated result files supplied with the project. The statistical statements, computational results, and implementation status are distinguished so that each conclusion can be read under its stated assumptions.

STATISTICAL FORMULATION

BAYESBREAK is a generalized hierarchical Bayesian multiple-changepoint method. For regular exponential-family observation models with proper conjugate priors, segment-specific parameters can be integrated out to obtain exact segment marginal likelihoods and posterior moments. These quantities enter sum-product dynamic programming for posterior distributions over ordered partitions. A separate max-sum recursion with backtracking returns a joint MAP partition. The method also treats irregular designs, common or group-specific changepoints across multiple sequences, finite latent groups, and explicitly identified numerical segment integration for nonconjugate models.

Status terminology

Status	Meaning in this volume
ESTABLISHED	A complete proof is given under the stated assumptions.
EXACT DERIVATION	The expression follows by direct algebra or exact finite summation.
REAL RESULT	A populated value or figure is an archived executed computation.
PARTIAL	An executed result addresses only part of a stated question or lacks a required comparison or uncertainty analysis.
CONDITIONAL	A conclusion follows only if an additional assumption, such as a uniform numerical error bound, is established.
TARGET	A proof, implementation correction, or experiment remains to be completed and is not used as an established result.
EXCLUDED	The computation was executed, but its observation model, coordinate axis, reference quantity, or support rule is incompatible with the scientific interpretation under discussion.

ARCHIVED COMPUTATIONS EXCLUDED FROM STATED CONCLUSIONS

No populated archived numerical value is silently changed. Two computations are retained in the result history but excluded from the analyses for which they were originally reported: RES-BB-CMP-002, whose array-CGH comparator used incompatible coordinate axes, and RES-BB-RD-007Q, whose methylation held-out calculation used a Gaussian predictive routine for Beta-valued observations and assigned out-of-range coordinates to the last segment. A corrected calculation must receive a new result identifier.

Reading routes

- **Statistical formulation and exact inference:** Chapters 1, 3, 4, 5, and 6.
- **Irregular designs and hierarchical extensions:** Chapters 7, 8, 9, and 10.
- **Prediction, implementation, and empirical studies:** Chapters 11, 12, 13, 14, and Appendices B–E.
- **Literature and limitations:** Chapters 2, 15, and Appendix A.

Source precedence

The editable manuscript and its archived numerical assets are the factual baseline. The author's explicit decisions fix the title and main narrative. Mathematical corrections are stated in place; implementation defects and unresolved experiments are reported without rewriting the archived outputs. External results are attributed to their sources and are not presented as BayesBreak results.

Generalized hierarchical Bayesian segmentation: problem, outputs, and scope

CHAPTER PATH

The research problem, the meaning of generality and hierarchy, the returned posterior quantities, and the conditions for exact computation.

LEARNING OBJECTIVES

- State the generalized hierarchical Bayesian segmentation problem and its principal outputs.
- Explain why the partition recursions are common across observation models once segment marginal likelihoods are available.
- Identify the roles of irregular designs, multiple sequences, known groups, and latent groups.
- Distinguish exact conjugate calculations from explicitly approximate nonconjugate calculations.

Multiple-changepoint analysis partitions an ordered set of observations into contiguous segments within which a statistical parameter is constant or follows a specified segment model. Given responses $y_1, \dots, y_n$ observed at ordered coordinates $u_1 < \dots < u_n$, the inferential questions include the number of segments, the changepoint locations, the segment-specific parameters, and uncertainty in all of these quantities. Bayesian formulations place a prior on the partition and segment parameters and therefore yield posterior probabilities for competing segmentations rather than only a single penalized estimate [3, 4, 14, 19].

The central statistical structure is a product-partition model. Conditional on a partition, observations in different segments are independent after segment parameters are introduced. When a segment parameter can be integrated out, each candidate segment $(i, j]$ contributes a segment marginal likelihood. A prior that factorizes over segments or transitions then permits exact finite summation over all admissible ordered partitions by dynamic programming. This yields marginal likelihoods, posterior distributions for the segment count, ordered-changepoint marginals, changepoint-event probabilities, and posterior moments of segment functionals [3, 14, 19].

BAYESBREAK develops this calculation as a generalized hierarchical Bayesian segmentation method. The term *generalized* refers to the following mathematical property: the recursion over partitions does not depend on the particular observation family once the required segment marginal likelihoods and posterior moments have been computed. For a regular exponential

family with a proper conjugate prior and finite normalizing constants, these segment quantities are available by exact integration from sufficient statistics. Gaussian observations with an unknown segment mean, Poisson counts with exposure, and Binomial observations with known trial counts are examples. A nonconjugate segment model may use numerical integration or another stated approximation, but exact partition summation for an approximate score array is not the same as exact Bayesian inference for the intended observation model.

The term *hierarchical* refers to probability models that couple sequences or groups through shared partition structure or group-level parameters. The book treats: (i) a single ordered sequence; (ii) irregularly spaced observations through design-dependent segment cohesion and boundary hazards; (iii) multiple aligned sequences with common changepoints and conditionally independent segment likelihoods; (iv) known groups of sequences; and (v) a finite latent-group criterion in which sequences are assigned to candidate changepoint templates. The first four are Bayesian models with explicit joint distributions under their stated assumptions. The finite latent-group extension is presented according to its actual optimization criterion and is not relabelled as a normalized mixture likelihood.

Principal contributions.

- **Exponential-family segment integration.** A common notation separates sufficient statistics, exposure or trial descriptors, and optional likelihood powers. Proper conjugate priors give exact segment marginal likelihoods and posterior moments through ratios of normalizing constants.
- **Posterior computation over ordered partitions.** Forward and backward sum-product recursions compute fixed-count marginal likelihoods, $p(k | y)$, ordered-changepoint marginals, changepoint-event probabilities, and model-averaged segment functionals. A distinct max-sum recursion with backtracking computes the joint MAP partition.
- **Irregular-design priors.** Segment cohesion and interior-boundary hazard are separated mathematically. This permits priors based on physical segment length or local design spacing without confusing design geometry with exposure in the observation likelihood.
- **Multiple-sequence hierarchies.** For aligned conditionally independent sequences with a common partition, sequence-specific segment marginal likelihoods multiply. Known groups use the same calculation within each group. A finite-candidate consistency result states conditions under which the posterior concentrates on the common partition as the number of informative sequences increases.
- **Grouped and latent-group segmentation.** Known group memberships lead to groupwise shared-changepoint posteriors. For unknown memberships, a finite latent-group objective alternates auxiliary allocation weights with exact responsibility-weighted max-sum updates of the group templates; monotonicity follows from a Jensen minorization argument.
- **Numerical segment integration and prediction.** For nonconjugate observation models, the effect of a uniform segmentwise log marginal-likelihood error is propagated through the partition sums. Posterior prediction is defined by the fitted observation family and coordinate support, and changepoint metrics use one-to-one matching.

1.1 Research question and returned quantities

The research question is whether one generalized Bayesian formulation can combine observation-model-specific segment integration with posterior computation over contiguous partitions while

retaining irregular-design and hierarchical multiple-sequence structure. For a sequence y , the principal returned quantities are

$$\{p(y | k), p(k | y), p(t_p = h | y, k), p(b_h = 1 | y), \mathbb{E}[m(\theta_s) | y], \hat{t}^{\text{MAP}}(k)\}.$$

These quantities have different meanings. In particular, a marginal mode of each changepoint coordinate need not form the joint MAP partition, and a posterior mean curve is not a selected piecewise-constant partition.

KEY IDEA

The observation model determines the segment marginal likelihoods and posterior moments. The partition model determines how these quantities are summed over ordered segmentations. Generality comes from separating these two mathematical calculations while keeping their probability models explicit.

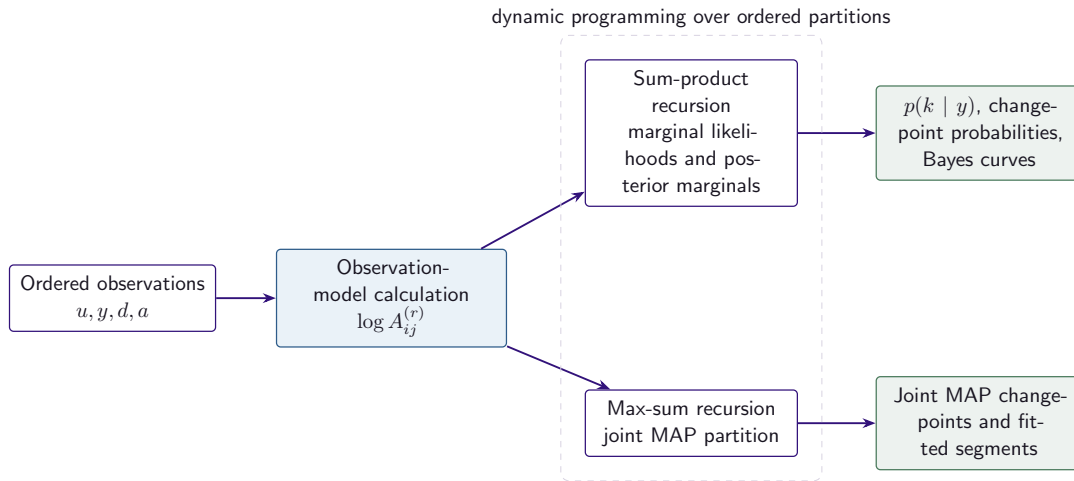


Figure 1.1. Observation-model calculations and inference over ordered partitions. Sum-product and max-sum recursions use the same prior-adjusted segment scores but return different posterior or optimization quantities.

1.2 Conditions and exclusions

The exact results concern offline segmentation on a finite ordered candidate set with a prior and likelihood that factor into segment or transition contributions. The method is not an online run-length filter, a general nonparametric detector, or a proof that arbitrary dependence can be represented without enlarging the state space. Exact Bayesian inference requires exact segment marginal likelihoods. When these quantities are approximated, the resulting partition calculation is exact only for the supplied approximate array; comparison with the intended posterior requires additional numerical error analysis.

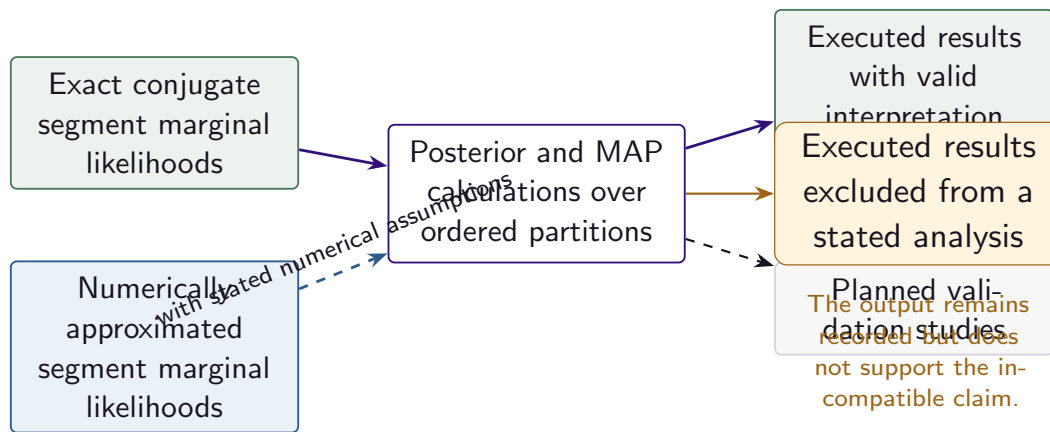


Figure 1.2. Distinction among exact calculations, numerical approximations, executed empirical results, planned validation, and executed results excluded from an incompatible interpretation.

Statistical foundations and relation to prior work

CHAPTER PATH

The established literature on exponential families, product partitions, exact Bayesian changepoint recursion, irregular designs, multiple sequences, latent groups, and numerical approximation.

2.1 Exponential families and conjugate analysis

For a regular exponential family, likelihood contributions can be represented through sufficient statistics, a natural parameter, and a log-partition function. Conjugate priors for exponential families preserve the natural form of the posterior and express integrated likelihoods through normalizing-constant ratios [5, 12]. This is established Bayesian theory, not a new property of `BAYESBREAK`. Its role here is to provide exact segment marginal likelihoods and posterior moments for every candidate interval when the required normalizers are finite. Exposure in Poisson sampling and trial counts in Binomial sampling enter the likelihood through family-specific sufficient summaries; they are not generic likelihood weights.

2.2 Product-partition models and exact Bayesian changepoint inference

Product-partition models assign a probability to a partition through products of segment cohesion terms and combine this prior with segmentwise integrated likelihoods [3, 4]. Exact Bayesian multiple-changepoint recursions exploit the same factorization to sum over possible previous changepoints and compute posterior distributions without enumerating every partition explicitly [14]. Bayesian piecewise-constant regression further shows how partition averaging yields posterior regression curves, changepoint probabilities, and marginal likelihoods [19].

Accordingly, this book does not claim to introduce product partitions, conjugate segment integration, or dynamic programming for Bayesian changepoint models. `BAYESBREAK` uses these established results as the foundation for a broader model specification covering multiple observation families, design-dependent priors, multiple aligned sequences, and group structure. The sum-product recursion for posterior marginals and the max-sum recursion for a joint MAP partition are stated separately because they solve different algebraic problems even when they use the same segment scores.

2.3 Frequentist segmentation and alternative Bayesian computation

Frequentist methods optimize additive segment costs or multiscale criteria. Segment-neighborhood and optimal-partitioning algorithms provide exact minimizers for fixed segment counts or penalties [2, 21]; pruned dynamic programming and PELT reduce practical cost under additional conditions [22, 30]. Total-variation and fused-lasso estimators use convex regularization, including group penalties for multiple sequences [7, 31, 35]. SMUCE and wild binary segmentation provide multiscale or recursive detection procedures with different inferential targets [16, 17]. Circular binary segmentation is widely used for array-CGH analysis [25]. These procedures are relevant comparators but do not, without additional construction, return the same posterior distribution over partitions.

When segment parameters cannot be marginalized or the dependence structure is not amenable to a finite recursion, reversible-jump Markov chain Monte Carlo and other trans-dimensional sampling methods provide a general Bayesian alternative [11, 18]. Bayesian online changepoint detection instead filters a run-length distribution as data arrive and solves a different online problem [1]. BAYESBREAK is an offline finite-partition method.

2.4 Irregular designs and design-dependent partition priors

Product-partition models have been generalized so that cohesion depends on covariates or other predictors [24, 26]. In changepoint problems, irregular observation coordinates affect the prior plausibility of segment lengths and boundary locations. This book therefore distinguishes a segment-cohesion factor $c_u(i, j)$ from an interior-boundary hazard $h_u(j)$. The distinction prevents two different prior constructions—a prior on segment length and a local hazard of a changepoint—from being described as the same object. It also keeps design spacing separate from exposure or trial information in the observation likelihood.

2.5 Hierarchical and multiple-sequence changepoint models

Hierarchical Bayesian changepoint models have a long history [8]. Methods for simultaneous changepoints in multiple data sequences use hierarchical priors to borrow information about the propensity for changes across sequences [13]. Joint random partition models provide more flexible dependence among partitions across several ordered processes [29]. Other work permits dependence between successive segments or richer within-segment regression structures [15, 32].

The exact multiple-sequence calculation developed here is deliberately narrower: aligned sequences share the same candidate partition, are conditionally independent given that partition, and may have sequence-specific segment parameters. Their segment marginal likelihoods then multiply. This is an exact hierarchical model under those assumptions, not a claim to encompass every form of partial synchrony or dependence among sequence-specific changepoints. Known groups repeat the calculation within each group.

2.6 Latent groups and optimization

Model-based clustering of longitudinal or ordered data with common structural changes is relevant prior work for clustering sequences through changepoint patterns [9]. Finite-mixture identifiability results concern normalized component distributions [34], and the EM algorithm concerns likelihoods or objectives for which the auxiliary-function construction is valid [10].

The latent-group calculation in this book is stated more narrowly. It defines a finite objective

$$\mathcal{F}(\pi, \tau) = \sum_s \log \left\{ \sum_g \pi_g S_s(\tau_g) \right\},$$

where $S_s(\tau_g)$ is a positive sequence-template score. A Jensen minorizer yields auxiliary allocation weights; exact maximization of the minorizer over template partitions uses a responsibility-weighted max-sum recursion. The proof establishes monotone ascent of this stated criterion. It does not establish that $S_s(\tau_g)$ is a normalized component density, identify a finite mixture, or average over uncertainty in the templates.

2.7 Nonconjugate segment models

Laplace approximation, variational bounds, Pólya–Gamma augmentation, expectation propagation, and numerical quadrature are established tools for nonconjugate Bayesian calculations [20, 23, 27, 36]. BAYESBREAK uses them, when appropriate, to approximate individual segment marginal likelihoods. The general theorem in Chapter 10 is conditional: if every admissible segment log marginal likelihood is within ε of a reference value, then partition marginal likelihoods and posterior odds obey explicit bounds. The theorem does not supply a method-specific value of ε .

2.8 Position of the present work

The contribution is a generalized hierarchical Bayesian segmentation formulation built from established components whose assumptions are kept explicit. It combines:

1. exact segment integration for a class of conjugate exponential-family observation models;
2. sum-product posterior computation and separate max-sum MAP optimization over ordered partitions;
3. priors for irregular designs that distinguish segment cohesion from boundary hazard;
4. exact common-changepoint pooling for aligned conditionally independent sequences and known groups;
5. a finite latent-group template criterion with a proved monotone alternating algorithm;
6. conditional propagation of numerical segment-score error; and
7. family-specific posterior prediction and clearly defined changepoint metrics.

The novelty claim is attached to this combined generalized method and its derivations, not to the prior existence of dynamic programming, exponential-family conjugacy, product partitions, or EM.

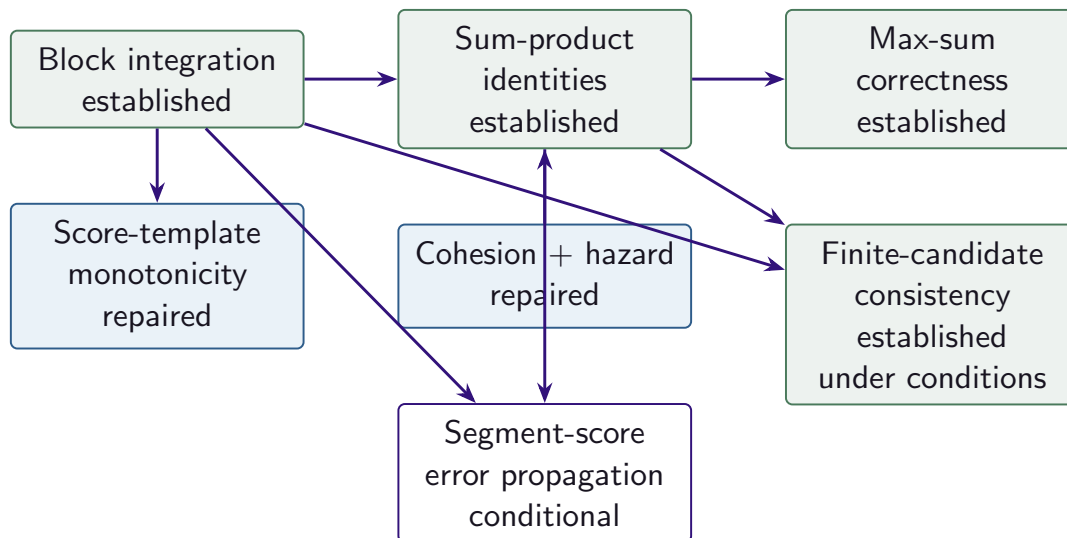


Figure 2.1. Logical dependence of the principal results. Exact partition calculations require finite segment marginal likelihoods and factorized prior contributions. The nonconjugate result additionally requires a uniform segmentwise error bound.

2.9 Recommended reading sequence

A reader new to the subject should begin with exponential-family conjugacy, product-partition models, and exact Bayesian multiple-changepoint recursion; then proceed to Bayesian piecewise-constant regression, design-dependent product partitions, and hierarchical multiple-sequence methods. The literature annotations in Appendix A state the specific role of every reference retained in the book. Numerical implementation additionally requires stable log-sum-exp evaluation and the approximation methods used for nonconjugate segment models [6].

Formal setup, notation, and model specification

CHAPTER PATH

Ordered segmentation, exposure-aware local likelihoods, posterior targets, and the exact factorization required by the partition dynamic program.

3.1 Notation and symbols

This section fixes notation used throughout the book. We use lower-case Roman letters for indices, reserve $t = (t_0, \dots, t_k)$ for boundary vectors, and adopt the half-open block notation $(i, j] := \{i + 1, \dots, j\}$ for $0 \leq i < j \leq n$.

For irregular designs, physical segment lengths are defined using candidate boundary coordinates $u_0 < u_1 < \dots < u_n$ rather than by implicitly referencing an undefined observation location x_0 . The physical length of block $(i, j]$ is $\Delta_x(i, j) := u_j - u_i$. On a regular index grid set $u_i = i$; for interval data the u_i are the observed interval endpoints. Point observations without meaningful physical interval endpoints use the index-uniform prior or provide an external left endpoint u_0 .

Conjugate-prior normalization convention. For exponential-family blocks we use the following (Diaconis–Ylvisaker) convention for the conjugate prior normalizer. For hyperparameters (α, β) ,

$$\begin{aligned} p(\theta \mid \alpha, \beta) &= \exp\left\{\eta(\theta)^\top \alpha - \beta A(\theta) - \log Z(\alpha, \beta)\right\}, \\ Z(\alpha, \beta) &:= \int \exp\left\{\eta(\theta)^\top \alpha - \beta A(\theta)\right\} d\theta. \end{aligned} \tag{3.1}$$

With this convention, Z is the integral itself (not its reciprocal) and block evidences take the ratio form $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})/Z(\alpha_0, \beta_0)$ (Theorem 4.1 and Chapter 4).

Table 3.1. Notation and symbols used throughout the book. This is a notation glossary; no source CSV is associated with it.

Symbol	Meaning
n	Number of observations on the ordered design.
x_i	Design point / location of observation y_i .

u_i	Candidate boundary coordinate used for physical segment lengths, with $u_0 < \dots < u_n$.
$\Delta_x(i, j)$	Physical length of block $(i, j]$, defined as $u_j - u_i$.
y_i	Observation at location x_i (single-sequence case).
d_i	Family-specific exposure, trial-count, offset, or precision descriptor when required by the observation model. Optional power/reliability weights are denoted a_i ; neither role is inferred from irregular spacing.
$k; k_{\max}$	Number of segments; maximum segment count considered by the DP.
$t = (t_0, \dots, t_k)$	Boundary vector, with $0 = t_0 < t_1 < \dots < t_k = n$.
$I_q(t)$	Index set of segment q : $\{t_{q-1} + 1, \dots, t_q\}$.
$(i, j]$	Candidate block $\{i + 1, \dots, j\}$.
$S(y, d), E(d), \eta(\theta), A(\theta), h(y, d)$	Family-specific statistic contribution, exposure multiplier, natural-parameter map, log-partition function, and base measure for the chosen exponential-family block model.
$m_\star(\theta) = m(\theta; d_\star)$	Declared segment functional on the reporting scale, evaluated at a fixed reference descriptor $d_\star$ when the observation mean depends on exposure or trial structure. Examples include a Gaussian mean, a Poisson rate at unit exposure, and a Binomial success probability.
(α, β)	Conjugate hyperparameters with normalizer $Z(\alpha, \beta)$ defined in (3.1).
(α_0, β_0)	Prior hyperparameters before observing block data.
S_{ij}	Block statistic sum $\sum_{t=i+1}^j a_t S(y_t, d_t)$.
W_{ij}	Block log-partition multiplier $\sum_{t=i+1}^j a_t E(d_t)$.
H_{ij}	Powered base-measure sum $\sum_{t=i+1}^j a_t \log h(y_t, d_t)$.
$A_{ij}^{(0)}$	Block evidence on $(i, j]$.
$A_{ij}^{(r)}$	Block moment numerator $A_{ij}^{(0)} \mathbb{E}[m_\star(\theta)^r \mid y_{i+1:j}]$ (for integer $r \in \{1, 2, \dots\}$; conjugate-case definition). In the non-conjugate Laplace discussion of Section 10, the superscript is a test function rather than a moment order; those objects are denoted $M_{ij}^{[g]}$ to avoid collision.
$\tilde{A}_{ij}^{(r)}$	Prior-adjusted block quantity $A_{ij}^{(r)} \gamma_x(i, j)$, where γ_x combines cohesion and an interior-boundary hazard when appropriate.
$L_{k,j}$	Prefix evidence for $y_{1:j}$ with exactly k segments.
$R_{k,i}$	Suffix evidence for $y_{i+1:n}$ with exactly k segments.
C_k	Normalizer of $p(t \mid k)$ under a segment-factorized partition prior.
$p(k \mid y)$	Posterior over segment counts.
$p(t_p = h \mid y, k)$	Marginal posterior of the p th boundary given k .
$p(b_i = 1 \mid y, k)$	Boundary-event marginal at interior index i , equal to $\sum_{p=1}^{k-1} p(t_p = i \mid y, k)$. This is the calibration target in Section 13.
$\hat{t}^{\text{MAP}}(k)$	Joint MAP segmentation given k , obtained by max-sum DP and backtracking.

$\hat{\mu}_q^{(r)}$	Posterior moment of the segment- q mean parameter conditional on a chosen segmentation.
$\hat{f}_t^{\text{Bayes}}$	Bayes regression curve at index t (posterior mean of the latent piecewise-constant signal).
$s \in \{1, \dots, S\}$	Sequence / subject index in multi-sequence settings.
$g \in \{1, \dots, G\}$	Group index.
r_{sg}	Auxiliary allocation weight for sequence s and latent group g in the finite latent-group criterion.
$\mathcal{M}_g$	Fitted group-specific model exported for prediction.
ℓ_g	Log predictive score of new data under group g .
π_g	Prior group weight used for prediction or in the score-template objective.
ε	Uniform per-block log-evidence error tolerance for non-conjugate block approximations; appears as the hypothesis controlling the posterior-odds stability bound in Section 10.
$\hat{A}_{ij}^{(0)}$	Approximate (non-conjugate) block evidence; satisfies $ \log \hat{A}_{ij}^{(0)} - \log A_{ij}^{(0)} \leq \varepsilon$ under the stability assumption.
ϕ_t	Known per-observation precision used in the Beta-block likelihood for $y_t \in (0, 1)$.
ω_t	Pólya–Gamma augmentation variable for logistic / negative-binomial mean-field block approximations.
$M_{ij}^{[g]}$	Block test-function numerator $\int g(\theta) p(y_{i+1:j} \theta) p(\theta) d\theta$, used in the Laplace-approximation discussion of Section 10 where the superscript denotes a test function rather than a moment order.
$f_{\text{Lap}}, f_{\text{JJ}}, f_{\text{EP}}, f_{\text{PG}}, f_{\text{Q}}$	Local block-evidence approximations: Laplace, Jaakkola–Jordan variational bound, expectation propagation, Pólya–Gamma mean field, and one-dimensional quadrature. Each instantiates $\hat{A}_{ij}^{(0)}$ and inherits the same DP layer.

Remarks. Two recurring distinctions: *design irregularity* may enter through the partition prior (§7.1); d_i enters the likelihood only when the observation model supplies genuine exposures or known precisions. *Boundary marginals* and the *joint MAP segmentation* are different posterior summaries (§5) and must not be conflated.

BayesBreak addresses *offline* Bayesian segmentation of ordered data. Throughout, the fundamental unknown is a contiguous partition of the index set $\{1, \dots, n\}$ together with a segment parameter attached to each block. This section states the objects of interest in a model-agnostic way; Section 3.1 introduces the detailed notation used later.

Unless stated otherwise, all inferential statements in the book use the following standing assumption.

Assumption 3.1 (Offline segmentation support and block independence). The observations are ordered on a fixed set of candidate boundary locations and the segment-count set

$\{1, \dots, k_{\max}\}$ is finite. Structural admissibility of a candidate block is determined before observing the response values by the candidate grid, declared segment-length restrictions, and prior support. Every structurally admissible block has finite nonnegative block evidence; a zero evidence contributes zero to the likelihood-weighted partition sum but does not change the prior support or its normalizer C_k . Structurally excluded blocks have local prior factor zero and are stored with log-score $-\infty$. Segment parameters are conditionally independent across blocks given a segmentation, and the conditional partition prior factorizes over segments as in Assumption 3.6. For approximate non-conjugate blocks, the same statement applies to the supplied approximate score array, while comparison with the exact non-conjugate model requires the additional error hypothesis of Assumption 10.6.

3.1.1 Single-sequence segmentation and Bayes regression

We observe ordered pairs $\{(x_i, y_i)\}_{i=1}^n$ with strictly increasing design points $x_1 < \dots < x_n$. The design can be irregular. When the observation model includes known exposures, trial counts, or precision weights, we also allow deterministic nonnegative quantities w_i ; unless explicitly stated otherwise, point-observation models use $w_i \equiv 1$.

A segmentation into k contiguous blocks is represented by a boundary vector

$$0 = t_0 < t_1 < \dots < t_{k-1} < t_k = n, \quad (3.2)$$

where block $q \in \{1, \dots, k\}$ is the index set

$$I_q(t) := \{t_{q-1} + 1, \dots, t_q\}. \quad (3.3)$$

Definition 3.2 (Segmentation space and structural support). For fixed n and k , let

$$\mathcal{T}_{k,n} := \{t = (t_0, \dots, t_k) : 0 = t_0 < t_1 < \dots < t_k = n\}.$$

The *structural support* $\mathcal{T}_{k,n}^+ \subseteq \mathcal{T}_{k,n}$ consists of those boundary vectors for which every block satisfies the declared grid and length restrictions and has positive local prior factor $\gamma_x(t_{q-1}, t_q)$. Conditional on the design and model specification, this set is fixed independently of the observed response values and of $A_{ij}^{(0)}$. A structurally admissible partition may have zero likelihood for the observed data; it then contributes zero to the posterior partition sum, while the prior normalizer C_k continues to sum over the full structural support.

The corresponding segment parameter is denoted $\theta_q \in \Theta$. Conditionally on $(k, t, \theta_{1:k})$, observations factorize across blocks:

$$p(y_{1:n} \mid k, t, \theta_{1:k}) = \prod_{q=1}^k \prod_{i \in I_q(t)} p(y_i \mid \theta_q, w_i), \quad (3.4)$$

where $p(y_i \mid \theta_q, w_i)$ is interpreted according to the chosen family. For example, depending on the family, w_i encodes exposure in Poisson models, trial count in Binomial models, or known precision in Gaussian models. Irregular spacing by itself does not force a non-unit w_i ; design geometry can instead enter through the partition prior.

For any candidate block $(i, j]$ with $0 \leq i < j \leq n$, the key primitive is the integrated

single-block evidence

$$\mathcal{L}(i, j; \alpha_0, \beta_0) := p(y_{i+1:j} \mid \alpha_0, \beta_0) = \int \prod_{t=i+1}^j p(y_t \mid \theta, w_t) p(\theta \mid \alpha_0, \beta_0) d\theta, \quad (3.5)$$

where $p(\theta \mid \alpha_0, \beta_0)$ is the segment prior with hyperparameters (α_0, β_0) (see Section 3.1). When $\mathcal{L}(i, j)$ is available in closed form, the entire offline posterior can be computed exactly by dynamic programming. When it is only approximated, BayesBreak isolates the approximation at the block level and reuses the same global DP.

Definition 3.3 (Boundary-event variables, Bayes curve, and exported estimator). For a boundary vector $t \in \mathcal{T}_{k,n}$, define the interior boundary indicator $b_i(t) := \sum_{p=1}^{k-1} \mathbf{1}\{t_p = i\}$ for $i = 1, \dots, n-1$. Because the boundaries are strictly ordered, $b_i(t) \in \{0, 1\}$. For fixed k , the Bayes regression curve at index i is the posterior mean of the observation-scale segment mean covering i after marginalizing over $t \in \mathcal{T}_{k,n}^+$ and over segment parameters. The exported segmentation estimator is a deterministic boundary vector, usually the joint MAP vector $\hat{t}^{\text{MAP}}(k)$, paired with segment-level posterior summaries for downstream prediction. These two objects need not coincide because the Bayes curve averages over boundary uncertainty whereas the exported estimator chooses one segmentation.

The inferential targets are:

- (i) the marginal evidence at fixed segment count,

$$p(y_{1:n} \mid k) = \sum_{t \in \mathcal{T}_{k,n}^+} p(t \mid k) \prod_{q=1}^k \mathcal{L}(t_{q-1}, t_q);$$

- (ii) the posterior $p(k, t \mid y)$ over segment counts and boundaries and its marginals $p(k \mid y)$, $p(t_p = h \mid y, k)$, and the boundary-event marginal $p(b_i = 1 \mid y, k) = \sum_{p=1}^{k-1} p(t_p = i \mid y, k)$ (the calibration target of §13);
- (iii) posterior summaries of the latent piecewise-constant signal, namely $p(\theta_i \mid y)$ and $\mathbb{E}[m_\star(\theta_i) \mid y]$ with $m_\star$ the observation-scale mean parameter (the Bayes regression curve; conditional on the selected $\hat{k}$ unless stated otherwise);
- (iv) point estimators for downstream use. In particular the *joint* MAP segmentation $\hat{t}^{\text{MAP}}(k) \in \arg \max_t p(t \mid y, k)$ differs from the vector of marginal boundary modes $\arg \max_h p(t_p = h \mid y, k)$ and is recovered by max-sum DP. The prior on k and the normalizer $\log C_k$ enter the objective only when also optimizing over k .

The three posterior summaries entering downstream analysis are isolated as labelled objects so that later sections can refer to them without re-deriving notation.

Definition 3.4 (Posterior summaries: marginal boundaries, segment count, and joint MAP).

Fix a sequence $y = y_{1:n}$, a maximum segment count $k_{\max}$, and Assumption 3.1. Define:

- (a) the *marginal boundary posterior* $\text{MB}_i(k) := p(b_i = 1 \mid y, k)$ for interior indices $i \in \{1, \dots, n-1\}$ and admissible $k \in \{1, \dots, k_{\max}\}$;
- (b) the *posterior over segment count* $\text{PK}(k) := p(k \mid y) \propto p(k) p(y \mid k)$ for $k \in \{1, \dots, k_{\max}\}$;
- (c) the *joint MAP segmentation at fixed k* $\hat{t}^{\text{MAP}}(k) := \arg \max_{t \in \mathcal{T}_{k,n}^+} p(t \mid y, k)$, with ties broken by a deterministic rule (Section 5.0.5).

These three objects are returned by distinct DP recursions (sum-product, sum-product with k -aggregation, and max-sum, respectively), and they are not in general consistent with each other: MB_i and $\hat{t}^{\text{MAP}}(k)$ can place boundary mass at different indices (Remark 5.3), and the mode of PK need not coincide with the cardinality of any single thresholded set of marginal boundaries.

3.1.2 Multi-sequence structure

Many applications provide multiple related sequences observed on a common ordered grid or on a common set of candidate boundary locations. Let sequence $s \in \{1, \dots, S\}$ be denoted by $y_{1:n}^{(s)}$, with optional weights $w_i^{(s)}$ and subject-specific segment parameters $\theta_q^{(s)}$ when boundaries are shared. We use $z_s \in \{1, \dots, G\}$ for group membership, $\mathcal{S}_g := \{s : z_s = g\}$ for known groups, and r_{sg} for latent-group responsibilities. BayesBreak treats three progressively richer settings.

Definition 3.5 (Multi-sequence segmentation classes). A shared-boundary replicate model has one boundary vector t and subject-specific segment parameters $\theta_{1:k}^{(s)}$. A known-group model has observed labels z_s and one boundary vector $t^{(g)}$ per group. A latent-template model has unobserved labels z_s and a finite set of templates $\tau_g = (k_g, t^{(g)})$ optimized under the finite latent-group criterion of Section 9.3. These three classes preserve the block-evidence interface; fully Bayesian averaging over both unknown labels and all group segmentations is a different target and is not claimed for the score-template model.

Shared-boundary replicates. All sequences share a common segmentation $t \in \mathcal{T}_{k,n}^+$, but each sequence has its own within-block parameters $\theta_{1:k}^{(s)}$. The common-grid requirement is an assumption on the candidate boundary coordinates; subject-specific missingness or exposures are handled inside the block evidences. Conditional on t , the sequences are independent:

$$p(\{y^{(s)}\}_{s=1}^S \mid t, \{\theta^{(s)}\}) = \prod_{s=1}^S \prod_{q=1}^k \prod_{i \in I_q(t)} p(y_i^{(s)} \mid \theta_q^{(s)}, w_i^{(s)}). \quad (3.6)$$

Integrating out the subject-specific segment parameters yields pooled block evidences obtained by multiplying subject-level block evidences.

Known groups. Sequences are partitioned into known groups $g \in \{1, \dots, G\}$. Each group has its own segmentation $t^{(g)} \in \mathcal{T}_{k,n}^+$, while sequences within a group share that segmentation and retain their own segment parameters. Conditional independence is within group and across subjects given $t^{(g)}$.

Latent groups. Group labels are unknown. A fully Bayesian formulation averages over both assignments and group-specific segmentations. BayesBreak adopts a tractable *template-mixture* formulation: each group is associated with a single latent boundary template $\tau_g = (k_g, t^{(g)})$, and coordinate ascent alternates between soft assignments (normalized auxiliary weights for the current templates) and exact responsibility-weighted max-sum DP updates of those templates. The finite latent-group criterion is non-decreasing under the exact alternating updates (Theorem 9.1); this is a finite-template optimization model rather than exact Bayesian averaging over latent-group segmentations.

3.1.3 Computational scope

For fixed $k_{\max}$, the core BayesBreak workflow consists of:

- (a) precomputing all block evidences and required moment numerators;
- (b) running sum-product DP recursions for evidences and posterior marginals; and
- (c) optionally running a max-sum/backtracking recursion for a joint MAP segmentation.

When block evaluation is constant time after prefix-sum preprocessing, this yields $\mathcal{O}(n^2)$ block precomputation and $\mathcal{O}(k_{\max}n^2)$ DP time. These worst-case costs make the exact method most natural for moderate sequence lengths (hundreds to low thousands), where exact posterior quantities are still computationally feasible and often worth the cost.

3.1.4 Prediction as a first-class inferential target

Beyond posterior inference on the training sequence(s), BayesBreak treats *prediction for new data* as an additional inferential target at the same level as evidence, posteriors, and point estimators. Given a fitted model $\mathcal{M}_g$ (one per group) and new observations $(X^{\text{new}}, y^{\text{new}})$, the relevant quantities are (i) group-conditional posterior-predictive likelihoods $p(y^{\text{new}} | \mathcal{M}_g)$, (ii) group membership posteriors $p(g | y^{\text{new}})$, and (iii) signal predictions $\hat{f}_g^{\text{MAP}}(x^*)$ or $\hat{f}_g^{\text{Bayes}}(x^*)$ at query points x^* . Section 11.1 develops the prediction layer in detail, covering pointwise, set-valued, and vector-valued new data, and separates exported-segmentation scoring from optional full resegmentation of new data.

3.2 Exponential-family segment likelihood with exposure and power weights

Let d_t collect family-specific known information, such as Poisson exposure, Binomial trial count, or Gaussian precision. Let $a_t \geq 0$ be an optional power or reliability weight that, when used, multiplies the complete log contribution. The canonical local model is

$$p(y_t | \theta, d_t)^{a_t} = h(y_t, d_t)^{a_t} \exp \left\{ a_t S(y_t, d_t)^\top \eta(\theta) - a_t E(d_t) A(\theta) \right\}. \quad (3.7)$$

Thus

$$S_{ij} = \sum_{t=i+1}^j a_t S(y_t, d_t), \quad W_{ij} = \sum_{t=i+1}^j a_t E(d_t), \quad H_{ij} = \sum_{t=i+1}^j a_t \log h(y_t, d_t).$$

This notation is deliberately more general than a scalar “weight”: a Binomial trial count changes both the statistic and the log-partition coefficient, while a power weight scales the entire observation contribution.

Family	$S(y, d)$	$E(d)$	Role of d
Gaussian, known precision ω	ωy	ω	precision/exposure in the quadratic family
Poisson, exposure e	y	e	mean is $e\lambda$; base measure contains $e^y/y!$
Binomial, trials m	y	m	log-partition is multiplied by the trial count
Unweighted point observation	$T(y)$	1	d is empty and $a = 1$

Table 3.2. Valid specializations of the exposure-aware segment likelihood. Optional a_t is a separate likelihood-power weight.

3.3 Ordered partitions and factorized priors

Assumption 3.6 (Factorizing ordered-partition prior). For fixed k and candidate-boundary coordinates x , the prior on $t = (0 = t_0 < \dots < t_k = n)$ has the form

$$p(t \mid k, x) = C_k(x)^{-1} \prod_{q=1}^k c_x(t_{q-1}, t_q) \prod_{q=1}^{k-1} h_x(t_q), \quad (3.8)$$

where $c_x(i, j) \geq 0$ is a segment cohesion and $h_x(j) \geq 0$ is an interior-boundary hazard. The terminal endpoint n receives no hazard factor. Define $\gamma_x(i, j) = c_x(i, j)h_x(j)^{\mathbf{1}_{\{j < n\}}}$.

The normalizer is

$$C_k(x) = \sum_{t \in \mathcal{T}_{k,n}^+} \prod_{q=1}^k c_x(t_{q-1}, t_q) \prod_{q < k} h_x(t_q), \quad (3.9)$$

and the prior-adjusted local arrays are

$$\tilde{A}_{ij}^{(r)} = A_{ij}^{(r)} \gamma_x(i, j). \quad (3.10)$$

This factorization subsumes an index-uniform prior, a length cohesion, a local boundary-intensity model, or their product without confusing their interpretations.

Integrated block evidence and posterior moments

CHAPTER PATH

One local interface covers exact conjugate families and clearly labeled deterministic quadrature without changing the global partition engine.

Plate model (single segment). For a block $(i, j]$ and conjugate hyperparameters (α_0, β_0) , the generative model is

$$\theta \sim p(\theta \mid \alpha_0, \beta_0), \quad y_t \mid \theta \stackrel{\text{ind}}{\sim} p(y_t \mid \theta, d_t) \text{ as in (3.7)}, \quad t \in \{i+1, \dots, j\}. \quad (4.1)$$

The block evidence $A_{ij}^{(0)}$ is obtained by marginalizing θ out of (4.1).

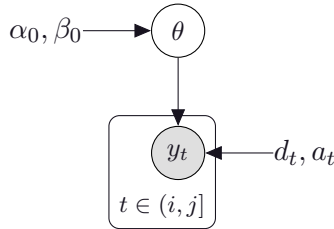


Figure 4.1. Plate model for a single conjugate EF block. Shaded nodes are observed; unshaded are latent; point nodes are fixed hyperparameters and family descriptors or power weights. Marginalizing θ gives $A_{ij}^{(0)}$.

4.1 Conjugate block integration

Assume the segment likelihood in Equation (3.7) and the conjugate prior

$$p(\theta \mid \alpha_0, \beta_0) = \exp\{\eta(\theta)^\top \alpha_0 - \beta_0 A(\theta) - \log Z(\alpha_0, \beta_0)\}.$$

Theorem 4.1 (Exposure-aware conjugate block integration). *For a candidate block $(i, j]$, if $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij}) < \infty$, then*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})}{Z(\alpha_0, \beta_0)}. \quad (4.2)$$

The posterior remains conjugate with hyperparameters $(\alpha_0 + S_{ij}, \beta_0 + W_{ij})$. For any measurable observation-scale target $m_\star(\theta)$ whose r th posterior moment exists,

$$A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[m_\star(\theta)^r \mid y_{i+1:j}, d_{i+1:j}, a_{i+1:j}]. \quad (4.3)$$

Proof. Multiplying the powered likelihood contributions gives

$$\exp\{H_{ij}\} \exp\{\eta(\theta)^\top S_{ij} - W_{ij}A(\theta)\}.$$

After multiplication by the prior, the terms depending on θ are exactly the conjugate kernel with updated hyperparameters $\alpha_0 + S_{ij}$ and $\beta_0 + W_{ij}$. Integrating this kernel yields its normalizer and hence Equation (4.2). Division by that evidence gives the updated posterior. Finally, inserting $m_\star(\theta)^r$ into the same integral factors it as the evidence times the corresponding posterior moment, provided the moment is finite. $\square$

Remark 4.2 (Scope of “exact”). The theorem is exact for the stated powered/exposure-aware model. It does not authorize treating an arbitrary sample weight as exposure, a Binomial trial count as a likelihood power, or an approximate numerical integral as a conjugate closed form.

4.2 Family-specific derivations

We now instantiate (4.2)–(4.3) for three closed-form conjugate families (Gaussian, Poisson, and Binomial), an overdispersed count model with a conjugate distributional parameter, and one low-dimensional quadrature example for $(0, 1)$ -valued responses. In each case we give (i) the segment model and prior, (ii) posterior hyperparameters, (iii) the block evidence $A_{ij}^{(0)}$, and (iv) first two posterior moments on the scale used by the Bayes-curve interface. For non-Gaussian families this scale must be stated explicitly: natural-parameter, distribution-parameter, and observation-mean moments are not interchangeable. Implementation routines are provided immediately after each derivation (Algorithms 1–5).

4.2.1 Gaussian with known variance (heteroscedastic via weights)

Proposition 4.3 (Gaussian-block evidence). *Suppose $y_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w_t)$ with known variance σ^2 and positive weights $w_t > 0$, and a Gaussian conjugate prior $\mu \sim \mathcal{N}(\nu, \rho^2)$. Define the block summaries $n_{ij} = j - i$, $W_{ij} = \sum_{t=i+1}^j w_t$, $S_{ij} = \sum_{t=i+1}^j w_t y_t$, $Q_{ij} = \sum_{t=i+1}^j w_t (y_t - \nu)^2$, and $\kappa_{ij} = \rho^2 W_{ij} + \sigma^2$. Then*

$$\mu \mid y_{(i,j]} \sim \mathcal{N}\left(\frac{\rho^2 S_{ij} + \sigma^2 \nu}{\kappa_{ij}}, \frac{\sigma^2 \rho^2}{\kappa_{ij}}\right),$$

and the block evidence admits the closed form

$$A_{ij}^{(0)} = (2\pi\sigma^2)^{-n_{ij}/2} \left(\prod_{t=i+1}^j w_t^{1/2} \right) \left(1 + \frac{\rho^2}{\sigma^2} W_{ij} \right)^{-1/2} \exp\left\{ -\frac{Q_{ij}}{2\sigma^2} + \frac{(S_{ij} - \nu W_{ij})^2}{2(\sigma^2 W_{ij} + \sigma^4/\rho^2)} \right\}.$$

The corresponding first and second moment numerators are $A_{ij}^{(1)} = A_{ij}^{(0)} \mathbb{E}[\mu \mid y_{(i,j)}]$ and $A_{ij}^{(2)} = A_{ij}^{(0)} (\mathbb{V}[\mu \mid y_{(i,j)}] + \mathbb{E}[\mu \mid y_{(i,j)}]^2)$.

Proof. Multiplying the weighted Gaussian likelihood by the Gaussian prior produces a quadratic kernel in μ . Completing the square gives posterior precision $\rho^{-2} + W_{ij}/\sigma^2$, posterior mean ν_B , and the stated normalizing-factor ratio. Multiplying the evidence by the first two moments of $\mathcal{N}(\nu_B, \rho_B^2)$ yields the moment numerators. $\square$

The proof specializes Theorem 4.1 to the Gaussian–Gaussian conjugate pair with weighted base measure.

Implementation. Algorithm 1 gives a numerically stable implementation of the Gaussian block log-evidence and posterior moments using cumulative weighted sums. The routine is written so that the DP layer consumes only $\log \mathcal{L}_{ij}$ and does not depend on Gaussian-specific algebra.

Algorithm 1: GaussianBlock($y, w, i, j, \nu, \rho^2, \sigma^2$)

Input: Block indices $(i, j]$; observations $y_{i+1:j}$; weights $w_{i+1:j}$

- 1 $n \leftarrow j - i$; $W \leftarrow \sum_{t=i+1}^j w_t$; $S \leftarrow \sum_{t=i+1}^j w_t y_t$; $Q \leftarrow \sum_{t=i+1}^j w_t (y_t - \nu)^2$;
- 2 $\kappa \leftarrow \rho^2 W + \sigma^2$;
- 3 $\log A0 \leftarrow -\frac{1}{2}n \log(2\pi) - \frac{1}{2} \sum_{t=i+1}^j \log(\sigma^2/w_t) - \frac{Q}{2\sigma^2} + \frac{(S-\nu W)^2}{2(\sigma^2 W + \sigma^4/\rho^2)} - \frac{1}{2} \log(1 + \frac{\rho^2}{\sigma^2} W)$;
- 4 $A0 \leftarrow \exp(\log A0)$;
- 5 $\mu_{\text{mean}} \leftarrow (\rho^2 S + \sigma^2 \nu)/\kappa$; $\mu_{\text{var}} \leftarrow \sigma^2 \rho^2/\kappa$;

Output: $A0, \mu_{\text{mean}}, \mu_{\text{var}}$

4.2.2 Poisson with exposure

Proposition 4.4 (Poisson-block evidence). *For $y_t \mid \lambda \sim \text{Pois}(\lambda w_t)$ with $w_t > 0$ and a Gamma conjugate prior $\lambda \sim \text{Gamma}(a_0, b_0)$, the block posterior is $\lambda \mid y_{(i,j]} \sim \text{Gamma}(a_0 + C_{ij}, b_0 + W_{ij})$, where $C_{ij} = \sum_{t=i+1}^j y_t$ and $W_{ij} = \sum_{t=i+1}^j w_t$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{b_0^{a_0} \Gamma(a_0 + C_{ij})}{\Gamma(a_0) (b_0 + W_{ij})^{a_0 + C_{ij}}}, \quad H_{ij} = \sum_{t=i+1}^j [y_t \log w_t - \log(y_t!)],$$

and moment numerators take the standard form $A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[\lambda^r \mid y_{(i,j)}]$ for $r \in \{1, 2, \dots\}$.

Proof. The block likelihood is $\{\prod_t e_t^{y_t}/y_t!\} \lambda^{C_{ij}} e^{-W_{ij}\lambda}$. Multiplication by the $\text{Gamma}(a_0, b_0)$ density gives a Gamma kernel with shape $a_0 + C_{ij}$ and rate $b_0 + W_{ij}$. Integrating the kernel gives the evidence; its ordinary moments give the stated moment numerators. $\square$

The proof specializes Theorem 4.1 to the Gamma–Poisson conjugate pair.

Implementation. Algorithm 2 summarizes the Poisson-with-exposure block evidence computation (and associated posterior summaries) in a form that matches the BayesBreak block-evidence interface.

Algorithm 2: PoissonBlock(y, w, i, j, a_0, b_0)**Input:** Block $(i, j]$; counts $y_{i+1:j}$; exposures $w_{i+1:j}$

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $W \leftarrow \sum_{t=i+1}^j w_t$;
- 2 $\log A0 \leftarrow \sum_{t=i+1}^j y_t \log w_t - \sum_{t=i+1}^j \log(y_t!) + a_0 \log b_0 + \log \Gamma(a_0 + C) - \log \Gamma(a_0) - (a_0 + C) \log(b_0 + W)$;
- 3 $A0 \leftarrow \exp(\log A0)$;
- 4 $\lambda_{\text{mean}} \leftarrow (a_0 + C)/(b_0 + W)$; $\lambda_{\text{var}} \leftarrow (a_0 + C)/(b_0 + W)^2$;

Output: $A0, \lambda_{\text{mean}}, \lambda_{\text{var}}$ **4.2.3 Binomial with Beta prior**

Proposition 4.5 (Binomial-block evidence). *For $y_t \mid p \sim \text{Binom}(m_t, p)$ with known trial counts m_t and a Beta conjugate prior $p \sim \text{Beta}(a_0, b_0)$, the block posterior is $p \mid y_{(i,j]} \sim \text{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$, where $C_{ij} = \sum_{t=i+1}^j y_t$ and $M_{ij} = \sum_{t=i+1}^j m_t$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{B(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})}{B(a_0, b_0)}, \quad H_{ij} = \sum_{t=i+1}^j \log \binom{m_t}{y_t},$$

with moment numerators $A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[p^r \mid y_{(i,j)}]$.

Proof. Collecting success and failure powers gives $p^{C_{ij}}(1-p)^{M_{ij}-C_{ij}}$ times the product of binomial coefficients. Multiplication by the Beta prior gives a $\text{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$ kernel. The ratio of Beta normalizers is the evidence, and Beta moments yield the remaining formulas. $\square$

The proof specializes Theorem 4.1 to the Beta–Binomial conjugate pair.

Implementation. Algorithm 3 provides pseudocode for Binomial/Bernoulli block evidence evaluation under Beta conjugacy, along with posterior mean/variance computations used for segment-wise moments and MAP/Bayes curve summaries.

Algorithm 3: BinomialBlock(y, m, i, j, a_0, b_0)**Input:** Block $(i, j]$; successes $y_{i+1:j}$; trials $m_{i+1:j}$

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $M \leftarrow \sum_{t=i+1}^j m_t$;
- 2 $\log A0 \leftarrow \sum_{t=i+1}^j \log \binom{m_t}{y_t} + \log B(a_0 + C, b_0 + M - C) - \log B(a_0, b_0)$;
- 3 $A0 \leftarrow \exp(\log A0)$;
- 4 $p_{\text{mean}} \leftarrow (a_0 + C)/(a_0 + b_0 + M)$;
- 5 $p_{\text{var}} \leftarrow \frac{(a_0 + C)(b_0 + M - C)}{(a_0 + b_0 + M)^2 (a_0 + b_0 + M + 1)}$;

Output: $A0, p_{\text{mean}}, p_{\text{var}}$

4.2.4 Negative-Binomial block (overdispersed counts) with fixed dispersion

Proposition 4.6 (Negative-Binomial-block evidence). *Fix dispersion $r > 0$ and assume $y_t \mid p \sim \text{NegBin}(r, p)$ with Beta conjugate prior $p \sim \text{Beta}(a_0, b_0)$. The block posterior is $p \mid y_{(i,j]} \sim \text{Beta}(a_0 + N_{ij}, b_0 + C_{ij})$, where $C_{ij} = \sum_{t=i+1}^j y_t$ and $N_{ij} = \sum_{t=i+1}^j r$. The block evidence is*

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{B(a_0 + N_{ij}, b_0 + C_{ij})}{B(a_0, b_0)}, \quad H_{ij} = \sum_{t=i+1}^j \log \binom{y_t + r - 1}{y_t},$$

and the moment numerators $A_{ij}^{(r)}$ on the observation-mean scale are obtained from the Beta posterior moments via the count-mean transformation $m_*(p) = r_*(1 - p)/p$.

Proof. Under the displayed parameterization, the block likelihood is the product of combinatorial terms times $p^{N_{ij}}(1 - p)^{C_{ij}}$. Multiplication by the Beta prior therefore updates its two shape parameters to $a_0 + N_{ij}$ and $b_0 + C_{ij}$. Integration gives the Beta-normalizer ratio. The observation-scale moments follow by integrating $r_*(1 - p)/p$ and its square, which requires the stated posterior-shape conditions. $\square$

Remark 4.7 (Observation-scale vs. parameter-scale moments). The Beta posterior moments of p are useful diagnostics, but the Bayes-curve target on the observation scale is the count mean $m_*(p) = r_*(1 - p)/p$ for a future count with dispersion r_* . With $a_B = a_0 + N_{ij}$ and $b_B = b_0 + C_{ij}$, $\mathbb{E}[m_*(p) \mid (i, j)] = r_* b_B / (a_B - 1)$ when $a_B > 1$, and $\mathbb{V}[m_*(p) \mid (i, j)] = r_*^2 b_B (a_B + b_B - 1) / \{(a_B - 1)^2 (a_B - 2)\}$ when $a_B > 2$. Moment numerators $A_{ij}^{(r)}$ therefore multiply $A_{ij}^{(0)}$ by the observation-mean moments, not by raw Beta moments of p , unless the exported quantity is explicitly labelled parameter-scale. When r is unknown, a semi-conjugate EM or one-dimensional quadrature in r can be layered on top.

Implementation. Algorithm 4 summarizes the Negative-Binomial block computation in the same interface as Algorithms 1–3.

Algorithm 4: NegBinBlock(y, r, i, j, a_0, b_0)

Input: Block $(i, j]$; counts $y_{i+1:j}$; dispersion r (possibly per- t)

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $N \leftarrow \sum_{t=i+1}^j r_t$;
- 2 $\log A0 \leftarrow \sum_{t=i+1}^j \log \binom{y_t + r_t - 1}{y_t} + \log B(a_0 + N, b_0 + C) - \log B(a_0, b_0)$;
- 3 $A0 \leftarrow \exp(\log A0)$;
- 4 $a_B \leftarrow a_0 + N$; $b_B \leftarrow b_0 + C$;
- 5 $\text{mean_per_disp} \leftarrow b_B / (a_B - 1)$ if $a_B > 1$;
 $\text{var_per_disp} \leftarrow b_B (a_B + b_B - 1) / \{(a_B - 1)^2 (a_B - 2)\}$ if $a_B > 2$;

Output: $A0$, posterior summaries for p , and observation-mean moments

$r_* \times \text{mean_per_disp}$ when a prediction dispersion r_* is specified

4.2.5 Beta-distributed observations with known precision

Proposition 4.8 (Beta-block evidence as a one-dimensional integral). *Assume $y_t \in (0, 1)$ with latent mean μ constant within a block, known precisions $\phi_t > 0$, and prior $\mu \sim \text{Beta}(a_0, b_0)$. Define*

$$\begin{aligned}\ell_{ij}(\mu) &:= \sum_{t=i+1}^j \log \text{BetaPDF}(y_t \mid \phi_t \mu, \phi_t(1 - \mu)), \\ \log \pi(\mu) &:= \log \text{BetaPDF}(\mu \mid a_0, b_0).\end{aligned}$$

Then the exact block evidence and moment numerators are the one-dimensional integrals

$$A_{ij}^{(r)} = \int_0^1 \mu^r \exp\{\ell_{ij}(\mu) + \log \pi(\mu)\} d\mu, \quad r \in \{0, 1, 2, \dots\}.$$

A Q -node Gauss–Legendre rule on $(0, 1)$ gives the deterministic approximation

$$\hat{A}_{ij}^{(r)} = \sum_{q=1}^Q w_q^{\text{GL}} \mu_q^r \exp\{\ell_{ij}(\mu_q) + \log \pi(\mu_q)\}.$$

Its convergence and error rate depend on the endpoint behavior and regularity of the transformed integrand. No monotonicity in Q and no routine-wide algebraic rate are asserted here.

Proof. The unknown block parameter is one-dimensional and supported on $(0, 1)$. Multiplication of the Beta-observation likelihood by the Beta prior gives the displayed nonnegative integrand. Its integral is exactly the block evidence; multiplication by μ^r gives the moment numerator. The finite weighted sum is the Gauss–Legendre approximation after the standard affine map from $(-1, 1)$ to $(0, 1)$. $\square$

Remark 4.9 (Quadrature error assessment). The block log-posterior $\ell_{ij}(\mu) + \log \pi(\mu)$ is not generally log-concave and may have non-smooth endpoint behavior after exponentiation. Implementations therefore compare nested node counts or an independently more accurate reference calculation, evaluate in log space, and inspect endpoint contributions. A certified error statement must be tied to the chosen transformation, node rule, and integrand regularity.

Implementation. Algorithm 5 implements this one-dimensional quadrature routine. It is useful for continuous proportions and other $(0, 1)$ -valued signals, and is a deterministic non-conjugate block integral rather than a conjugate closed form.

Algorithm 5: BetaObsBlock($y, \phi, i, j, a_0, b_0, \{\mu_g, w_g\}_{g=1}^G$)

Input: Block $(i, j]$; observations $y_{i+1:j} \in (0, 1)$; known precisions $\phi_{i+1:j}$; Beta prior (a_0, b_0) ; quadrature nodes $\{\mu_g, w_g\}$

```

1  $S_0 \leftarrow 0, S_1 \leftarrow 0, S_2 \leftarrow 0;$ 
2 for  $g = 1$  to  $G$  do
3    $\log \ell_g \leftarrow \sum_{t=i+1}^j \log \text{BetaPDF}(y_t \mid \phi_t \mu_g, \phi_t (1 - \mu_g));$ 
4    $\log \pi_g \leftarrow \log \text{BetaPDF}(\mu_g \mid a_0, b_0);$ 
5    $v_g \leftarrow \exp(\log \ell_g + \log \pi_g);$ 
6    $S_0 \leftarrow S_0 + w_g v_g; \quad S_1 \leftarrow S_1 + w_g v_g \mu_g; \quad S_2 \leftarrow S_2 + w_g v_g \mu_g^2;$ 
7  $A_0 \leftarrow S_0; \quad \mu_{\text{mean}} \leftarrow S_1/S_0; \quad \mu_{\text{var}} \leftarrow S_2/S_0 - (\mu_{\text{mean}})^2;$ 
Output:  $A_0, \mu_{\text{mean}}, \mu_{\text{var}}$ 

```

4.2.6 Summary of block families

Table 4.1 collects the block families derived in this section in a single reference view. The conjugate rows expose closed-form ratios of normalizers, while the Beta-observation row supplies deterministic quadrature values. In both cases the DP layer of Section 5 only requires a compatible block-evidence array and is invariant to the family used to fill that array. Numerical-implementation guidance common to all families (log-space arithmetic, `logsumexp` normalization, prefix-sum ordering, and sanity checks on the DP outputs) is collected in the implementation checklist of Section 6.0.6; every numerical routine in this section and in Section 10 should be read against that checklist.

Family	Prior	Posterior	Block evidence $A_{ij}^{(0)}$	Mean $\mathbb{E}[m_*(\theta) \mid (i, j)]$
Gaussian($\mu, \sigma^2/w$)	$\mathcal{N}(\nu, \rho^2)$	$\mathcal{N}(\nu_B, \rho_B^2)$	closed-form normal ratio	$(\rho^2 S + \sigma^2 \nu)/\kappa$
Poisson(λw)	Gamma(a_0, b_0)	Gamma($a_0 + C, b_0 + W$)	$\propto b_0^{a_0} \Gamma(a_0 + C)/(b_0 + W)^{a_0 + C}$	$(a_0 + C)/(b_0 + W)$
Binomial(m, p)	Beta(a_0, b_0)	Beta($a_0 + C, b_0 + M - C$)	$B(a_0 + C, b_0 + M - C)/B(a_0, b_0)$	$(a_0 + C)/(a_0 + b_0 + M)$
NegBin(r, p)	Beta(a_0, b_0)	Beta(a_B, b_B)	$B(a_B, b_B)/B(a_0, b_0)$	$r_* b_B/(a_B - 1)$
Beta-obs(μ, ϕ_t)	Beta(a_0, b_0)	(1D quadrature)	Gauss–Legendre nodes	from quadrature

Table 4.1. Block families derived in Section 4.2. Base-measure factor $\exp\{H_{ij}\}$ is implicit in the conjugate rows. The first three rows are closed-form conjugate examples; the Negative-Binomial row distinguishes parameter-scale Beta moments from the observation-mean target; and the Beta-observation row is non-conjugate but remains compatible with the DP layer through one-dimensional quadrature. This is a derivation/reference table; no source CSV is associated with it.

4.2.7 Posterior-predictive formulas for common conjugate families (optional)

For groups that export segment-level conjugate posteriors, the segment posterior-predictive (11.1) reduces to standard closed forms: Gaussian (known variance) with $\mu \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2)$ and $y \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w)$ gives $y \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2 + \sigma^2/w)$; Poisson–Gamma with $\lambda \mid \text{train} \sim \text{Gamma}(a_B, b_B)$ and exposure w gives a negative-binomial predictive in y ; Binomial–Beta with $p \mid \text{train} \sim \text{Beta}(a_B, b_B)$ and trial count m gives a Beta-Binomial,

$$p(y \mid m, \text{train}) = \binom{m}{y} \frac{B(a_B + y, b_B + m - y)}{B(a_B, b_B)}.$$

For M new Gaussian points within one segment, the joint posterior-predictive has rank-one-plus-diagonal covariance and admits $\mathcal{O}(M)$ log-density evaluation via Sherman–Morrison.

Sum-product partition inference

CHAPTER PATH

Forward/backward identities, evidence, segment-count uncertainty, boundary marginals, and model-averaged Bayes curves.

For $k \in \{0, \dots, k_{\max}\}$ and $j \in \{0, \dots, n\}$, let L_{kj} denote the evidence of the prefix $y_{1:j}$ under a segmentation into k segments (after integrating out segment parameters), and let R_{ki} denote the evidence of the suffix $y_{i+1:n}$ under k segments. We adopt the conventions

$$L_{0,0} = 1, \quad L_{0,j} = 0 \ (j > 0), \quad R_{0,n} = 1, \quad R_{0,i} = 0 \ (i < n).$$

5.0.1 Forward/backward recursions

Given single-block evidences $A_{ij}^{(0)}$, the forward and backward recursions are

$$L_{k+1,j} = \sum_{h=k}^{j-1} L_{k,h} A_{hj}^{(0)}, \quad R_{k+1,i} = \sum_{h=i+1}^{n-k} A_{ih}^{(0)} R_{k,h}, \quad (5.1)$$

for $k \geq 0$, $1 \leq j \leq n$, and $0 \leq i \leq n-1$, where sums are restricted to valid indices.

Proposition 5.1 (Forward-backward duality and total-evidence identity). *Under Assumption 3.1 and Assumption 3.6, with prior-adjusted block scores $\tilde{A}_{ij}^{(0)} = A_{ij}^{(0)} \gamma_x(i, j)$ (5.2), let $\{\tilde{L}_{k,j}\}$ and $\{\tilde{R}_{k,i}\}$ denote the prefix and suffix sum-product messages of (5.1). Then for every admissible $k \in \{1, \dots, k_{\max}\}$:*

- (i) Boundary conditions. $\tilde{L}_{0,0} = 1$, $\tilde{L}_{0,j} = 0$ for $j > 0$, $\tilde{R}_{0,n} = 1$, and $\tilde{R}_{0,i} = 0$ for $i < n$. Furthermore $\tilde{L}_{k,j} = 0$ whenever $j < k$ and $\tilde{R}_{k,i} = 0$ whenever $i > n - k$.
- (ii) Total-evidence identity. $\tilde{L}_{k,n} = \tilde{R}_{k,0} = \sum_{t \in T_{k,n}^+} \prod_{q=1}^k \tilde{A}_{t_{q-1}t_q}^{(0)}$, i.e. the forward-evidence terminal and the backward-evidence initial value both equal the same prior-adjusted partition sum.
- (iii) Boundary marginal product form. For interior boundary position $h \in \{p, \dots, n - (k - p)\}$ and $1 \leq p \leq k - 1$,

$$p(t_p = h \mid y, k) = \frac{\tilde{L}_{p,h} \tilde{R}_{k-p,h}}{\tilde{L}_{k,n}},$$

and the boundary-event marginal of Definition 3.4(a) satisfies $\text{MB}_h(k) = \sum_{p=1}^{k-1} p(t_p = h \mid y, k)$.

Proof. Boundary conditions follow from definition; the support constraints $j \geq k$ and $i \leq n - k$ restate that a k -segment prefix requires at least k observations and a k -segment suffix requires at least k observations. For (ii), iterate (5.1) forward from $\tilde{L}_{0,0}$: by induction on k , $\tilde{L}_{k,j} = \sum_{t_0 < \dots < t_k = j} \prod_{q=1}^k \tilde{A}_{t_{q-1}t_q}^{(0)}$, so $\tilde{L}_{k,n}$ enumerates all k -segmentations of $\{1, \dots, n\}$. The same induction backward from $\tilde{R}_{0,n}$ gives $\tilde{R}_{k,i} = \sum_{i=t_0 < \dots < t_k = n} \prod_{q=1}^k \tilde{A}_{t_{q-1}t_q}^{(0)}$, and $\tilde{R}_{k,0}$ enumerates the same set. Both equal the asserted partition sum. For (iii), split each segmentation at the p th boundary: a segmentation with $t_p = h$ is uniquely described by its first p blocks (counted by $\tilde{L}_{p,h}$) and its last $k - p$ blocks (counted by $\tilde{R}_{k-p,h}$). Their product, divided by $\tilde{L}_{k,n}$, is the conditional probability that $t_p = h$. Summing over $p \in \{1, \dots, k - 1\}$ yields the boundary-event marginal because the boundaries are strictly ordered, so for fixed h at most one p satisfies $t_p = h$ in any given t . $\square$

The identity $\tilde{L}_{k,n} = \tilde{R}_{k,0}$ is also one of the DP invariants used in the runtime correctness checklist of Section 6.0.6; Proposition 5.1 states the algebraic basis of that numerical check.

Implementation. Algorithm 6 presents the algebraic sum-product recursions corresponding to (5.1). It is a mathematical specification rather than the recommended floating-point implementation: Section 6.0.2 gives the implementable log-space version using `logsumexp`. The algebraic routine takes a triangular block-evidence array and returns the forward messages, backward messages, and segment-count posteriors needed for boundary and moment extraction.

Algorithm 6: Generic algebraic DP for EF segmentation (prefix/suffix form)

Input: Block evidences $\tilde{A}_{ij}^{(0)}$ for $0 \leq i < j \leq n$; $k_{\max}$; C_k (default $C_k = \binom{n-1}{k-1}$)

- 1 Initialize $\tilde{L}_{0,0} = 1$, $\tilde{L}_{0,j>0} = 0$; $\tilde{R}_{0,n} = 1$, $\tilde{R}_{0,i<n} = 0$;
- 2 **for** $k = 0$ **to** $k_{\max} - 1$ **do**
- 3 **for** $j = 1$ **to** n **do**
- 4 $\tilde{L}_{k+1,j} \leftarrow \sum_{h=k}^{j-1} \tilde{L}_{k,h} \tilde{A}_{hj}^{(0)}$;
- 5 **for** $i = 0$ **to** $n - 1$ **do**
- 6 $\tilde{R}_{k+1,i} \leftarrow \sum_{h=i+1}^{n-k} \tilde{A}_{ih}^{(0)} \tilde{R}_{k,h}$;
- 7 Compute $P(k | y) \propto p(k) \tilde{L}_{kn} / C_k$ (cf. (5.4)) and choose $\hat{k}$;
- 8 **for** $p = 1$ **to** $\hat{k} - 1$ **do**
- 9 For $h \in \{p, \dots, n - (\hat{k} - p)\}$ set $P(t_p = h | y, \hat{k}) \leftarrow \tilde{L}_{p,h} \tilde{R}_{\hat{k}-p,h} / \tilde{L}_{\hat{k}n}$;
- 10 Record the marginal mode $\arg \max_h P(t_p = h | y, \hat{k})$ as a diagnostic only;

Output: $\{\tilde{L}_{kj}\}$, $\{\tilde{R}_{ki}\}$, $P(k | y)$, $\hat{k}$, and boundary-marginal diagnostics

5.0.2 Incorporating segment-factorized boundary priors

Suppose the boundary prior given k factorizes over segment lengths as in (3.8):

$$p(t | k) = \frac{1}{C_k} \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)), \quad C_k := \sum_{t: 0=t_0 < \dots < t_k=n} \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)).$$

Then we can absorb g into block evidences by defining

$$\tilde{A}_{ij}^{(0)} := A_{ij}^{(0)} \gamma_x(i, j), \quad \tilde{A}_{ij}^{(r)} := A_{ij}^{(r)} \gamma_x(i, j). \quad (5.2)$$

Running the DP (5.1) with $\tilde{A}_{ij}^{(0)}$ computes $\tilde{L}_{kn}$, the *unnormalized* evidence sum over all segmentations weighted by g . The normalizer C_k can be computed by the same DP with $A_{ij}^{(0)} \equiv 1$:

$$C_k = \sum_t \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)) = L_{kn}^{(g)}, \quad \text{where} \quad L_{k+1,j}^{(g)} = \sum_{h=k}^{j-1} L_{k,h}^{(g)} \gamma_x(h, j), \quad L_{0,0}^{(g)} = 1. \quad (5.3)$$

In the index-uniform case $g \equiv 1$, (5.3) reduces to $C_k = \binom{n-1}{k-1}$.

5.0.3 Posteriors over k , boundary marginals, and joint MAP segmentations

Let $p(k)$ be a prior on segment count. Under index-uniform $p(t | k)$ we have $C_k = \binom{n-1}{k-1}$ and

$$P(y | k) = \frac{L_{kn}}{C_k}, \quad P(k | y) \propto p(k) \frac{L_{kn}}{C_k}.$$

For general segment-factorized $p(t | k)$, replace L by $\tilde{L}$ and C_k by (5.3). In all cases we can write compactly

$$P(k | y) \propto p(k) \frac{\tilde{L}_{kn}}{C_k}, \quad k = 1, \dots, k_{\max}. \quad (5.4)$$

Let $\hat{k} = \arg \max_k P(k | y)$ denote the MAP segment count.

Conditional on k , the marginal posterior for the p th boundary t_p is

$$P(t_p = h | y, k) = \frac{\tilde{L}_{p,h} \tilde{R}_{k-p,h}}{\tilde{L}_{kn}}, \quad h \in \{p, \dots, n - (k - p)\}, \quad (5.5)$$

where $(\tilde{L}, \tilde{R})$ are computed from $\tilde{A}^{(0)}$ as in (5.1). The denominator is the unnormalized evidence under the length-weighted partition prior; the normalizer C_k cancels in (5.5) because the same C_k multiplies every boundary vector in the numerator and denominator. (When $g \equiv 1$, tildes can be dropped.)

Remark 5.2 (Boundary-event marginal). A related object we use repeatedly is the *boundary-event marginal*

$$P(b_i = 1 | y, k) := \sum_{p=1}^{k-1} P(t_p = i | y, k) = \sum_{p=1}^{k-1} \frac{\tilde{L}_{p,i} \tilde{R}_{k-p,i}}{\tilde{L}_{kn}}, \quad 1 \leq i \leq n - 1, \quad (5.6)$$

the posterior probability that interior index i coincides with *some* boundary, without conditioning on which boundary. The objects $P(t_p = h | y, k)$ and $P(b_i = 1 | y, k)$ are distinct: the former is a probability distribution over h for each fixed p (and sums to 1 over h), while the latter is a per-index Bernoulli probability summing to $k - 1$ over i . Calibration experiments in Section 13 target $P(b_i = 1 | y, k)$; a full derivation is given in Appendix 5.

A useful diagnostic is the *marginal boundary mode*

$$\tilde{t}_p^{\text{marg}}(k) \in \arg \max_h P(t_p = h | y, k).$$

However, maximizing the marginals independently need not yield an ordered boundary vector and, in general, does *not* recover the joint MAP segmentation.

Remark 5.3 (A small counterexample). Consider $n = 5$ and $k = 3$, so the two interior boundaries satisfy $1 \leq t_1 < t_2 \leq 4$. For example, let the joint posterior put masses 0.30, 0.28, 0.22, 0.20 on the four ordered configurations (1, 4), (2, 3), (2, 4), (3, 4), respectively. The marginal modes are then $t_1 = 2$ and $t_2 = 4$, so the independently selected marginal-mode vector is (2, 4), even though the joint MAP configuration is (1, 4). This is exactly the kind of discrepancy the max-sum DP resolves.

The joint MAP segmentation is the solution of

$$\hat{t}^{\text{MAP}}(k) \in \arg \max_{0=t_0 < \dots < t_k=n} \left\{ \sum_{q=1}^k \log \tilde{A}_{t_{q-1}t_q}^{(0)} \right\}, \quad (5.7)$$

with the convention that any k -specific prior terms such as $\log p(k) - \log C_k$ are added only when selecting among different values of k . Equation (5.7) is solved exactly by a max-sum DP with backpointers. Inline, the max-sum recursion reads

$$M_{k+1,j} = \max_{h \in \{k, \dots, j-1\}} \{M_{k,h} + \log \tilde{A}_{hj}^{(0)}\}, \quad h^*(k+1, j) \in \arg \max_h \{M_{k,h} + \log \tilde{A}_{hj}^{(0)}\}, \quad (5.8)$$

with initial conditions $M_{0,0} = 0$, $M_{0,j} = -\infty$ for $j > 0$, and backtracking from (k, n) as detailed in Section 6. In the sequel, whenever we report an *exported MAP segmentation*, we mean the joint optimizer $\hat{t}^{\text{MAP}} := \hat{t}^{\text{MAP}}(\hat{k})$ rather than a vector of marginal boundary modes.

5.0.4 Segment-level and regression-curve posterior moments

For segment-wise posterior moments attached to a *fixed* segmentation $t = (t_0, \dots, t_k)$, let segment $q \in \{1, \dots, k\}$ correspond to block $(i, j] = (t_{q-1}, t_q]$. Then

$$\mu_q^{(r)}(t) = \frac{\tilde{A}_{ij}^{(r)}}{\tilde{A}_{ij}^{(0)}} = \frac{A_{ij}^{(r)}}{A_{ij}^{(0)}}, \quad r = 1, 2, \dots, \quad (5.9)$$

where the final equality holds because the prior factor $\gamma_x(i, j)$ cancels. In particular, for an exported joint MAP segmentation $\hat{t}^{\text{MAP}}$, the segment summaries are obtained by setting $t = \hat{t}^{\text{MAP}}$ in (5.9).

To compute the Bayes regression curve within a fixed k (typically $k = \hat{k}$), define, for each block $(i, j]$ and moment order r ,

$$F_{ij}^{(r)}(k) := \frac{1}{\tilde{L}_{kn}} \sum_{m=1}^k \tilde{L}_{m-1,i} \tilde{A}_{ij}^{(r)} \tilde{R}_{k-m,j}.$$

Then the r th posterior moment of the latent mean at index t given k is obtained by summing $F_{ij}^{(r)}(k)$ over all candidate blocks $(i, j]$ that contain the index t (i.e., all blocks with $i < t \leq j$), so that each candidate segmentation contributes exactly once through the block that actually covers t :

$$\hat{\mu}_t'^{(r)} := \mathbb{E}[(\mu_t')^r \mid y, k] = \sum_{i < t \leq j} F_{ij}^{(r)}(k), \quad t = 1, \dots, n. \quad (5.10)$$

In particular, $\hat{\mu}_t' := \hat{\mu}_t'^{(1)}$ is the Bayes regression curve and $\mathbb{V}[\mu_t' \mid y, k] = \hat{\mu}_t'^{(2)} - (\hat{\mu}_t'^{(1)})^2$. If a fully k -marginalized curve is desired, compute the same conditional curve for each admissible k and average the resulting moments using $P(k \mid y)$ from (5.4). The empirical sections report curves conditional on the selected $\hat{k}$ unless explicitly stated otherwise.

Proposition 5.4 (Block-covering decomposition for Bayes-curve moments). *Fix k and an index $t \in \{1, \dots, n\}$. Under the support and factorization hypotheses of Theorem 5.5, the conditional posterior moment of the latent mean at t is obtained by summing the block contributions $F_{ij}^{(r)}(k)$ over exactly those candidate blocks $(i, j]$ with $i < t \leq j$. No segmentation contributes more than once to the sum in (5.10).*

Proof. Every admissible ordered boundary vector partitions $\{1, \dots, n\}$ into disjoint blocks, hence contains a unique block $(i, j]$ satisfying $i < t \leq j$. Conditioning on that covering block splits the remaining boundaries into an $(m - 1)$ -segmentation of the prefix $y_{1:i}$ and a $(k - m)$ -segmentation of the suffix $y_{j+1:n}$ for some segment position m . The prefix, block, and suffix contributions multiply as $\tilde{L}_{m-1,i} \tilde{A}_{ij}^{(r)} \tilde{R}_{k-m,j}$, and summing over m and over all covering blocks gives (5.10). Disjointness of the covering block proves that no segmentation is double counted. $\square$

Theorem 5.5 (Correctness of the DP). *Assume finite block evidences on all admissible blocks, a finite segment-count range, and a segment-factorized conditional partition prior satisfying Assumption 3.6 with the boundary-coordinate convention of Section 3.1. Under the EF-conjugate single-segment model of Section 4.1, or under any other block model whose exact evidence array is supplied, the recursions (5.1) compute exact marginal evidences for all prefixes/suffixes: $\tilde{L}_{kj}$ and $\tilde{R}_{ki}$ are exact evidence sums over all k -segmentations of the corresponding prefix/suffix. Moreover:*

1. *The posterior $P(k | y)$ given by (5.4) is exact.*
2. *The boundary marginal $P(t_p = h | y, k)$ given by (5.5) is exact.*
3. *The segment-wise moments (5.9) and regression-curve moments (5.10) are exact posterior moments conditional on k .*

The separate max-sum recursion and backtracking theorem for the joint MAP partition is stated and proved immediately after its statement in Theorem 6.1.

Proof. We argue for general $\tilde{A}_{ij}^{(0)}$; $g \equiv 1$ is the untilded specialization.

Step 1: $\tilde{L}_{kj}$ is the prefix evidence sum. For any k -segmentation $(t_0, \dots, t_k)$ of $y_{1:j}$ with $t_0 = 0$, $t_k = j$, integrating out segment parameters yields $p(y_{1:j} | t, k) = \prod_q \tilde{A}_{t_{q-1}, t_q}^{(0)}$. Summing over admissible t gives $\tilde{L}_{kj} = \sum_{t: t_k=j} \prod_{q=1}^k \tilde{A}_{t_{q-1}, t_q}^{(0)}$.

Step 2: forward recursion. Conditioning a $(k+1)$ -segmentation of $y_{1:j}$ on its last boundary $h \in \{k, \dots, j-1\}$ splits it into a k -segmentation of $y_{1:h}$ and the final block $(h, j]$; summing over h gives the forward recursion in (5.1). The backward recursion follows by the same argument applied to suffixes.

Step 3: posterior over k . $P(y | k) = \tilde{L}_{kn}/C_k$ by definition of C_k ; Bayes' rule yields (5.4).

Step 4: boundary posteriors. Conditioning on $t_p = h$ splits the sequence into a p -segmentation of $y_{1:h}$ and a $(k-p)$ -segmentation of $y_{h+1:n}$, contributing $\tilde{L}_{p,h} \tilde{R}_{k-p,h}$; normalizing by $\tilde{L}_{kn}$ yields (5.5).

Step 5: moments. Theorem 4.1 gives $A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[m_{\star}(\theta)^r \mid (i, j)]$; the length factor $\gamma_x(i, j)$ cancels in the ratio, so (5.9) holds. Equation (5.10) follows by grouping segmentation contributions through the unique block $(i, j]$ that contains index t . $\square$

Posterior-evidence sanity checks. After every DP run, verify four invariants: normalized segment-count weights sum to one; the full-sequence forward and backward evidences agree, $\tilde{L}_{kn} = \tilde{R}_{k0}$; boundary-event marginals satisfy $\sum_{i=1}^{n-1} P(b_i = 1 \mid y, k) = k - 1$; and the max-sum backtracking score equals the stored terminal max-sum value. In log-space arithmetic these checks are evaluated with a declared tolerance, typically a small multiple of machine epsilon times the number of accumulated summands. A failure of the first two checks is treated as a normalization or admissibility mismatch; a failure of the boundary-sum check indicates an indexing or support error; and a failure of the backtracking check indicates inconsistent tie-breaking or stale backpointers.

Theorem 5.6 (Time and space complexity). *Assume single-block evidences $A_{ij}^{(0)}$ are available for all $0 \leq i < j \leq n$. Computing all forward and backward messages via (5.1) requires $\mathcal{O}(k_{\max} n^2)$ arithmetic operations. The triangular block-evidence matrix requires $\mathcal{O}(n^2)$ memory, while all forward and backward layers require $\mathcal{O}(k_{\max} n)$ memory. Max-sum values and backpointers each require $\mathcal{O}(k_{\max} n)$ additional memory. Boundary-event probabilities for every k and candidate index require $\mathcal{O}(k_{\max} n)$ output storage. Ordered-boundary distributions for every triple (k, p, h) require $\mathcal{O}(k_{\max}^2 n)$ output storage, or $\mathcal{O}(kn)$ for one fixed segment count k . For R requested reporting moments, retaining all block moment numerators requires $\mathcal{O}(Rn^2)$ memory and the final Bayes curves require $\mathcal{O}(Rn)$; explicitly retaining every block-cover contribution over all k can require $\mathcal{O}(k_{\max} n^2)$ additional workspace but is not necessary when those contributions are accumulated sequentially.*

Proof. For each fixed k , the forward recursion computes $L_{k+1,j}$ for $j = 1, \dots, n$, and each $L_{k+1,j}$ sums at most $j - k$ terms. Thus the forward work per k is $\sum_{j=1}^n \mathcal{O}(j) = \mathcal{O}(n^2)$. The backward recursion is analogous. Repeating for $k = 0, \dots, k_{\max} - 1$ gives $\mathcal{O}(k_{\max} n^2)$ time. A message or backpointer array has one entry per (k, j) state. Boundary-event output has one value per (k, h) , whereas the full ordered-boundary output has one value per (k, p, h) and therefore $\sum_{k \leq k_{\max}} \mathcal{O}(kn) = \mathcal{O}(k_{\max}^2 n)$ entries. The remaining statements follow by counting the triangular block arrays, reporting-moment arrays, and optional block-cover workspace. $\square$

Corollary 5.7 (Decision rules for full-vector and coordinatewise losses). *For the loss $\ell_{01}(t, t') = \mathbb{1}\{t \neq t'\}$ on the full boundary vector, the Bayes estimator is the joint MAP segmentation $\hat{t}^{\text{MAP}}(k)$ from (5.7). For a separable coordinatewise loss and an unconstrained Cartesian action space, each coordinate can instead be chosen by minimizing its marginal posterior risk; under coordinatewise 0–1 loss this gives the marginal modes. If the reported action must belong to the strictly ordered partition space $\mathcal{T}_{k,n}^+$, coordinatewise marginal modes can be inadmissible, and the Bayes action must minimize posterior expected loss over that constrained set. It can differ from both the coordinatewise marginal modes and the joint MAP partition.*

Proof. The posterior risk of t' under full-vector 0–1 loss is $1 - p(t = t' \mid y, k)$, so any joint posterior mode minimizes it. Under a separable loss on an unconstrained Cartesian action space, the posterior risk is a sum of coordinatewise risks, each minimized using the corresponding marginal posterior. Restricting the action to strictly ordered boundary vectors couples the

coordinates, so independent coordinatewise minimization no longer solves the constrained decision problem. $\square$

Remark 5.8 (Memory/time trade-offs for large n). Keeping only the current and previous k -layers reduces message memory to $\mathcal{O}(n)$, in addition to any stored block array, when only terminal fixed- k evidences are required. Posterior marginals and MAP paths then require retained layers or a documented recomputation pass. Checkpoint spacing can trade retained layers against recomputation, but its exact cost depends on which summaries are requested and is therefore an implementation choice rather than a universal bound. When the n^2 block-evidence matrix is prohibitive and every segment score is available in constant time from prefix summaries, scores can instead be evaluated on demand during the DP sweep. This uses less block storage but performs up to $\mathcal{O}(k_{\max}n^2)$ segment evaluations because the same segment can be revisited at different DP layers.

Remark 5.9 (Evidence across segment counts need not be monotone). The fixed- k evidence $P(y | k) = \tilde{L}_{kn}/C_k$ is not monotone in general. Adding a boundary changes both the set of admissible partitions and the product of block evidences, while the normalized conditional prior contributes the normalizer C_k . Consequently, diagnose segment-count behavior through the normalized evidences $P(y | k)$ and posterior weights $P(k | y)$ rather than through an unnormalized pathwise “add one boundary” heuristic.

5.0.5 End-to-end BayesBreak inference routine

Collecting the ingredients from Sections 4.1 and 5—block-evidence precomputation, exact sum-product DP over partitions, and max-sum backtracking for exported segmentations—yields an end-to-end routine suitable for direct implementation. Algorithm 7 summarizes this workflow in a modular way: the only model-specific input is a function that evaluates (or approximates) block evidences $\log \mathcal{L}_{ij}$ and, when needed, block moment numerators.

Algorithm 7: BayesBreak($y, x, w, \text{model}, \text{hyper}, k_{\max}, p(k), g$)

Input: Observations $y_{1:n}$ at locations $x_{1:n}$ (optional); observation-model weights/exposures $w_{1:n}$ (default determined by the chosen likelihood, often $w_i \equiv 1$);

model family $\text{model} \in \{\text{Gaussian}, \text{Poisson}, \text{Binomial}, \text{BetaObs}, \dots\}$;

hyperparameters hyper ; maximum segment count $k_{\max}$; prior $p(k)$; length factor $g(\cdot)$ (default $g \equiv 1$).

Output: Posterior $P(k | y)$; MAP segment count $\hat{k}$; boundary marginals $P(t_p | y, \hat{k})$; exported joint MAP segmentation $\hat{t}^{\text{MAP}}$; segment-level moments; Bayes regression curve; (optional) evidence $P(y)$.

1 **1. Precompute block evidences and moments;**

2 **for** $0 \leq i < j \leq n$ **do**

3 Compute block summaries (S_{ij}, W_{ij}, H_{ij}) using prefix sums (or running sums);

4 Call a family-specific routine to obtain $A_{ij}^{(0)}, A_{ij}^{(1)}, A_{ij}^{(2)}$;

5 Set $\tilde{A}_{ij}^{(r)} \leftarrow A_{ij}^{(r)} \gamma_x(i, j)$ for $r \in \{0, 1, 2\}$ (cf. (5.2));

6 Compute C_k under the same admissibility and length-prior convention used in the score array; for $g \equiv 1$, use $C_k = \binom{n-1}{k-1}$, and otherwise use (5.3) (DP with $A_{ij}^{(0)} \equiv 1$);

7 **2. Run the sum-product dynamic program;**

8 Compute $\tilde{L}_{kj}, \tilde{R}_{ki}$ via (5.1) for $k = 0, \dots, k_{\max}$ using $\tilde{A}_{ij}^{(0)}$;

9 Compute $P(k | y) \propto p(k) \tilde{L}_{kn}/C_k$ via (5.4); set $\hat{k} \leftarrow \arg \max_k P(k | y)$;

10 **3. Boundary marginals;**

11 **for** $p = 1$ **to** $\hat{k} - 1$ **do**

12 For all $h \in \{p, \dots, n - (\hat{k} - p)\}$, set $P(t_p = h | y, \hat{k}) \leftarrow \tilde{L}_{p,h} \tilde{R}_{\hat{k}-p,h} / \tilde{L}_{\hat{k}n}$ (cf. (5.5));

13 **4. Export a joint MAP segmentation;**

14 Run the max-sum/backpointer recursion of Section 6 using $\log \tilde{A}_{ij}^{(0)}$ and the chosen count $\hat{k}$;

15 Backtrack to obtain $\hat{t}^{\text{MAP}} = (0 = \hat{t}_0^{\text{MAP}} < \dots < \hat{t}_{\hat{k}}^{\text{MAP}} = n)$;

16 **5. Segment-level posterior moments on the exported segmentation;**

17 **for** $q = 1$ **to** $\hat{k}$ **do**

18 Let $(i, j) \leftarrow (\hat{t}_{q-1}^{\text{MAP}}, \hat{t}_q^{\text{MAP}}]$;

19 For $r \in \{1, 2\}$ set $\hat{\mu}_q^{(r)} \leftarrow \tilde{A}_{ij}^{(r)} / \tilde{A}_{ij}^{(0)}$ (cf. (5.9));

20 **6. Bayes regression curve within $\hat{k}$;**

21 Using $\tilde{L}, \tilde{R}, \tilde{A}^{(r)}$ and (5.10), compute $\hat{\mu}'_i$ and $\mathbb{V}[\mu'_i | y, \hat{k}]$ for $i = 1, \dots, n$;

22 **7. Optional: pooling/groups;**

23 If multiple subjects or groups are present, pool block evidences (Theorem 8.2) and/or run the finite latent-group template procedure of Chapter 9, then repeat Steps 2–6.

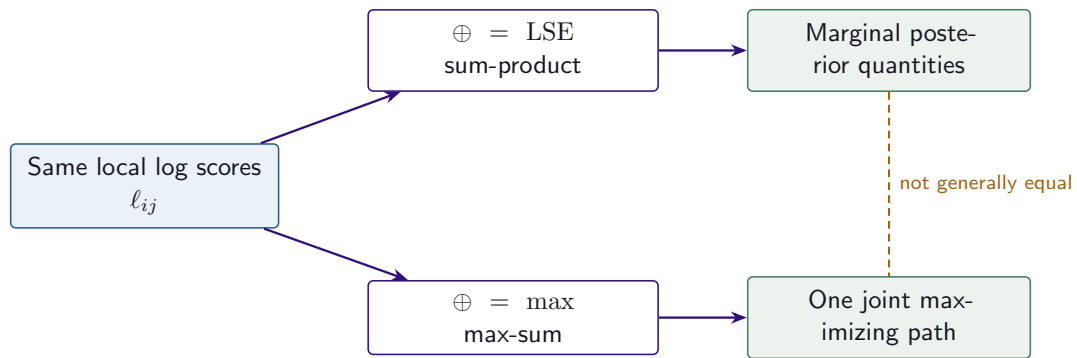


Figure 5.1. Same local scores, different semirings. Sum-product produces marginal posterior quantities; max-sum produces one joint maximizing path.

Joint MAP recovery, algorithms, and numerical stability

CHAPTER PATH

Implementation-grade recursions, an in-place correctness proof, complexity, tie behavior, and log-space diagnostics.

ESTIMATOR SEPARATION

The vector of marginal boundary modes is not generally a valid joint optimizer and can even contain crossing or duplicate positions under naive assembly. BayesBreak exports the joint MAP path only from max-sum DP with backtracking. Sum-product outputs remain posterior summaries.

Section 5 derives the exact dynamic program (DP) that sits at the core of BayesBreak. The present section complements that development with implementation-ready pseudocode and a collection of numerical-stability considerations. The primary purpose is pragmatic: to make explicit what must be computed, stored, and normalized in order to reproduce the theoretical quantities (evidences, boundary posteriors, Bayes curves) without relying on undocumented conventions.

6.0.1 Block-evidence precomputation via prefix sums

For conjugate exponential-family blocks, Theorem 4.1 shows that the single block evidence and moments can be expressed as functions of aggregated sufficient statistics. This enables an *exact* $\mathcal{O}(n^2)$ precomputation of all candidate blocks $(i, j]$ with $0 \leq i < j \leq n$. Throughout this section we use the body convention: a block $(i, j]$ has length $j - i$ and contains the indices $i + 1, \dots, j$.

In practice, it is numerically preferable to store *log* block evidences,

$$\ell_{ij}^{(0)} := \log \tilde{A}_{ij}^{(0)}, \quad (6.1)$$

and similarly for positive moment numerators required for the Bayes regression curve. If a chosen moment numerator can be zero or negative, implementations use tagged signed-log storage rather than silently taking a logarithm: a signed quantity m_{ij} is represented as (s_{ij}, a_{ij}) with $s_{ij} \in \{-1, 0, 1\}$ and $a_{ij} = \log |m_{ij}|$ when $s_{ij} \neq 0$, while $s_{ij} = 0$ records exact zero. Each `BlockRoutine` must return a finite log evidence for admissible blocks, $-\infty$ for excluded

blocks, optional positive log-moment numerators, and optional signed-log moment records with this zero convention. Downstream DP code must reject untagged nonpositive moment numerators rather than treating them as log values. Algorithm 8 expresses the precomputation at an abstract level, with a single call to a `BlockRoutine` that implements the family-specific closed forms given in Section 4.2.

Algorithm 8: `PrecomputeBlocks( $y, x, w, \text{BlockRoutine}$ )`

Input: Ordered observations $\{(x_i, y_i, w_i)\}_{i=1}^n$ (with $w_i \equiv 1$ if unweighted); family-specific routine `BlockRoutine` implementing Theorem 4.1.

Output: Matrices of log-block evidences $\ell_{ij}^{(0)} = \log \tilde{A}_{ij}^{(0)}$ on the index set $\{0 \leq i < j \leq n\}$ and (optional) moment numerators, stored as $\log \tilde{A}_{ij}^{(r)}$ only when they are positive and otherwise in a signed-log or linear representation.

- 1 **1. Prefix summaries;**
 - 2 Compute prefix cumulative sufficient statistics and weights required by `BlockRoutine`:
store $W_j := \sum_{t=1}^j w_t$, $S_j := \sum_{t=1}^j w_t T(y_t)$, and any additional family-specific terms (e.g., $Q_j := \sum_{t=1}^j w_t y_t^2$ for Gaussian). Set $W_0 = S_0 = Q_0 = 0$.
 - 3 **2. Enumerate blocks;**
 - 4 **for** $i \leftarrow 0$ **to** $n - 1$ **do**
 - 5 **for** $j \leftarrow i + 1$ **to** n **do**
 - 6 Extract block summaries for $y_{i+1:j}$ via $W_{ij} = W_j - W_i$, $S_{ij} = S_j - S_i$,
 $Q_{ij} = Q_j - Q_i$ (constant time); `BlockRoutine` returns $\log A_{ij}^{(0)}$ and any
moment numerators $\log A_{ij}^{(r)}$; Add any length factor $\log g(\Delta_x(i, j))$ and hazard
constants as in (5.2); Store $\ell_{ij}^{(0)} \leftarrow \log \tilde{A}_{ij}^{(0)}$ and, if needed, $\ell_{ij}^{(r)} \leftarrow \log \tilde{A}_{ij}^{(r)}$.
-

Memory notes. Storing all $\ell_{ij}^{(0)}$ uses $\Theta(n^2)$ memory. When n is large but $k_{\max}$ is moderate, the memory footprint is reduced by streaming blocks in the inner DP loop (computing $\ell_{ij}^{(0)}$ on the fly from prefix statistics), at cost of recomputing some block summaries. The DP theory remains unchanged.

As sanity checks, verify that invalid blocks are represented by $-\infty$ log-scores, that $\log C_k$ is computed with the same admissibility rules as the DP, that forward and backward full-sequence evidences agree, and that boundary-event marginals sum to $k - 1$ conditional on fixed k .

6.0.2 Log-space dynamic programming

The forward/backward DP in Section 5 sums products of block evidences. To avoid under-flow/overflow, work in log space and use a stable `logsumexp` primitive. Algorithm 9 presents a log-space version of the recursions in (5.1)–(5.4).

Algorithm 9: $\text{DPLog}(\ell_{ij}^{(0)}, k_{\max}, \log p(k), \log C_k)$

Input: Log block evidences $\ell_{ij}^{(0)} = \log \tilde{A}_{ij}^{(0)}$ on the index set $\{0 \leq i < j \leq n\}$ (block $(i, j]$); maximum segments $k_{\max}$; log prior on k , $\log p(k)$; log normalizers $\log C_k$ computed under the same admissible-block and length-prior convention as the score array. For the index-uniform fixed- k prior with $g \equiv 1$, $C_k = \binom{n-1}{k-1}$ unless the caller has intentionally absorbed a normalized prior into the block or count scores.

Output: Forward messages $\log \tilde{L}_{k,j}$, backward messages $\log \tilde{R}_{k,i}$, log evidence per k , and normalized posterior $p(k | y)$.

- 1 **1. Initialize;**
- 2 Set $\log \tilde{L}_{0,0} \leftarrow 0$ and $\log \tilde{L}_{0,j} \leftarrow -\infty$ for $j \geq 1$; set $\log \tilde{R}_{0,n} \leftarrow 0$ and $\log \tilde{R}_{0,i} \leftarrow -\infty$ for $i \leq n-1$.
- 3 **2. Forward recursion;**
- 4 **for** $k \leftarrow 1$ **to** $k_{\max}$ **do**
- 5 **for** $j \leftarrow k$ **to** n **do**
- 6 $\log \tilde{L}_{k,j} \leftarrow \text{logsumexp}_{h \in \{k-1, \dots, j-1\}} (\log \tilde{L}_{k-1,h} + \ell_{h,j}^{(0)})$;
- 7 **3. Backward recursion;**
- 8 **for** $k \leftarrow 1$ **to** $k_{\max}$ **do**
- 9 **for** $i \leftarrow n-k$ **to** 0 (decreasing) **do**
- 10 $\log \tilde{R}_{k,i} \leftarrow \text{logsumexp}_{h \in \{i+1, \dots, n-k+1\}} (\ell_{i,h}^{(0)} + \log \tilde{R}_{k-1,h})$;
- 11 **4. Evidence and posterior over k ;**
- 12 For each $k \in \{1, \dots, k_{\max}\}$ set $\log p(y | k) \leftarrow \log \tilde{L}_{k,n} - \log C_k$; compute unnormalized log posterior scores $\log \pi_k \leftarrow \log p(k) + \log p(y | k)$; set $p(k | y) \leftarrow \exp\{\log \pi_k - \text{logsumexp}_{k'}(\log \pi_{k'})\}$.

The backward upper endpoint $n - k + 1$ is the last split that leaves $k - 1$ nonempty suffix blocks to cover indices $h + 1, \dots, n$. Empty summation ranges are interpreted as $-\infty$ in log space, so edge cases are handled by the same admissibility convention as the forward recursion.

Boundary marginals and Bayes curves. Once $\log \tilde{L}$ and $\log \tilde{R}$ are available, the boundary marginal in (5.5) can be computed stably by subtracting the common normalizer $\log \tilde{L}_{k,n}$. Similarly, Bayes regression moments (Proposition 5.4; cf. (5.10)) can be computed by replacing $\ell_{ij}^{(0)}$ with compatible moment numerators and reusing the same log-space DP bookkeeping. Signed or nonpositive moment targets require the signed-log or scaled-linear convention noted in Section 6.0.1. A signed-log accumulator stores each target as a sign bit and a log-magnitude; sums of signed terms are evaluated by separately applying logsumexp to positive and negative parts and subtracting the two scaled results only at the final reporting step. This convention is required for any routine that returns moment numerators rather than marginal likelihoods. The backward-loop bound $h \leq n - k + 1$ in Algorithm 9 enforces the nonempty-suffix convention: after choosing the next endpoint h for the block $(i, h]$, the remaining $k - 1$ blocks must occupy at least one observation each in $(h, n]$. Thus the summation is empty, and the message is $-\infty$, whenever the remaining suffix cannot support the requested number of blocks.

6.0.3 MAP segmentation and backtracking

The DP used for evidences can be augmented to recover a MAP segmentation in $\mathcal{O}(k_{\max}n^2)$ time. During the forward recursion, one stores, for each (k, j) , the maximizing split point

$$h^*(k, j) \in \arg \max_{h \in \{k-1, \dots, j-1\}} \{M_{k-1, h} + \ell_{h, j}^{(0)}\}, \quad (6.2)$$

where $M_{k, j}$ is the max-sum message defined in (5.8). Algorithm 10 computes $M_{k, j}$ and the backpointers; Algorithm 11 then backtracks from $(\hat{k}, n)$ to recover the boundary vector $\hat{b} = (\hat{b}_0, \dots, \hat{b}_{\hat{k}})$.

Theorem 6.1 (Correctness of max-sum DP for joint MAP segmentation). *For any choice of segment count $\hat{k} \in \{1, \dots, k_{\max}\}$, Algorithms 10 and 11 jointly return the (or a) boundary vector $\hat{b} = (\hat{b}_0, \dots, \hat{b}_{\hat{k}})$ that attains*

$$\hat{b} \in \arg \max_{t \in T_{\hat{k}, n}} \left\{ \log p(\hat{k}) - \log C_{\hat{k}} + \sum_{q=1}^{\hat{k}} \ell_{t_{q-1}, t_q}^{(0)} \right\},$$

where $T_{\hat{k}, n}$ is the set of admissible boundary vectors of length $\hat{k}$ on n observations under the segment-factorized partition prior of Section 5. In particular, $\hat{b}$ is a joint MAP segmentation conditional on the chosen $\hat{k}$, and any tie among maximizers is broken deterministically by the back-pointer convention encoded in $h^*(k, j)$. The runtime is $\mathcal{O}(k_{\max}n^2)$ and the working space is $\mathcal{O}(k_{\max}n)$.

Proof. For $q = 1$, $V_{1, j} = \ell_{0, j}$ is the best score among the sole one-block partition of $(0, j]$. Assume $V_{q-1, h}$ equals the best score over all admissible $(q-1)$ -block partitions ending at h . Every admissible q -block partition ending at j has a unique penultimate endpoint h and score equal to a $(q-1)$ -block prefix score plus $\ell_{h, j}$. Maximizing over h therefore gives $V_{q, j} = \max_h \{V_{q-1, h} + \ell_{h, j}\}$. Induction proves optimality at (k, n) . Storing one deterministically tie-broken maximizing predecessor at each state and following those predecessors backward reconstructs a path attaining $V_{k, n}$; hence the returned boundary vector is a joint MAP partition for fixed k . $\square$

Algorithm 10: MAPForward($\ell_{ij}^{(0)}, k_{\max}$)

Input: Log block evidences $\ell_{ij}^{(0)}$ on $\{0 \leq i < j \leq n\}$; $k_{\max}$.

Output: Max-sum messages $M_{k, j}$ and backpointers $h^*(k, j)$ for $k \leq k_{\max}$.

```

1 Set  $M_{0,0} \leftarrow 0$ ,  $M_{0,j} \leftarrow -\infty$  for  $j \geq 1$ ;
2 for  $k \leftarrow 1$  to  $k_{\max}$  do
3   for  $j \leftarrow k$  to  $n$  do
4      $h^*(k, j) \leftarrow \arg \max_{h \in \{k-1, \dots, j-1\}} \{M_{k-1, h} + \ell_{h, j}^{(0)}\}$ ;
5      $M_{k, j} \leftarrow M_{k-1, h^*(k, j)} + \ell_{h^*(k, j), j}^{(0)}$ ;
```

Algorithm 11: MAPBacktrack($h^*, \hat{k}$)**Input:** Backpointers $h^*(k, j)$ as in (6.2); chosen segment count $\hat{k}$.**Output:** MAP boundary vector $\hat{b}_0 = 0 < \hat{b}_1 < \dots < \hat{b}_{\hat{k}} = n$.

- 1 Set $\hat{b}_{\hat{k}} \leftarrow n$ and $q \leftarrow \hat{k}$. **while** $q \geq 1$ **do**
- 2 $\hat{b}_{q-1} \leftarrow h^*(q, \hat{b}_q);$
- 3 $q \leftarrow q - 1;$
- 4 **Return** $\hat{b}$.

6.0.4 Pooling across sequences and groups

For replicated or multi-subject data with shared boundaries (Section 8), Theorem 8.2 implies that subject-level block evidences multiply. In log space, pooling is therefore a simple sum:

$$\ell_{ij}^{(0, \text{pooled})} = \sum_{s=1}^S \ell_{ij}^{(0, s)}. \quad (6.3)$$

The same applies to known groups, with the sum taken over subjects in a group. The resulting pooled matrices can be passed unchanged to Algorithm 9.

Algorithm 12: Pooled-DP for shared-boundary multi-subject inference (log-space)

Input: Per-subject log-block-evidence arrays $\{\ell_{ij}^{(0, s)}\}_{s=1}^S$ for $0 \leq i < j \leq n$; local prior factors $\log c_x(i, j)$ and interior hazards $\log h_x(j)$; $k_{\max}$; group-label set $\{z_s\}_{s=1}^S$ if pooling by known group, else $z_s \equiv 1$ for global pooling.

Output: Per-group pooled posteriors $P_g(k | y)$, $P_g(t_p | y, k)$, MAP segmentations $\hat{t}_g^{\text{MAP}}(\hat{k}_g)$, and segment-conditional parameter posteriors.

- 1 **for** each distinct group label $g \in \{z_s\}$ **do**
- 2 Let $\mathcal{S}_g = \{s : z_s = g\};$
- 3 $\ell_{ij, g}^{(0, \text{pool})} \leftarrow \sum_{s \in \mathcal{S}_g} \ell_{ij}^{(0, s)}$ for every admissible $(i, j);$
 // The local prior factor is absorbed once at the pooled level.
- 4 $\tilde{\ell}_{ij, g}^{(0, \text{pool})} \leftarrow \ell_{ij, g}^{(0, \text{pool})} + \log \gamma_x(i, j);$
- 5 Run Algorithm 9 with score $\tilde{\ell}_{ij, g}^{(0, \text{pool})}$ to obtain $P_g(k | y)$, $P_g(t_p | y, k)$, $\hat{k}_g$;
- 6 Run the max-sum routine (Algorithms 10–11) on the same score to obtain $\hat{t}_g^{\text{MAP}}(\hat{k}_g);$
- 7 Conditional on $\hat{t}_g^{\text{MAP}}$, recover subject-specific segment-parameter posteriors from each $s \in \mathcal{S}_g$ block update;
- 8 **return** $\{P_g(k | y), P_g(t_p | y, k), \hat{t}_g^{\text{MAP}}, \text{per-subject parameter posteriors}\}_g;$
 // Complexity: $\Theta(\sum_g (|\mathcal{S}_g|n^2 + k_{\max}n^2))$ time and $\Theta(n^2 + k_{\max}n)$ working
 memory per group when subject sums are streamed.
 // Numerical: pool in log space and use logsumexp; carry per-subject
 log-blocks to floating-point underflow safely; sum $-\infty$
 contributions are absorbed at the pooled step.

6.0.5 Complexity summary and stability

Component	Time	Memory	Numerical note
Conjugate block precomputation	$\Theta(n^2)$	$\Theta(n^2)$	prefix summaries; invalid blocks stored as $-\infty$
Exact sum-product messages	$\Theta(k_{\max}n^2)$	$\Theta(k_{\max}n)$ in addition to the block matrix	logsumexp normalization
Boundary and Bayes-curve extraction	$\Theta(k_{\max}n^2)$ for streamed fixed- k summaries	output dependent; up to $\Theta(k_{\max}^2n)$ for every ordered-boundary table	common evidence normalizer
Max-sum recursion and backtracking	$\Theta(k_{\max}n^2)$	$\Theta(k_{\max}n)$ for values and backpointers	deterministic tie rule
Non-conjugate precomputation	$\Theta(n^2T_{\text{block}})$	$\Theta(n^2)$ plus routine state	routine convergence/failure flags
Shared-boundary pooling	$\Theta(Sn^2 + k_{\max}n^2)$ after subject blocks	$\Theta(Sn^2)$ or streamed subject sums	sum subject log scores
Latent-group template ascent	per iteration $\Theta(GSn^2 + Gk_{\max}n^2)$	$\Theta(Sn^2 + Gk_{\max}n)$	log-responsibility normalization
Prediction scoring	$\Theta(Gm)$ after segment assignment; $\Theta(Gk_{\max}m^2)$ for resegmentation	problem dependent	log predictive scores

Table 6.1. Implementation-level complexity summary. This is a theoretical accounting table; no source CSV is associated with it.

Proposition 6.2 (BayesBreak time and space complexity). *Suppose the chosen exponential-family block model admits constant-time block-evidence evaluation from cumulative sufficient statistics (Sections 4.1–4.2), and let n denote the number of observations and $k_{\max}$ the maximum segment count considered by the DP. Then:*

1. *Block-evidence precomputation (Algorithm 8) runs in $\Theta(n^2)$ time and stores $\Theta(n^2)$ values.*
2. *The forward and backward DP (Algorithm 9) runs in $\Theta(k_{\max}n^2)$ time and stores $\Theta(k_{\max}n)$ messages. Boundary-event summaries for every k require $\Theta(k_{\max}n)$ output storage; all ordered-boundary distributions for every (k, p, h) require $\Theta(k_{\max}^2n)$ output storage. Explicit storage of every block-cover contribution can require $\Theta(k_{\max}n^2)$, but it is not required for streamed accumulation.*
3. *The max-sum recursion plus backtracking (Algorithms 10–11) runs in $\Theta(k_{\max}n^2)$ time and $\Theta(k_{\max}n)$ working space, including one predecessor per DP state.*
4. *For non-conjugate block models with per-block approximation cost T_{block} (e.g. Laplace, JJ, EP, PG mean field, 1D quadrature), block precomputation costs $\Theta(n^2T_{\text{block}})$ and the rest of the DP is unchanged.*

Proof. There are $n(n+1)/2 = \Theta(n^2)$ admissible blocks. When cumulative sufficient statistics are available, each block score requires constant time and one stored scalar, proving (i). For every $k \leq k_{\max}$ and endpoint j , the forward recurrence scans at most j predecessors. Hence its operation count is bounded above by $\sum_{k=1}^{k_{\max}} \sum_{j=k}^n j = O(k_{\max}n^2)$ and bounded below by a constant multiple of the same order when all blocks are admissible, giving $\Theta(k_{\max}n^2)$. Storing one message per (k, j) is $\Theta(k_{\max}n)$. Boundary-event output has one value per (k, h) , while the complete ordered-boundary output has one value per (k, p, h) and therefore $\Theta(k_{\max}^2n)$ entries. Block-cover terms can be accumulated without retaining every (k, i, j) contribution. The max-sum recurrence visits the same predecessor sets, stores one value and one predecessor per state, and backtracking takes only $O(k_{\max})$, proving (iii). Finally, if one local block evaluation costs T_{block} , evaluating all $\Theta(n^2)$ blocks costs $\Theta(n^2T_{\text{block}})$; the already constructed array is then opaque to the unchanged partition recursions. This proves (iv). $\square$

Two numerical practices are essential in realistic data sets: (i) always store evidences and DP messages in log space and normalize via logsumexp; (ii) standardize/center sufficient statistics when available (e.g., for Gaussian blocks) to reduce catastrophic cancellation in moment computations. These practices do not affect correctness, but they materially improve stability when n is in the hundreds or thousands.

6.0.6 A numerical-implementation checklist

The following checklist distills the common pitfalls we have encountered and translates them into implementation tests. Each item should be enforced either by a unit test, an assertion in the runtime path, or a recorded metadata check in the experiment record.

1. **Log-space storage test.** Store all block evidences, moment numerators, and DP messages as log values from the start, with explicit tags for signed quantities. Convert to linear space only for display.
2. **Stable-logsumexp test.** Implement logsumexp as $\text{lse}(x) = m + \log \sum \exp(x_i - m)$ with $m = \max_i x_i$ [6]. Cache the maximum once and reuse for both numerator and denominator when normalizing.
3. **Prefix-array test.** Compute prefix sums of w_t , $w_t T(y_t)$, and (for Gaussian) $w_t y_t^2$ *before* enumerating blocks. Store them as double-precision arrays of length $n + 1$ initialized at index 0. Block summaries then cost one subtraction per block.
4. **DP-invariant test.** After the forward recursion, verify that $\log \tilde{L}_{0,0} = 0$ and $\log \tilde{L}_{k,j} = -\infty$ for $j < k$; after the backward recursion, verify the analogous boundary conditions on $\log \tilde{R}$. Verify that $\log \tilde{L}_{k,n} = \log \tilde{R}_{k,0}$ up to floating-point tolerance, that $\sum_k p(k | y) = 1$, that the fixed- k boundary marginal sums to $k - 1$, and that the reported C_k matches the prior convention used to form the block scores.
5. **Moment-scaling test.** When computing Bayes regression curves via (5.10), subtract a block-specific shift (e.g., the block posterior-mean estimate) before exponentiating to reduce dynamic range, then add the shift back at the end.
6. **Degenerate-block test.** Guard against $W_{ij} = 0$ (missing-data blocks) by returning $\log A_{ij}^{(0)} = -\infty$ or a user-specified fallback. Guard against Q_{ij} numerically negative (from catastrophic cancellation) by clamping to zero.
7. **Reproducibility-metadata test.** Fix a seed for EM initializations and record it in the run metadata; record hyperparameters, $k_{\max}$, the count prior $p(k)$, length factor g , approximation method and settings, data-preprocessing hashes, and the exact version of every family-specific `BlockRoutine` used.
8. **Unit tests tied to theory.** Test the required block-routine outputs, the forward/backward identity, MAP backtracking, fixed- k boundary sums, signed-log moment reconstruction, and every reported complexity mode on small instances where exhaustive enumeration is possible.

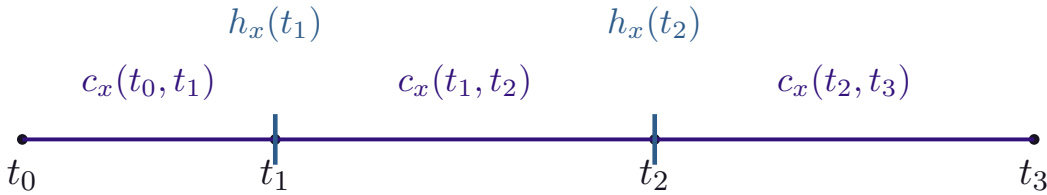
Ordered priors: segment cohesion and boundary hazard

CHAPTER PATH

Irregular designs enter through explicitly separated local factors, retaining exact dynamic programming without a false Poisson-process equivalence.

7.1 Two distinct geometric beliefs

A segment cohesion $c_x(i, j)$ scores the span of a complete block. An interior-boundary hazard $h_x(j)$ scores the candidate split at j . They answer different questions and cannot be justified by the same Poisson-process argument unless the model explicitly contains both factors.



$p(t \mid k, x) \propto \prod_q c_x(t_{q-1}, t_q) \prod_{q < k} h_x(t_q)$; the terminal endpoint receives no boundary-hazard factor.

Figure 7.1. Segment cohesion labels complete intervals, whereas a boundary hazard labels interior endpoints only. Both remain local and compatible with dynamic programming.

Theorem 7.1 (Local prior factors preserve exact partition DP). *Under Assumption 3.6, replace every block evidence by $\tilde{A}_{ij}^{(r)} = A_{ij}^{(r)} \gamma_x(i, j)$, where $\gamma_x(i, j) = c_x(i, j) h_x(j) \mathbf{1}_{\{j < n\}}$. The forward and backward recursions then sum exactly the unnormalized joint weights $p(y, t \mid k, x)$, and division by $C_k(x)$ yields the proper fixed- k evidence.*

Proof. For any ordered partition t , multiplication of its local factors gives

$$\prod_{q=1}^k A_{t_{q-1}, t_q}^{(0)} \gamma_x(t_{q-1}, t_q) = \left\{ \prod_q A_{t_{q-1}, t_q}^{(0)} \right\} \left\{ \prod_q c_x(t_{q-1}, t_q) \right\} \left\{ \prod_{q < k} h_x(t_q) \right\}.$$

The terminal indicator removes $h_x(n)$. This is precisely the likelihood times the unnormalized prior in Equation (3.8). The ordinary partition recursion enumerates every admissible path once, so it returns their sum. Dividing by the prior normalizer completes the claim. $\square$

Proposition 7.2 (Fixed-count interval occupancy from a Poisson boundary process). *Let potential interior boundaries arise from an inhomogeneous Poisson process of intensity $\lambda(u)$ over disjoint physical coordinate intervals $(u_{j-1}, u_j]$, and let $\Lambda_j = \int_{u_{j-1}}^{u_j} \lambda(v) dv$. If the process is reduced to the occupancy indicators of these intervals and one conditions on exactly m distinct occupied candidate intervals, then a subset B with $|B| = m$ has probability proportional to*

$$\prod_{j \in B} \frac{1 - e^{-\Lambda_j}}{e^{-\Lambda_j}} = \prod_{j \in B} (e^{\Lambda_j} - 1).$$

Thus the compatible fixed-count local boundary factor is $h_x(j) \propto e^{\Lambda_j} - 1$. For constant intensity and equal-width intervals this factor is constant, so the induced fixed-count boundary prior is index-uniform when $c_x \equiv 1$.

Proof. Counts in disjoint Poisson-process intervals are independent. Their occupancy indicators are therefore independent Bernoulli variables with $p_j = 1 - e^{-\Lambda_j}$. For a subset B of occupied intervals,

$$\Pr(B) = \prod_{j \in B} p_j \prod_{j \notin B} (1 - p_j) = \left\{ \prod_j (1 - p_j) \right\} \prod_{j \in B} \frac{p_j}{1 - p_j}.$$

After conditioning on $|B| = m$, the first factor is common to all subsets and the local odds are $p_j/(1 - p_j) = e^{\Lambda_j} - 1$. Equal integrated intensities give equal odds and hence a uniform prior over subsets of the fixed size. $\square$

Remark 7.3 (What the proposition does not imply). The proposition weights candidate boundary intervals after conditioning on their number. Without that conditioning, both occupied and unoccupied Bernoulli factors remain in the probability. It does not imply that multiplying each segment likelihood by its physical length is a Poisson boundary-process prior. A length cohesion $c_x(i, j) = g(u_j - u_i)$ is a separate modeling choice.

Proposition 7.4 (Coarse-to-fine likelihood consistency for inherited partitions). *Suppose a fine grid refines a coarse grid without changing observations or the physical contents of any inherited coarse block, and the block model's sufficient summaries are additive. Then the likelihood block evidence $A_{ij}^{(r)}$ for every inherited block is the same whether computed on the coarse representation or by aggregating the fine representation. Prior invariance additionally requires the chosen c_x and h_x to assign the same factors to inherited boundaries.*

Proof. Additivity makes S_{ij} , W_{ij} , and H_{ij} identical for an inherited physical block, so the block integral and moments are identical. The prior statement follows only if its local factors are also representation invariant; it is logically separate from likelihood equality. $\square$

7.2 Recommended configurations

Use $c_x = h_x = 1$ for a truly index-uniform fixed-count prior. Use $c_x(i, j) = g(u_j - u_i)$ when the scientific belief concerns segment duration. Use $h_x(j)$ when external geometry or intensity changes the prior plausibility of a boundary at candidate location j . Combined models are valid, but sensitivity to both components must be reported.

Exact shared-boundary pooling

CHAPTER PATH

Finite-sample block-evidence products, known groups, and a valid finite-candidate consistency theorem.

Plate model (shared boundaries, subject-specific parameters). With a shared segmentation $t = (t_0, \dots, t_k)$ and subject-specific conjugate priors, the generative model nests an inner *segment* plate inside an outer *subject* plate:

$$t \sim p(t | k), \quad \theta_q^{(s)} \stackrel{\text{ind}}{\sim} p(\theta | \alpha_0^{(s)}, \beta_0^{(s)}), \quad y_i^{(s)} | \theta_{q(i)}^{(s)}, w_i^{(s)} \stackrel{\text{ind}}{\sim} p(y | \theta_{q(i)}^{(s)}, w_i^{(s)}), \quad (8.1)$$

where $q(i)$ is the segment index containing observation i under t .

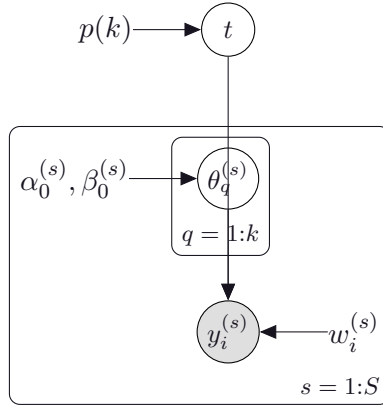


Figure 8.1. Plate model for multi-subject pooling under shared boundaries. The boundary vector t is shared across subjects; each subject has its own segment parameters $\theta_q^{(s)}$ and exposures $w_i^{(s)}$.

Suppose we observe S independent subjects, each with its own observations $\{(x_i^{(s)}, y_i^{(s)})\}$ and exposure weights $w_i^{(s)}$, but we wish to model them as sharing a common segmentation (boundary locations) while allowing subject-specific segment parameters. For simplicity of exposition, assume a common index grid $i = 1, \dots, n$ (e.g., after alignment/union of grids); subject-specific missing observations are handled by setting $w_i^{(s)} = 0$ at missing indices.

Let $(i, j]$ denote a block and write subject-specific block sums as $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$ with subject-specific prior hyperparameters $(\alpha_0^{(s)}, \beta_0^{(s)})$.

Assumption 8.1 (Common-grid shared-boundary support and conditional independence). Subjects are aligned to a common ordered candidate-boundary grid, and every shared boundary vector $t = (t_0, \dots, t_k)$ used by the pooled DP lies in the intersection of the subject-specific admissible supports after missingness/exposure conventions are applied. Conditional on this shared segmentation, subjects $s = 1, \dots, S$ are independent, each subject has its own segment parameters $\theta_{1:k}^{(s)}$ with subject-specific conjugate priors, and subject-level block evidences are finite and strictly positive on the pooled support.

Theorem 8.2 (Exact pooling of block evidences). *Under Assumption 8.1, the pooled block evidence for $(i, j]$ factorizes as*

$$A_{ij,\text{pool}}^{(0)} = \prod_{s=1}^S \exp\{H_{ij}^{(s)}\} \frac{Z(\alpha_0^{(s)} + S_{ij}^{(s)}, \beta_0^{(s)} + W_{ij}^{(s)})}{Z(\alpha_0^{(s)}, \beta_0^{(s)})}.$$

Equivalently, $\log A_{ij,\text{pool}}^{(0)} = \sum_s \log A_{ij,s}^{(0)}$. Running the DP recursions (5.1)–(5.10) with $A_{ij}^{(0)}$ replaced by $A_{ij,\text{pool}}^{(0)}$ yields the exact posterior over shared boundaries. Conditional on any selected or averaged boundary configuration, the subject-specific segment-parameter posteriors are then recovered from the corresponding subject-level conjugate updates.

Proof. **Step 1: subject-wise block evidences.** For each subject s , Theorem 4.1 yields the subject-specific block evidence $A_{ij,s}^{(0)}$ computed from $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$.

Step 2: independence implies product pooling. Conditional on shared boundaries, subjects are independent and have independent segment parameters. Therefore, the joint likelihood (with segment parameters integrated out) for the block is the product of the subject-wise block evidences: $A_{ij,\text{pool}}^{(0)} = \prod_s A_{ij,s}^{(0)}$, giving the stated form.

Step 3: DP correctness under pooled evidences. The DP in Section 5 depends only on the block evidence matrix. Replacing $A_{ij}^{(0)}$ by $A_{ij,\text{pool}}^{(0)}$ therefore computes the exact shared-boundary posterior for the pooled model; parameter posteriors remain available conditionally on boundaries from the subject-specific block updates. $\square$

8.1 Groups with known membership

When subject group labels $z_s \in \{1, \dots, G\}$ are observed, BayesBreak applies the pooled DP of Theorem 8.2 independently within each group: for group g with subject index set $\mathcal{S}_g$, pool per-subject block evidences $\{A_{ij,s}^{(0)}\}_{s \in \mathcal{S}_g}$ into a group-specific pooled evidence $A_{ij,g}^{(0)}$, then run the DP recursions (5.1)–(5.10) with $A_{ij}^{(0)} = A_{ij,g}^{(0)}$. This yields group-specific $P_g(k | y)$, boundary posteriors $P_g(t_p | y, k)$, and segment/regression-curve moments at total cost G times the single-group cost. This is the supervised counterpart of the latent-template mixture in Section 9.3.

8.2 Finite-candidate concentration across independent subjects

Theorem 8.3 (Shared-boundary posterior consistency on a finite candidate set). *Let $\mathcal{T}$ be a finite set of candidate segmentations and let $t^* \in \mathcal{T}$ be the data-generating candidate. For subject s , let $M_s(t) > 0$ be the integrated score used by the shared-boundary model.*

Assume subjects are independent, the prior satisfies $p(t^*) > 0$, and for every $t \neq t^*$:

- (i) $\mathbb{E}|\log M_s(t^*) - \log M_s(t)| < \infty$;
- (ii) a strong law holds for the centered log-score differences; and
- (iii) $\liminf_{S \rightarrow \infty} S^{-1} \sum_{s=1}^S \mathbb{E}[\log M_s(t^*) - \log M_s(t)] = \delta_t > 0$.

Then $p(t^* | y^{(1:S)}) \rightarrow 1$ almost surely, and every joint MAP segmentation equals t^* eventually almost surely.

Proof. For any $t \neq t^*$, the posterior ratio is

$$\frac{p(t | y^{(1:S)})}{p(t^* | y^{(1:S)})} = \frac{p(t)}{p(t^*)} \exp \left\{ - \sum_{s=1}^S [\log M_s(t^*) - \log M_s(t)] \right\}.$$

By the strong-law assumption and the positive limiting average expectation gap, the normalized sum in brackets has positive liminf δ_t almost surely. Hence the ratio converges to zero exponentially on the S scale. Because $\mathcal{T}$ is finite, summing these ratios over all wrong candidates still converges to zero. Normalization gives posterior concentration at t^* ; the MAP conclusion follows because all wrong posterior ratios are eventually below one. $\square$

Remark 8.4 (Why one informative subject is insufficient). A single fixed informative subject does not imply a positive *average* score gap as the number of subjects grows. The theorem requires information to accumulate through a positive limiting average gap; this repairs the earlier asymptotic statement without altering exact finite-sample pooling.

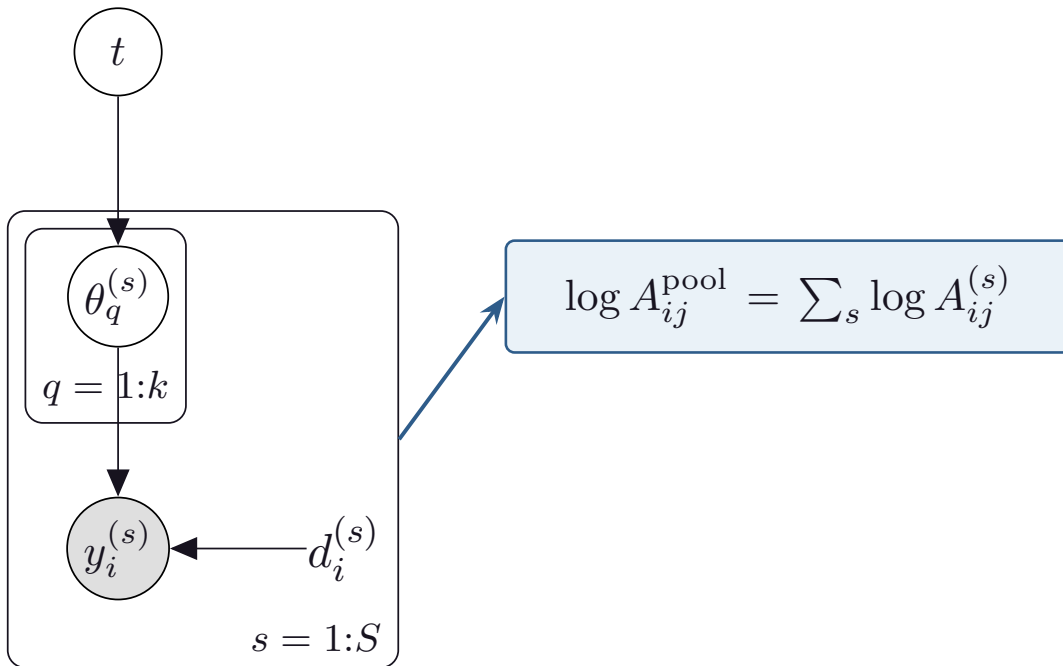


Figure 8.2. Conditional independence across subjects converts each pooled block score into a product of subject-level block evidences, or a sum in log space.

Grouped and latent-group models

CHAPTER PATH

Known-group shared-changepoint analysis and a finite latent-group criterion with Jensen minorization and exact template updates over a finite partition set.

9.1 Known groups

Suppose each sequence s has an observed group label $g(s) \in \{1, \dots, G\}$. Within a group, sequences may share a common partition while retaining sequence-specific segment parameters. Under conditional independence, the group-specific marginal likelihood for a candidate segment is the product of the sequence-specific segment marginal likelihoods. The sum-product and max-sum recursions of Chapters 5 and 6 then apply separately to each group. This is a direct hierarchical extension of the shared-changepoint model in Chapter 8.

9.2 Finite latent-group criterion

When group labels are unknown, let $\tau_g = (k_g, t^{(g)})$ denote a deterministic candidate changepoint template for group g . The archived implementation uses the sequence–template score

$$S_s(\tau) = \frac{p(k)}{C_k} \prod_{q=1}^k A_{t_{q-1}, t_q}^{(0,s)} \gamma_u(t_{q-1}, t_q), \quad (9.1)$$

with the convention that an impossible segment gives score zero. This subject-template score is not assumed to integrate to one over the sample space of sequence s . For weights $\pi_g \geq 0$ with $\sum_g \pi_g = 1$, define

$$\mathcal{F}(\pi, \tau) = \sum_{s=1}^S \log \left\{ \sum_{g=1}^G \pi_g S_s(\tau_g) \right\}. \quad (9.2)$$

Equation (9.2) is the optimization criterion studied in this chapter. The results below concern this finite criterion and do not require interpreting $S_s(\tau_g)$ as a normalized mixture component.

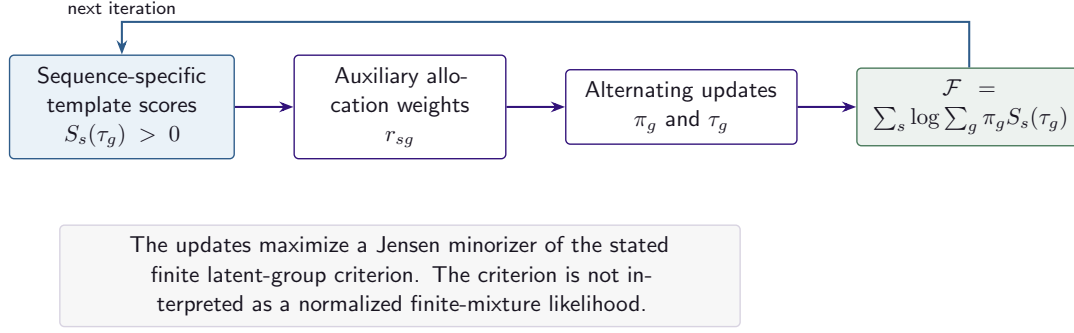


Figure 9.1. Alternating optimization of the finite latent-group criterion. Auxiliary allocation weights make a Jensen lower bound tight; group-template updates maximize the resulting weighted partition score over the finite candidate set.

9.3 Jensen minorization and alternating updates

For any nonnegative weights r_{sg} satisfying $\sum_g r_{sg} = 1$, the log-sum inequality gives

$$\mathcal{F}(\pi, \tau) \geq \mathcal{Q}(r, \pi, \tau) := \sum_{s,g} r_{sg} [\log \pi_g + \log S_s(\tau_g) - \log r_{sg}]. \quad (9.3)$$

Equality holds at

$$r_{sg} = \frac{\pi_g S_s(\tau_g)}{\sum_h \pi_h S_s(\tau_h)}. \quad (9.4)$$

For fixed r , maximization over the group weights gives $\pi_g = S^{-1} \sum_s r_{sg}$. Let $n_g = \sum_s r_{sg}$. Substituting Equation (9.1) into $\mathcal{Q}$ shows that, for a candidate segment count k , the group- g template contribution is

$$n_g \{\log p(k) - \log C_k\} + \sum_{q=1}^k \ell_{t_{q-1}, t_q}^{(g)}, \quad \ell_{ij}^{(g)} = \sum_{s=1}^S r_{sg} \log A_{ij}^{(0,s)} + n_g \log \gamma_u(i, j). \quad (9.5)$$

Thus the update is an exact max-sum recursion at fixed k , followed by comparison across k using the displayed count offset. The convention $0 \log 0 = 0$ applies when $r_{sg} = 0$, while a positive responsibility on a zero sequence–template score makes that candidate value $-\infty$.

Theorem 9.1 (Monotonicity of exact latent-group updates). *Consider an iteration that (i) sets r according to Equation (9.4), (ii) maximizes $\mathcal{Q}$ over π , and (iii) exactly maximizes each responsibility-weighted template criterion using the local score and segment-count offset in Equation (9.5) over its declared finite candidate set. Then the iteration cannot decrease $\mathcal{F}$.*

Proof. At the current parameter values, Equation (9.4) makes the Jensen bound tight: $\mathcal{F}_{\text{old}} = \mathcal{Q}(r, \pi_{\text{old}}, \tau_{\text{old}})$. Exact maximization of the minorizer gives

$$\mathcal{Q}(r, \pi_{\text{new}}, \tau_{\text{new}}) \geq \mathcal{Q}(r, \pi_{\text{old}}, \tau_{\text{old}}).$$

The log-sum inequality gives $\mathcal{F}_{\text{new}} \geq \mathcal{Q}(r, \pi_{\text{new}}, \tau_{\text{new}})$. Combining the inequalities yields $\mathcal{F}_{\text{new}} \geq \mathcal{F}_{\text{old}}$. $\square$

Algorithm 13: Alternating optimization for finite latent-group templates

Input: Sequence-specific segment-score arrays; number of groups G ; admissible segment counts; number of restarts; convergence tolerance.

```

1 foreach restart do
2   initialize  $\pi$  and templates  $\tau_{1:G}$ ;
3   evaluate the current value of  $\mathcal{F}$ ;
4   repeat
5     update  $r_{sg}$  by Equation (9.4);
6     update  $\pi_g \leftarrow S^{-1} \sum_s r_{sg}$ ;
7     for each group  $g$ , set  $n_g \leftarrow \sum_s r_{sg}$ , build the local scores in Equation (9.5), solve
       the max-sum recursion at every allowed  $k$ , and select the offset-adjusted
       maximizer using  $n_g \{\log p(k) - \log C_k\}$ ;
8     evaluate and append the new value of  $\mathcal{F}$ ;
9   until the increase in  $\mathcal{F}$  is at most the tolerance or the iteration limit is reached;
10  retain the restart with the largest final value in its recorded objective sequence;

```

Output: Group templates, group weights, auxiliary allocation weights, complete objective sequences, and restart ranking by the final objective value.

9.4 Non-uniqueness and implementation limitation

The criterion is invariant to permutation of the group labels. It may have duplicate or collapsed templates when G is larger than the effective number of groups, and multiple restarts do not guarantee a global maximum. These are optimization and identifiability limitations of the stated criterion.

The archived implementation contains an exit path that can return the objective value from the previous iteration rather than the last value in the objective sequence. In one of three diagnostic restarts, the difference was 0.003137360860080. The archived latent-group simulation remains a real computation, but CODE-BB-003 must correct the returned objective and restart ranking before those values are treated as release-quality optimization results.

ESTABLISHED RESULT AND NON-CLAIM

For Equation (9.2), the auxiliary allocation update makes the Jensen minorizer tight, the group-weight update maximizes that minorizer, and the finite template step is solved exactly by max-sum recursion when Equation (9.5) applies. The monotonicity theorem is complete. No theorem is claimed for identifiability of a normalized finite mixture or consistency of the estimated number of groups.

Nonconjugate block integration and numerical certification

CHAPTER PATH

Laplace, variational, expectation-propagation, Pólya–Gamma, and quadrature calculations for segment marginal likelihoods, with conditional propagation of segment-wise log-score error through the partition recursions.

EXACT GLOBAL ALGEBRA, APPROXIMATE LOCAL TARGET

Once an approximate block array is supplied, the partition recursions are evaluated exactly for that array. This is not the same as exact inference under the reference nonconjugate model. The gap is controlled only when a local certificate verifies Assumption 10.6 on all blocks reachable by the declared partition support.

Plate model (non-conjugate single segment with PG augmentation). The generative graph for a non-conjugate block $(i, j]$ is identical to Figure 4.1 with $\theta \sim \pi(\theta)$ replaced by a non-conjugate prior (e.g., Gaussian on a log or logit scale). For Pólya–Gamma-augmented binomial logistic blocks (Theorem 10.4), introducing the latent scale-mixture variables $\omega_t \sim \text{PG}(m_t, 0)$ restores conditional conjugacy and is reflected in the plate as a second observed-indexed latent:

$$\theta \sim \mathcal{N}(\nu, \rho^2), \quad \omega_t \stackrel{\text{ind}}{\sim} \text{PG}(m_t, 0), \quad y_t \mid \theta, \omega_t, m_t \stackrel{\text{ind}}{\sim} \text{Binom}(m_t, \sigma(\theta)) \text{ with (10.5), (10.1)}$$

where σ is the logistic function.

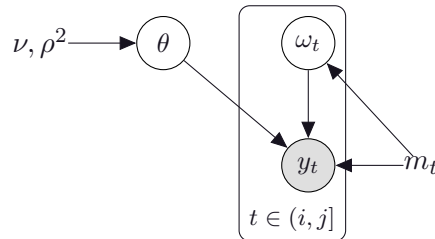


Figure 10.1. Plate model for a non-conjugate block with Pólya–Gamma augmentation. Conditional on ω_t , the block is quadratic in θ with a Gaussian posterior (Proposition 10.5); the same template applies to Laplace and variational routines with the ω_t node omitted.

In many applications the single-segment parameterization is not conjugate to the observation

model—for example, log-normal priors for positive means, heavy-tailed shrinkage priors, or logistic/negative-binomial links with structural offsets. This section develops a plug-in toolkit for computing approximate block evidences $A_{ij}^{(0)}$ and moments $A_{ij}^{(r)}$ when conjugacy is unavailable, while keeping the dynamic-programming layer of Section 5 unchanged. When approximate block evidences are used, the DP is exact for the approximate block-score model; additional error statements require explicit blockwise approximation control.

We focus on one-dimensional (or low-dimensional) canonical GLMs, where for a block $(i, j]$ we must approximate integrals of the form

$$A_{ij}^{(0)} = \int \left[\prod_{t=i+1}^j p(y_t | \theta) \right] p(\theta) d\theta, \quad A_{ij}^{(r)} = \int \left[\prod_{t=i+1}^j p(y_t | \theta) \right] p(\theta) m_{\star}(\theta)^r d\theta,$$

where $m_{\star}(\theta) = m(\theta; d_{\star})$ is the declared reporting functional at a fixed reference descriptor. We describe four complementary approaches: (i) first-order Laplace approximation; (ii) variational lower bounds (Jaakkola–Jordan [20] type for logistic regression and mean-field for more general GLMs); (iii) expectation propagation (EP); and (iv) Pólya–Gamma augmentation for logistic and negative-binomial blocks. Throughout we use the family-specific descriptor d_t and any separately declared power weight a_t and cumulative statistics (S_{ij}, W_{ij}, H_{ij}) .

10.0.1 Model and notation for GLM blocks

Consider a one-parameter canonical GLM on a block $(i, j]$:

$$\log p(y_t | \theta) = w_t \{y_t \theta - b(\theta)\} + \log h(y_t, w_t),$$

where b is the cumulant function and h collects base measures (including any w_t -dependent terms). The block log-likelihood is

$$\ell_{ij}(\theta) := \sum_{t=i+1}^j w_t \{y_t \theta - b(\theta)\} + H_{ij}, \quad H_{ij} := \sum_{t=i+1}^j \log h(y_t, w_t).$$

Let $\pi(\theta) = p(\theta)$ be an arbitrary prior (not necessarily conjugate), and write $\Psi_{ij}(\theta) := \ell_{ij}(\theta) + \log \pi(\theta)$ for the block log-posterior.

Derivatives are

$$\ell'_{ij}(\theta) = S_{ij} - W_{ij} b'(\theta), \quad \ell''_{ij}(\theta) = -W_{ij} b''(\theta),$$

and

$$\Psi'_{ij}(\theta) = \ell'_{ij}(\theta) + (\log \pi)'(\theta), \quad \Psi''_{ij}(\theta) = \ell''_{ij}(\theta) + (\log \pi)''(\theta).$$

10.0.2 Existence and uniqueness of block modes

Lemma 10.1 (Strict log-concavity and uniqueness). *Suppose b and $\log \pi$ are twice differentiable on a convex parameter domain, $b''(\theta) > 0$ for all θ (canonical GLM), the prior is log-concave ($(\log \pi)''(\theta) \leq 0$), and $W_{ij} > 0$. Then Ψ_{ij} is strictly concave. If, in addition, Ψ_{ij} is proper and attains its supremum on the parameter domain (for example by coercive tails or compact-domain truncation), the maximizer $\hat{\theta}_{ij}$ exists and is unique.*

Proof. Since $W_{ij} > 0$ and $b''(\theta) > 0$, we have $-W_{ij} b''(\theta) < 0$. Adding $(\log \pi)''(\theta) \leq 0$ yields $\Psi''_{ij}(\theta) < 0$ wherever the derivatives exist, so Ψ_{ij} is strictly concave. Strict concavity gives

uniqueness of any maximizer; existence is provided by the stated attainment assumption. $\square$

10.0.3 Laplace approximation for single-block evidence

Theorem 10.2 (First-order Laplace approximation with a uniform remainder). *Index a sequence of blocks by an effective information size $N = W_{ij} \rightarrow \infty$. Let $\hat{\theta}_N$ be the unique interior maximizer of Ψ_N and $H_N = -\Psi_N''(\hat{\theta}_N)$. Assume there are constants $c, C, \delta > 0$, independent of the block, such that: (i) $cN \leq -\Psi_N''(\theta) \leq CN$ throughout $|\theta - \hat{\theta}_N| \leq \delta$; (ii) Ψ_N is four times continuously differentiable on that neighborhood and $|\Psi_N^{(r)}(\theta)| \leq CN$ for $r = 3, 4$; and (iii) the integral outside the neighborhood is at most the local Gaussian leading term times $O(N^{-1})$. Then*

$$A_{ij}^{(0)} = e^{\Psi_N(\hat{\theta}_N)} \left(\frac{2\pi}{H_N} \right)^{1/2} \{1 + O(N^{-1})\}, \quad (10.2)$$

and hence

$$\log A_{ij}^{(0)} = \Psi_N(\hat{\theta}_N) + \frac{1}{2} \log(2\pi) - \frac{1}{2} \log H_N + O(N^{-1}). \quad (10.3)$$

For a bounded twice continuously differentiable test function g whose first two derivatives are uniformly bounded in the same neighborhood,

$$\frac{M_{ij}^{[g]}}{A_{ij}^{(0)}} = g(\hat{\theta}_N) + O(N^{-1}), \quad M_{ij}^{[g]} = \int g(\theta) e^{\Psi_N(\theta)} d\theta. \quad (10.4)$$

Proof. Set $u = \sqrt{H_N}(\theta - \hat{\theta}_N)$. Taylor's theorem through fourth order gives, on the local neighborhood,

$$\Psi_N(\theta) - \Psi_N(\hat{\theta}_N) = -\frac{1}{2}u^2 + a_N u^3 N^{-1/2} + b_N(u) u^4 N^{-1},$$

where a_N and $b_N(u)$ are uniformly bounded by assumptions (i)–(ii). On a growing central region $|u| \leq N^{1/12}$, expand the exponential. The term linear in u^3 integrates to zero against the symmetric Gaussian kernel; the remaining terms are bounded by an integrable Gaussian polynomial times N^{-1} . The part of the local neighborhood outside this central region is exponentially smaller by the quadratic term, and assumption (iii) controls the nonlocal tail. After the Jacobian $H_N^{-1/2}$, this proves the relative $1 + O(N^{-1})$ expansion. Applying the same argument to the numerator after the second-order Taylor expansion $g(\hat{\theta}_N + u/\sqrt{H_N}) = g(\hat{\theta}_N) + g'(\hat{\theta}_N)u/\sqrt{H_N} + O(u^2/N)$ and again using Gaussian symmetry yields the ratio statement. $\square$

Efficient Newton updates using block summaries. For canonical GLMs,

$$\Psi'_{ij}(\theta) = S_{ij} - W_{ij}b'(\theta) + (\log \pi)'(\theta), \quad \Psi''_{ij}(\theta) = -W_{ij}b''(\theta) + (\log \pi)''(\theta),$$

so Newton iteration is

$$\theta^{(m+1)} \leftarrow \theta^{(m)} - \frac{S_{ij} - W_{ij}b'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})}{-W_{ij}b''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})},$$

with cost independent of block length once S_{ij}, W_{ij} are available.

Implementation. Algorithm 14 implements the Laplace block-evidence approximation using a Newton or quasi-Newton solver for the block mode and the corresponding Hessian correction. The approximation is local but often highly accurate in moderate-to-large blocks. The remaining non-conjugate routines (Algorithms 15–18) follow the same template: a block-local optimizer feeds an approximate log-evidence into the DP.

Algorithm 14: GLMBlock.Laplace($y, w, i, j, b, \text{prior}$)

Input: $y_{i+1:j}$, weights $w_{i+1:j}$; cumulant $b(\theta)$; prior $\pi(\theta)$

- 1 Compute $S \leftarrow \sum_{t=i+1}^j w_t y_t$; $W \leftarrow \sum_{t=i+1}^j w_t$; $H \leftarrow \sum_{t=i+1}^j \log h(y_t, w_t)$;
- 2 Define $\Psi(\theta) = H + S\theta - Wb(\theta) + \log \pi(\theta)$ and its derivatives;
- 3 Initialize $\theta^{(0)}$ (e.g., solve $b'(\theta) = S/W$ or use prior mean);
- 4 **for** $m = 0, 1, \dots$ **until convergence do**
- 5 $g \leftarrow S - Wb'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})$;
- 6 $h \leftarrow -Wb''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})$;
- 7 **if** $|g|/|h| < \varepsilon_{\text{newton}}$ **then**
- 8 **break**
- 9 $\theta^{(m+1)} \leftarrow \theta^{(m)} - g/h$;
- 10 $\hat{\theta} \leftarrow \theta^{(m)}$; $H^* \leftarrow -\Psi''(\hat{\theta})$;
- 11 $\log \hat{A}_{ij}^{(0)} \leftarrow \Psi(\hat{\theta}) + \frac{1}{2} \log(2\pi) - \frac{1}{2} \log H^*$;

Output: $\hat{A}_{ij}^{(0)}$ and (optional) moments via (10.4)

10.0.4 Variational lower bounds for GLM blocks

When a deterministic lower bound is desirable, introduce a variational distribution $q(\theta)$ and apply Jensen:

$$\log A_{ij}^{(0)} = \log \int q(\theta) \frac{e^{\Psi_{ij}(\theta)}}{q(\theta)} d\theta \geq \int q(\theta) \Psi_{ij}(\theta) d\theta + \mathcal{H}[q] := \mathcal{L}_{ij}(q).$$

Jaakkola–Jordan bound for logistic blocks. For Bernoulli/logistic likelihoods, a standard quadratic bound yields a tractable Gaussian variational posterior. One convenient form (for sigmoid σ) is:

$$\log \sigma(z) \geq \log \sigma(\xi) + \frac{z - \xi}{2} - \lambda(\xi)(z^2 - \xi^2), \quad \lambda(\xi) = \frac{\tanh(\xi/2)}{4\xi},$$

with variational parameter $\xi > 0$. Applying this bound to the appropriate signed linear predictors yields a quadratic surrogate in θ , so the optimal Gaussian $q(\theta)$ and ξ admit closed-form coordinate updates.

Proposition 10.3 (Monotone improvement of variational block bounds). *Every exact coordinate-ascent update of the local variational parameters (for example ξ in a logistic block) or of a Gaussian $q(\theta)$ leaves the block evidence lower bound $\mathcal{L}_{ij}(q)$ non-decreasing. If the iterates converge to an interior fixed point and the objective is differentiable there, that fixed point is coordinatewise stationary. At every iteration, $\underline{A}_{ij}^{(0)} := \exp\{\mathcal{L}_{ij}(q)\}$ is a certified lower bound on $A_{ij}^{(0)}$.*

Proof. Each coordinate update is an exact maximization of $\mathcal{L}_{ij}(q)$ with respect to a subset of variables holding the others fixed, so the ELBO is non-decreasing. At a converged interior

fixed point, the first-order derivative along each updated coordinate block vanishes under the stated regularity, which is coordinatewise stationarity. Because $\mathcal{L}_{ij}$ is a Jensen lower bound on $\log A_{ij}^{(0)}$ at every iterate, exponentiating yields a valid evidence lower bound. $\square$

Algorithm 15: LogisticBlock_JJ($y, m, w, i, j, \text{prior}$) (variational lower bound)

Input: Binomial counts $y_t \sim \text{Binom}(m_t, p)$, logit link; prior $\pi(\theta)$

- 1 Initialize ξ and a Gaussian $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$;
 - 2 **repeat**
 - 3 Update the quadratic bound parameter $\xi \leftarrow \sqrt{\mathbb{E}_q[\theta^2]}$;
 - 4 Update Gaussian $q(\theta)$ by maximizing the block ELBO $\mathcal{L}_{ij}(q)$ (closed-form for quadratic surrogate);
 - 5 **until** *converged*;
 - 6 Return $\underline{A}_{ij}^{(0)} = \exp\{\mathcal{L}_{ij}(q)\}$ and moments from q .
-

10.0.5 Expectation propagation (EP) for GLM blocks

Expectation propagation (EP) [23] provides a deterministic approximation to the block posterior and block evidence by iteratively refining local site approximations and matching moments. In the formulation used here the prior is Gaussian, and each likelihood term $p(y_t | \theta)$ is approximated by a Gaussian site $\tilde{t}_t(\theta) = c_t \exp\{-\frac{1}{2}a_t\theta^2 + b_t\theta\}$. After convergence, the product of the Gaussian prior and sites is Gaussian and the approximate evidence is $\hat{A}_{ij}^{(0, \text{EP})} = \int \pi(\theta) \prod_{t=i+1}^j \tilde{t}_t(\theta) d\theta$. The constants c_t are part of the returned log evidence; dropping them changes the DP score. A non-Gaussian prior requires an additional site approximation or numerical integral and is not covered by this Gaussian-posterior description. EP is neither a bound nor guaranteed to converge.

Algorithm 16: GLMBlock_EP(y, w, i, j, prior)

Input: Canonical GLM; Gaussian prior $\pi(\theta)$; tolerance and maximum iterations

- 1 Initialize site parameters $\{a_t, b_t, c_t\}_{t=i+1}^j$; set $\tilde{p}(\theta) \propto \pi(\theta) \prod_t \tilde{t}_t(\theta)$;
 - 2 **repeat**
 - 3 **for** $t = i + 1$ **to** j **do**
 - 4 Form cavity $\tilde{p}^{\setminus t}(\theta) \propto \tilde{p}(\theta) / \tilde{t}_t(\theta)$ and verify positive cavity precision;
 - 5 Form tilted $p^{\text{tilt}}(\theta) \propto \tilde{p}^{\setminus t}(\theta) p(y_t | \theta)$;
 - 6 Moment-match p^{tilt} and update the site parameters (a_t, b_t, c_t) , damping if needed;
 - 7 **until** *moments stable or maximum iterations reached*;
 - 8 Return a convergence/failure flag, the approximate log evidence $\log \hat{A}_{ij}^{(0, \text{EP})} = \log \int \pi(\theta) \prod_{t=i+1}^j c_t \exp\{-a_t\theta^2/2 + b_t\theta\} d\theta$, and moments from $\tilde{p}$.
-

An implementation should export the convergence flag and the maximum site residual together with the block score. Downstream DP routines may then exclude failed EP blocks, retry them with stronger damping, or mark the whole EP approximation as uncertified for the stability analysis.

10.0.6 Pólya–Gamma augmentation for binomial-logistic blocks

Theorem 10.4 (Pólya–Gamma identity and conditional conjugacy). *For binomial logistic blocks with counts m_t and successes y_t , let $w_t \geq 0$ denote either one or a declared likelihood-power/replication weight, define $b_t = w_t m_t$ and $\kappa_t = w_t(y_t - \frac{m_t}{2})$, and introduce latent $\omega_t \sim \text{PG}(b_t, 0)$. Then the weighted logistic likelihood term admits the Gaussian scale-mixture representation*

$$\left\{ \frac{(e^\theta)^{y_t}}{(1 + e^\theta)^{m_t}} \right\}^{w_t} = 2^{-b_t} e^{\kappa_t \theta} \int_0^\infty e^{-\omega_t \theta^2 / 2} p(\omega_t) d\omega_t. \quad (10.5)$$

The binomial coefficient, when present, is independent of θ and must be retained in the base-measure contribution to the block evidence. Conditionally on $\omega_{i+1:j}$, the remaining block likelihood is quadratic in θ :

$$\log p(y_{i+1:j} \mid \theta, \omega) = -\frac{1}{2} \theta^2 \sum_{t=i+1}^j \omega_t + \theta \sum_{t=i+1}^j \kappa_t + \text{const}. \quad (10.6)$$

With a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the conditional posterior $p(\theta \mid y, \omega)$ is Gaussian.

Proof. Step 1: the identity. Equation (10.5) is the defining Pólya–Gamma augmentation of Polson et al. [27]: it expresses the binomial-logistic term as a Gaussian scale mixture in θ .

Step 2: condition on ω and collect quadratic terms. Given ω_t , each weighted term contributes $\kappa_t \theta - \frac{1}{2} \omega_t \theta^2$ up to constants, where the weight has already been absorbed into b_t and κ_t . Summing over t yields (10.6).

Step 3: combine with a Gaussian prior. Multiplying the quadratic-in- θ conditional likelihood by a Gaussian prior produces a Gaussian conditional posterior, because Gaussian priors are conjugate to Gaussian likelihoods. $\square$

Proposition 10.5 (Mean-field variational approximation under Pólya–Gamma). *Under the same setup as Theorem 10.4 and a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the mean-field factorization $q(\theta, \omega) = q(\theta) \prod_{t=i+1}^j q(\omega_t)$ admits the following closed-form coordinate-ascent updates: (i) $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$ with $\sigma_q^{-2} = \rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]$ and $\mu_q = \sigma_q^2(\rho^{-2}\nu + \sum_t \kappa_t)$; (ii) $q(\omega_t) = \text{PG}(b_t, c_t)$ with $c_t^2 = \mathbb{E}_q[\theta^2] = \mu_q^2 + \sigma_q^2$, giving $\mathbb{E}_q[\omega_t] = \frac{b_t}{2c_t} \tanh(c_t/2)$. These updates are monotone in the block ELBO.*

Proof. For a mean-field factor, the coordinate optimum satisfies

$$\log q^*(\theta) = \mathbb{E}_{q(\omega)}[\log p(y, \theta, \omega)] + \text{const}.$$

Combining Equation (10.6) with the Gaussian prior gives a quadratic log density whose precision and precision-weighted mean are $\rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]$ and $\rho^{-2}\nu + \sum_t \kappa_t$, respectively. This proves the stated Gaussian update. Similarly, $\log q^*(\omega_t) = \mathbb{E}_{q(\theta)}[\log p(y_t, \omega_t \mid \theta)] + \text{const}$ is the log density of an exponentially tilted $\text{PG}(b_t, 0)$ variable with tilt $c_t = \sqrt{\mathbb{E}_q[\theta^2]}$, so $q^*(\omega_t) = \text{PG}(b_t, c_t)$. Its standard mean is $b_t \tanh(c_t/2)/(2c_t)$, with the continuous limit $b_t/4$ at $c_t = 0$. Each update is the exact maximizer of the same block ELBO with the other factors fixed; therefore the ELBO is nondecreasing after every coordinate update. $\square$

Algorithm 17: GLMBlock_PG_VB(y, m, i, j, ν, ρ^2) (Gaussian prior, PG mean-field)

Input: Binomial/logistic block with trial counts m_t ; Gaussian prior $\mathcal{N}(\nu, \rho^2)$

- 1 Initialize $q(\omega_t)$ and $q(\theta) = \mathcal{N}(\mu, \sigma^2)$;
 - 2 **repeat**
 - 3 Set $b_t \leftarrow w_t m_t$ and $\kappa_t \leftarrow w_t(y_t - m_t/2)$ if likelihood-power weights are used;
 otherwise $w_t = 1$;
 - 4 $\sigma^{-2} \leftarrow \rho^{-2} + \sum_{t=i+1}^j \mathbb{E}_q[\omega_t]$;
 - 5 $\mu \leftarrow \sigma^2 \left(\rho^{-2} \nu + \sum_{t=i+1}^j \kappa_t \right)$;
 - 6 **foreach** t **do**
 - 7 Update $q(\omega_t) = \text{PG}(w_t m_t, c_t)$ using current $\mathbb{E}_q[\theta^2]$;
 - 8 **until** *ELBO converged*;
 - 9 Return a lower bound $\underline{A}_{ij}^{(0)}$ and approximate moments from $q(\theta)$.
-

Using approximate block evidences in BayesBreak. Once any block-evidence approximation is available (Laplace, JJ, EP, PG-VB, or another surrogate), BayesBreak uses it exactly as an input score array: build a triangular array of approximate block log-evidences and run the same DP recursions. The resulting posterior summaries are exact for that approximate score array, not automatically exact for the original non-conjugate model. Algorithm 18 provides a generic recipe for constructing this array in a model-agnostic way, emphasizing reuse of block-local optimizers and cached sufficient-statistic summaries.

Algorithm 18: Precompute block evidences for non-conjugate GLMs

Input: $\{(x_t, y_t, w_t)\}_{t=1}^n$, $\text{method} \in \{\text{Laplace, JJ, PG-VB, EP}\}$

- 1 **for** $i = 0$ **to** $n - 1$ **do**
 - 2 Initialize running summaries (as needed);
 - 3 **for** $j = i + 1$ **to** n **do**
 - 4 Update block summaries for $(i, j]$;
 - 5 Call chosen block routine to compute $\hat{A}_{ij}^{(0)}$ (and optional moments);
 - 6 Run DP (Algorithm 6) with resulting block evidences.
-

10.0.7 Propagation of block approximation error through the DP

Assumption 10.6 (Uniform per-block log-evidence error). Let $\hat{A}_{ij}^{(0)}$ denote the approximate block evidence produced by the chosen non-conjugate block routine (one of Laplace, JJ, EP, PG mean field, or 1-D quadrature) on candidate block $(i, j]$, and set $\delta_{ij} := \log \hat{A}_{ij}^{(0)} - \log A_{ij}^{(0)}$. There exists $\varepsilon > 0$ such that, for every candidate block $(i, j]$ with $0 \leq i < j \leq n$ reachable by some k -segmentation with $k \leq k_{\max}$,

$$|\delta_{ij}| \leq \varepsilon.$$

Assumption 10.6 is an additional numerical-analysis requirement, not a consequence of naming an approximation routine. It must be established independently for the chosen family, parameter domain, candidate-block support, transformation, initialization, convergence event, and tail treatment. Nested quadrature, a high-accuracy reference calculation, or a method-specific remainder theorem can provide such a certificate in a particular analysis. Convergence of an

optimizer, monotonicity of an ELBO, or an EP moment match alone does not provide the required two-sided uniform log-evidence bound.

Proposition 10.7 (Log-evidence and posterior-odds stability). *Under Assumption 10.6, for every DP state (k, j) with $k \leq k_{\max}$,*

$$e^{-k\varepsilon} L_{kj} \leq \widehat{L}_{kj} \leq e^{k\varepsilon} L_{kj}, \quad e^{-k\varepsilon} R_{kj} \leq \widehat{R}_{kj} \leq e^{k\varepsilon} R_{kj}.$$

Consequently:

1. For any $k, k' \leq k_{\max}$, the posterior odds for the segment count satisfy

$$\left| \log \frac{\widehat{P}(k | y)}{\widehat{P}(k' | y)} - \log \frac{P(k | y)}{P(k' | y)} \right| \leq (k + k')\varepsilon. \quad (10.7)$$

2. For any fixed $k \leq k_{\max}$ and any two candidate locations h, h' of boundary t_p ,

$$\left| \log \frac{\widehat{P}(t_p = h | y, k)}{\widehat{P}(t_p = h' | y, k)} - \log \frac{P(t_p = h | y, k)}{P(t_p = h' | y, k)} \right| \leq 2k\varepsilon. \quad (10.8)$$

Thus uniform blockwise log-evidence control yields uniform control of global posterior odds. Coarse absolute probability bounds follow by normalizing finite weights, as in Corollary 10.8; a margin condition is needed instead for preservation of a modal ranking.

Proof. Any term contributing to L_{kj} is a product of exactly k block evidences. Hence the corresponding approximate term differs by a multiplicative factor in $[e^{-k\varepsilon}, e^{k\varepsilon}]$. Summing over all paths yields the same sandwich bound for $\widehat{L}_{kj}$, and the suffix bound is identical. Equation (10.7) follows immediately because $P(k | y)$ is proportional to $p(k)L_{kn}/C_k$ and the prior and normalization constants are unchanged. For boundary odds, note that $P(t_p = h | y, k) \propto L_{p,h}R_{k-p,h}$. The numerator for each candidate location involves k block factors in total, so each candidate-specific weight is perturbed by at most $e^{\pm k\varepsilon}$; comparing two candidates therefore yields the factor $e^{\pm 2k\varepsilon}$ and hence (10.8). $\square$

Corollary 10.8 (Probability-error conversion for normalized finite weights). *Let $p_a = w_a / \sum_b w_b$ and $\widehat{p}_a = \widehat{w}_a / \sum_b \widehat{w}_b$ be two finite normalized distributions with $|\log \widehat{w}_a - \log w_a| \leq \eta$ for every state a . Then $e^{-2\eta} \leq \widehat{p}_a / p_a \leq e^{2\eta}$ for every a , and $\|\widehat{p} - p\|_{\text{TV}} \leq \min\{1, e^{2\eta} - 1\}$. Consequently, Assumption 10.6 implies a fixed- k boundary-distribution total-variation bound with $\eta = k\varepsilon$, and a segment-count posterior bound with $\eta = k_{\max}\varepsilon$. These absolute probability bounds are coarser than the odds bounds but require no additional margin condition.*

Proof. The normalizing denominator $\sum_b p_b \exp(\delta_b)$ lies in $[e^{-\eta}, e^{\eta}]$ when $\delta_a = \log \widehat{w}_a - \log w_a$. Hence each probability ratio lies in $[e^{-2\eta}, e^{2\eta}]$. Summing the positive parts of $\widehat{p}_a - p_a$ gives the exponential bound, and total variation is always at most one. $\square$

Corollary 10.9 (Preservation of boundary rankings under a margin condition). *Under the hypotheses of Proposition 10.7, if there is a log-odds margin $\Delta > 0$ separating the best candidate h^* from every alternative,*

$$\log \frac{P(t_p = h^* | y, k)}{P(t_p = h | y, k)} \geq \Delta \quad \text{for all } h \neq h^*,$$

then for every $\varepsilon < \Delta/(2k)$ the approximate posterior preserves the exact modal boundary location: $\arg \max_h \hat{P}(t_p = h \mid y, k) = h^*$.

Proof. By (10.8), the log-odds between h^* and any alternative h is perturbed by at most $2k\varepsilon < \Delta$, so the exact log-odds positivity is preserved. $\square$

Proof obligation 10.10 (Routine-specific block-error certification). No routine-wide uniform rate is asserted for Laplace, Jaakkola–Jordan, expectation propagation, Pólya–Gamma mean field, or adaptive quadrature without assumptions tailored to the family, parameter domain, block support, initialization, convergence event, and tails. A valid run must instead provide a reachable-block certificate that separates optimization residual, truncation or tail error, quadrature error, and comparison-to-reference error. The conditional global theorem below consumes the maximum certified log-score error; it does not manufacture that certificate.

Remark 10.11 (A small numerical illustration of the stability bound). Consider $n = 500$, $k_{\max} = 10$, and a non-conjugate block routine whose verified uniform per-block log-evidence error over all reachable blocks is $\varepsilon = 0.02$. Proposition 10.7 then guarantees segment-count log-odds error at most $(5 + 10) \times 0.02 = 0.30$ when comparing $k = 5$ to $k = 10$, and boundary log-odds error at most $2 \cdot 10 \cdot 0.02 = 0.40$ under any fixed $k \leq 10$. In terms of probability ratios, these correspond to factors of about $e^{0.30}$ and $e^{0.40}$. The numbers in this illustration are conditional on having already verified the uniform block-error bound; they are not a generic guarantee for any chosen approximation routine.

Remark 10.12 (When each non-conjugate routine struggles). The stability bound in Proposition 10.7 is tight only when the uniform blockwise error ε is actually small. Each non-conjugate routine has its own failure mode: Laplace (Theorem 10.2) requires posterior concentration near $\hat{\theta}_{ij}$, so it can be inaccurate on very short segments (W_{ij} small) or for heavy-tailed priors; mean-field variational Bayes (Proposition 10.5) can be overconfident and underestimate posterior variance, biasing moment numerators; EP can fail to converge for non-log-concave likelihoods; 1D quadrature is accurate for smooth 1D integrands but expensive in higher dimensions. For these reasons, monitor ε empirically per dataset rather than assuming it. A practical validation checklist is to compare a subset of blocks against quadrature or another trusted reference, report both maximum and high-quantile block errors, inspect errors along the exported MAP path, and assess sensitivity of $P(k \mid y)$ and boundary marginals.

Approximation-validation checklist. Before interpreting non-conjugate posterior summaries, compare a subset of block evidences against higher-accuracy quadrature or another trusted reference; report maximum and high-quantile block errors over reachable blocks; inspect errors on the MAP path; and assess sensitivity of $p(k \mid y)$, boundary marginals, and exported segmentations to the approximation method. The ranking corollary is meaningful only when the exact posterior margin is larger than the propagated approximation error.

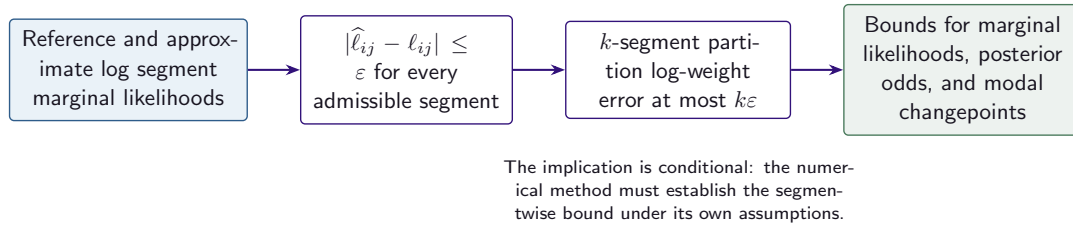


Figure 10.2. Conditional propagation from a reachable-block log-score tolerance to global evidence and posterior-odds bounds. The local numerical routine remains responsible for establishing the tolerance.

Prediction, coordinate support, and canonical metrics

CHAPTER PATH

Family-specific posterior prediction, explicit extrapolation, and evaluation rules that do not create false success from unmatched boundaries.

11.1 Posterior prediction

A prediction path must dispatch to the fitted observation family's posterior predictive. It may not fall through to a Gaussian default. The exported model also carries a coordinate support interval. Coordinates outside that interval raise an error by default; a named extrapolation policy may be used only when its behavior and provenance are recorded.

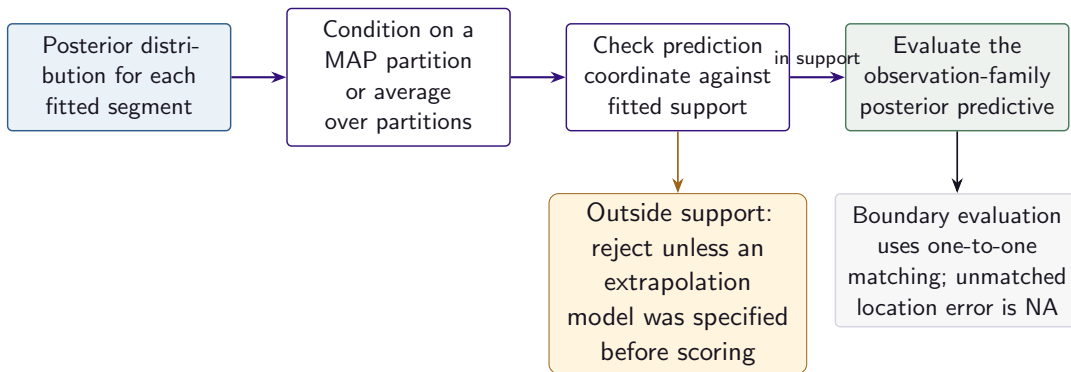


Figure 11.1. Posterior prediction and evaluation. The fitted observation family and coordinate support are checked before a proper score is evaluated.

Definition 11.1 (Canonical one-to-one boundary matching). Given predicted boundaries $\hat{\mathcal{B}}$, references $\mathcal{B}^*$, and tolerance τ , construct the bipartite graph containing edges $(\hat{b}, b^*)$ with $|\hat{b} - b^*| \leq \tau$. Select a matching with maximum cardinality; among all such matchings, select one with minimum total absolute distance, using a fixed deterministic tie rule. Precision, recall, and F_1 use the resulting matched count. No predicted or reference boundary is reused.

Definition 11.2 (Matched boundary-location error). For the canonical matching $\mathcal{M}_\tau$, define $\text{MAE}_\tau = |\mathcal{M}_\tau|^{-1} \sum_{(\hat{b}, b^*) \in \mathcal{M}_\tau} |\hat{b} - b^*|$ when at least one match exists. If no match

exists, the quantity is *not defined* and is reported as NA, never as zero.

Definition 11.3 (Held-out predictive log score). A held-out log score is valid only when the held-out observations are disjoint from training, the fitted family's posterior predictive is used, and every held-out coordinate is in support or is handled by a predeclared extrapolation policy. Failure of any condition makes the archived number inadmissible as a predictive score.

METHYLATION HELD-OUT COMPUTATION

The archived value -387.50040013308154 on $m = 381$ observations is real. It is not a valid Beta-observation held-out score because the implementation used a Gaussian predictive fallback and silently assigned coordinates beyond the training range to the final MAP block. It is retained as RES-BB-RD-007Q and excluded from predictive claims.

Having developed training-time posterior inference for partitions and segment parameters (Sections 5–10), we now turn to prediction for new (X, y) data. The prediction layer consumes posterior summaries (e.g., boundary posteriors, segment-parameter posteriors, and group evidences) and is therefore agnostic to whether block evidences were computed in closed form (Section 4.2) or via approximation (Section 10). We assume we have already trained (or otherwise obtained) a collection of G group-specific BayesBreak models $\{\mathcal{M}_g\}_{g=1}^G$, where each $\mathcal{M}_g$ comprises: (i) a fitted segmentation posterior (e.g. $P_g(k \mid y)$ and boundary posteriors), (ii) an exported MAP segmentation $\hat{t}^{(g)}$ with segment-level posterior hyperparameters, and (iii) optionally a Bayesian curve representation (e.g. posterior mean/variance at design points).

Typical outputs are group log scores, posterior group probabilities, optional unit-level responsibilities, exported MAP signal values, Bayes-curve values, and optional pointwise uncertainty summaries. These prediction interfaces define the outputs consumed by the planned real-data prediction diagnostics in Section 14.

We pause to fix two terms that have been used informally above and that recur throughout the rest of this section.

Definition 11.4 (Exported segmentation). An *exported segmentation* for a fitted group-specific model $\mathcal{M}_g$ is the tuple $(\hat{k}_g, \hat{t}^{(g)}, \{\hat{\theta}_q^{(g)}\}_{q=1}^{\hat{k}_g})$, where $\hat{k}_g$ is a chosen segment count (either an estimator of k such as the posterior mode, or a deterministic choice fixed at training time), $\hat{t}^{(g)} = (\hat{t}_0^{(g)}, \dots, \hat{t}_{\hat{k}_g}^{(g)})$ is the joint MAP boundary vector at $\hat{k}_g$ produced by Algorithms 10–11, and each $\hat{\theta}_q^{(g)}$ is a deterministic point summary of the segment- q posterior (typically the conjugate-family posterior mean or median of the expectation parameter). The exported segmentation is the signal-level analogue of a fitted model; it is what downstream prediction routines (P1 and P2) consume.

Definition 11.5 (Bayes curve). The *Bayes curve* for a fitted group-specific model $\mathcal{M}_g$ evaluated at design point x is the posterior mean of the latent piecewise-constant signal under the segment-marginalized posterior:

$$\hat{f}_g^{\text{Bayes}}(x) := \mathbb{E}[f(x) \mid y, \mathcal{M}_g] = \sum_{k=1}^{k_{\max}} P(k \mid y, \mathcal{M}_g) \mathbb{E}[f(x) \mid y, k, \mathcal{M}_g],$$

where the inner conditional expectation is taken over both the boundary vector t and the

segment parameters $\theta_{1:k}$, and where $f(x) = \theta_{q(x)}$ is the segment- q mean parameter for the segment $q(x)$ containing x under the boundary vector t .

Definition 11.6 (Segment-assignment map). For an exported boundary vector $\hat{t}^{(g)}$ on ordered coordinates $u_0^{(g)} < \dots < u_{n_g}^{(g)}$, a segment-assignment map is a declared function $a_g(x) \in \{1, \dots, \hat{k}_g\}$ assigning each query coordinate x to a training segment. The default interval-containment rule is $a_g(x) = q$ when $u_{\hat{t}_{q-1}^{(g)}}^{(g)} < x \leq u_{\hat{t}_q^{(g)}}^{(g)}$, with endpoint conventions recorded by the implementation. Queries outside the training range are either rejected or assigned by an explicitly declared extrapolation rule; silent extrapolation is not part of the default interface.

The *exported segmentation* (Definition 11.4) and the *Bayes curve* (Definition 11.5) are distinct objects with different operational meanings: the exported segmentation is a fixed piecewise-constant function with a discrete change-point structure that is convenient for downstream tasks (e.g. region labeling, hypothesis testing, set-valued prediction); the Bayes curve is a continuous mixture that integrates segmentation uncertainty and is the appropriate target when smooth predictive quantities are needed.

Prediction supports two primary tasks:

(P1) Group membership likelihoods/posteriors. Given new inputs (X, y) , compute for each group g an out-of-sample posterior-predictive likelihood $p(\text{new} \mid \mathcal{M}_g)$ (or its log), and convert these into posterior group membership probabilities $P(g \mid \text{new})$ under a prior π_g .

(P2) Signal prediction given group. Given query design points X and a specified (or inferred) group label g , return: (i) MAP piecewise-constant signal values $\hat{f}_g^{\text{MAP}}(X)$ including the corresponding breakpoints and segment levels, and/or (ii) Bayesian curve values $\hat{f}_g^{\text{Bayes}}(X)$ with pointwise uncertainty summaries when available.

Prediction output	Interpretation
ℓ_g	Group-specific log posterior-predictive score.
$P(g \mid \text{new})$	Group posterior after combining ℓ_g with prior weight π_g .
$\hat{f}_g^{\text{MAP}}(X)$	Piecewise-constant signal under the exported MAP segmentation.
$\hat{f}_g^{\text{Bayes}}(X)$	Bayes-curve posterior mean, optionally with pointwise uncertainty.
Unit responsibilities	Optional probabilities for set-valued units or grouped observations.

Table 11.1. Prediction-layer outputs exposed by fitted BayesBreak models. These interfaces support downstream scoring and calibration separately from the empirical result tables. This is an interface-definition table; no source CSV is associated with it.

Crucially, we distinguish *single point observations* (x_i, y_i) from *multivariate / set-valued observations* where each unit contains multiple internal pairs (x_{ur}, y_{ur}) , i.e. X_u and y_u are vectors. We also separately address *vector-valued responses* $y_i \in \mathbb{R}^d$ observed at each design point.

11.1.1 Prediction inputs: pointwise vs multivariate/set-valued observations

Definition 11.7 (Prediction-input classes). We distinguish three classes of new prediction inputs, jointly partitioning the observation-shape design space considered in this book.

(Case A) **Pointwise observations.** A new dataset

$$\mathcal{D}_{\text{pt}}^{\text{new}} = \{(x_i^{\text{new}}, y_i^{\text{new}}, w_i^{\text{new}})\}_{i=1}^m$$

with $x_1^{\text{new}} < \dots < x_m^{\text{new}}$, scalar (or low-fixed-dimensional) responses y_i^{new} , and optional exposure/precision weights w_i^{new} from the observation model. Each new observation lies on a single design point.

(Case B) **Multivariate/set-valued observations.** A new dataset

$$\mathcal{D}_{\text{mv}}^{\text{new}} = \left\{ ([a_u, b_u], \{(x_{ur}, y_{ur}, w_{ur})\}_{r=1}^{R_u}) \right\}_{u=1}^U,$$

where each *unit* u carries an interval support $[a_u, b_u]$ and a finite collection of internal observations indexed by $r = 1, \dots, R_u$. Prediction is performed at the unit level by integrating or aggregating over each unit's internal observations.

(Case C) **Vector-valued response at each design point.** A new dataset

$$\{(x_i^{\text{new}}, \mathbf{y}_i^{\text{new}}, w_i^{\text{new}})\}_{i=1}^m$$

where each $\mathbf{y}_i^{\text{new}} \in \mathbb{R}^d$ is a vector response (or a mixed discrete/continuous component vector). Two sub-cases are admitted: (C-i) a factorized EF across dimensions, and (C-ii) a joint multivariate EF with a single joint log-partition function and conjugate or approximate block routine.

The three cases are mutually exclusive in the data type they consume and jointly cover the observation shapes considered by the BayesBreak prediction interface.

The remainder of this subsection treats the three cases in turn; the corresponding algorithms appear in Section 11.1.8.

11.1.2 Posterior-predictive scoring for a fixed segment under EF conjugacy

For group membership inference, we require out-of-sample likelihoods. The most computationally efficient and interpretable approach is to score new data against the *exported* group curve (MAP segmentation or Bayesian curve). We first derive segment-level posterior-predictive likelihoods under EF conjugacy; these become building blocks for both Case A and Case B.

Fix a group g and consider a segment (or block) B of its exported MAP segmentation. Let the posterior for the segment parameter θ after training be the conjugate posterior

$$p(\theta \mid \alpha_B^{(g)}, \beta_B^{(g)}) \propto \exp\{\eta(\theta)^\top \alpha_B^{(g)} - \beta_B^{(g)} A(\theta)\},$$

where $(\alpha_B^{(g)}, \beta_B^{(g)})$ are the training-updated hyperparameters for that segment. Now let $\mathcal{Y}_B^{\text{new}}$ denote the collection of new observations whose design points fall into this segment.

Proposition 11.8 (EF posterior-predictive likelihood within a fixed segment). *Assume the weighted EF likelihood and conjugate prior conventions of §4.1 and §3.1, and let $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$ denote the EF block summaries of the new data restricted to segment B :*

$$S_B^{\text{new}} := \sum_{i \in B} w_i^{\text{new}} T(y_i^{\text{new}}), \quad W_B^{\text{new}} := \sum_{i \in B} w_i^{\text{new}}, \quad H_B^{\text{new}} := \sum_{i \in B} \log h(y_i^{\text{new}}, w_i^{\text{new}}).$$

Then the posterior-predictive likelihood of the new data in segment B under group g is

$$p(\mathcal{Y}_B^{\text{new}} \mid \mathcal{M}_g) = \exp\{H_B^{\text{new}}\} \frac{Z(\alpha_B^{(g)} + S_B^{\text{new}}, \beta_B^{(g)} + W_B^{\text{new}})}{Z(\alpha_B^{(g)}, \beta_B^{(g)})}. \quad (11.1)$$

Proof. The proof is identical to the single-block evidence derivation (Theorem 4.1) after replacing the prior hyperparameters (α_0, β_0) by the segment posterior hyperparameters $(\alpha_B^{(g)}, \beta_B^{(g)})$ learned from training data. $\square$

Interpretation. Equation (11.1) integrates out the segment parameter θ under the *posterior* induced by training, yielding a strictly Bayesian out-of-sample likelihood for new observations assigned to that segment. It is therefore suitable for group scoring and group posterior computation.

11.1.3 Group likelihoods for pointwise observations (Case A)

MAP-segmentation-based scoring. Let $\hat{t}^{(g)}$ denote the exported MAP boundaries for group g on the relevant domain. Assign each new design point x_i^{new} to its containing MAP segment $B = B(i; g)$ and compute segment-wise block summaries $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$. The group log-likelihood is then:

$$\ell_g^{\text{MAP}}(\mathcal{D}_{\text{pt}}^{\text{new}}) := \log p(\mathcal{D}_{\text{pt}}^{\text{new}} \mid \mathcal{M}_g) = \sum_{B \in \hat{t}^{(g)}} \log p(\mathcal{Y}_B^{\text{new}} \mid \mathcal{M}_g), \quad (11.2)$$

where each segment term uses (11.1). This yields $\mathcal{O}(m)$ evaluation time once segment membership is determined.

Bayesian-curve-based scoring (optional). If $\mathcal{M}_g$ exports a Bayesian curve representation that provides a local posterior over the mean parameter at each design point (or each small bin), then one can score each point under its local posterior predictive and sum over points. This can improve robustness near uncertain boundaries at additional computational cost.

Resegmentation scoring (optional, flexible). An alternative definition of group likelihood is to rerun the DP on the new data under group-specific hyperparameters (treating the new dataset as another draw from group g 's prior). This yields an evidence $P(\mathcal{D}_{\text{pt}}^{\text{new}} \mid g)$ that does not condition on the trained MAP boundaries. This option is useful when groups differ primarily in hyperparameters rather than in the learned curve geometry, but is computationally $\mathcal{O}(k_{\text{max}} m^2)$ per group.

11.1.4 Group likelihoods for multivariate/set-valued observations (Case B)

In Case B, each unit u contains multiple internal observations $\{(x_{ur}, y_{ur})\}_{r=1}^{R_u}$. Two output granularities are typically desired: (i) *unit-level* group likelihoods and group responsibilities, and (ii) a *global* group likelihood for the entire new dataset (product over units).

Unit-level predictive likelihood under a fixed group. Fix a group g . Partition the unit's internal design points by the MAP segments of group g . For each segment B , compute the block summaries $(S_{u,B}^{\text{new}}, W_{u,B}^{\text{new}}, H_{u,B}^{\text{new}})$ restricted to those internal points. Then define the unit likelihood as:

$$\ell_{ug} := \log p(\text{unit } u \mid \mathcal{M}_g) = \sum_{B \in \mathcal{I}(g)} \log p(\mathcal{Y}_{u,B}^{\text{new}} \mid \mathcal{M}_g), \quad (11.3)$$

where $\mathcal{Y}_{u,B}^{\text{new}}$ is the subset of unit u 's internal observations that fall into segment B and each segment term uses (11.1) with summaries for that subset.

Dataset likelihood and independence assumption.

Assumption 11.9 (Prediction-unit conditional independence). For Case B prediction, units are conditionally independent given the exported group model, and each unit contributes only through the segment-restricted predictive factors used in (11.3).

Under Assumption 11.9, the dataset likelihood under group g is:

$$\ell_g^{\text{MV}}(\mathcal{D}_{\text{mv}}^{\text{new}}) = \sum_{u=1}^U \ell_{ug}.$$

This is the direct multivariate generalization of (11.2) and corresponds to the earlier “fragment-level” intuition, but is fully application-agnostic.

11.1.5 Vector-valued responses (Case C): factorized and multivariate EF scoring

Suppose each observation is $\mathbf{y}_i = (y_{i1}, \dots, y_{id})$.

Factorized EF across dimensions (recommended default). Assume conditional independence across dimensions given segment parameters:

$$p(\mathbf{y}_i \mid \boldsymbol{\theta}) = \prod_{\ell=1}^d p_{\ell}(y_{i\ell} \mid \theta_{\ell}), \quad p(\boldsymbol{\theta}) = \prod_{\ell=1}^d p_{\ell}(\theta_{\ell}).$$

Then all block evidences and posterior-predictive terms factorize over ℓ , so log-likelihoods are sums over dimensions:

$$\ell_g(\mathbf{y}) = \sum_{\ell=1}^d \ell_{g,\ell}(y_{\cdot\ell}),$$

where each $\ell_{g,\ell}$ is computed by (11.2) or (11.3) for dimension ℓ . Computationally, this costs $\mathcal{O}(d)$ per observation (or per unit).

Multivariate EF (when available). If $(\mathbf{y}_i)$ admit a multivariate EF with conjugate prior on a joint parameter $\theta \in \mathbb{R}^p$, then the same ratio-of-normalizers structure applies with vector-valued

sufficient statistics. BayesBreak's DP layer remains unchanged; only the definition of S_{ij} and $Z(\alpha, \beta)$ changes.

11.1.6 From group likelihoods to posterior group membership probabilities

Single-dataset membership (one group per dataset). If the new dataset is assumed to come from a single (unknown) group g , define:

$$\ell_g := \log p(\text{new} \mid \mathcal{M}_g), \quad P(g \mid \text{new}) = \frac{\pi_g \exp(\ell_g)}{\sum_{g'=1}^G \pi_{g'} \exp(\ell_{g'})}.$$

This yields a *group membership likelihood vector* (the ℓ_g) and a *group membership posterior vector* (the $P(g \mid \text{new})$), matching the requested prediction behavior in an application-agnostic way.

Unit-wise membership (one group per unit). If each multivariate unit u is assumed to originate from one group, then define unit-wise responsibilities:

$$r_{ug} := P(g \mid \text{unit } u) = \frac{\pi_g \exp(\ell_{ug})}{\sum_{g'=1}^G \pi_{g'} \exp(\ell_{ug'})}, \quad (11.4)$$

where ℓ_{ug} is given by (11.3). This is precisely the out-of-sample analog of the E-step responsibilities in the latent-group model (Section 9.3).

11.1.7 MAP signal evaluation given group membership

Given a group label g and query design points $X^* = \{x_1^*, \dots, x_M^*\}$, BayesBreak returns either: (i) *MAP piecewise-constant predictions* or (ii) *Bayesian curve predictions*.

MAP piecewise-constant predictions. Let $\hat{t}^{(g)}$ be the exported MAP boundaries and let each MAP segment B provide the posterior mean of the mean parameter:

$$\hat{\mu}_B^{(g)} := \mathbb{E}[m_*(\theta) \mid \alpha_B^{(g)}, \beta_B^{(g)}].$$

Then the MAP signal at a query point x^* is:

$$\hat{f}_g^{\text{MAP}}(x^*) = \hat{\mu}_{B(x^*;g)}^{(g)},$$

where $B(x^*;g)$ denotes the MAP segment containing x^* .

Bayesian curve predictions. If the DP-derived Bayes regression curve is exported on a grid, return $\hat{f}_g^{\text{Bayes}}(x^*)$ by evaluating on the nearest grid point (or by piecewise-constant interpolation within each grid interval). When pointwise posterior variances are available, also return pointwise credible intervals; simultaneous bands require an additional construction and are not implied by the pointwise variances alone.

11.1.8 Algorithms for prediction

Algorithm 19: PredictGroup($\{\mathcal{M}_g\}_{g=1}^G, \mathcal{D}^{\text{new}}, \pi$)

Input: Trained group models $\{\mathcal{M}_g\}$; new data $\mathcal{D}^{\text{new}}$ (Case A or Case B); group prior weights π (default uniform)

Output: Log-likelihoods $\{\ell_g\}$ and posterior membership probabilities $\{P(g \mid \text{new})\}$; if Case B, also unit responsibilities $\{r_{ug}\}$

```

1 for  $g = 1$  to  $G$  do
2   if Case A (pointwise) then
3     Assign each  $(x_i^{\text{new}}, y_i^{\text{new}})$  to a MAP segment  $B(i; g)$  under  $\hat{t}^{(g)}$ ;
4     Accumulate segment summaries  $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$ ;
5     Compute  $\ell_g \leftarrow \sum_B [H_B^{\text{new}} + \log Z(\alpha_B^{(g)} + S_B^{\text{new}}, \beta_B^{(g)} + W_B^{\text{new}}) - \log Z(\alpha_B^{(g)}, \beta_B^{(g)})]$ ;
6   else
7     // Case B (multivariate/set-valued units)
8     for  $u = 1$  to  $U$  do
9       Partition unit  $u$  internal pairs by MAP segments of group  $g$ ; compute per-segment summaries;
10      Compute  $\ell_{ug}$  via (11.3) using (11.1);
11     $\ell_g \leftarrow \sum_{u=1}^U \ell_{ug}$ ;
12 Compute  $P(g \mid \text{new}) \propto \pi_g \exp(\ell_g)$  and normalize over  $g$ ;
13 if Case B then
14   Compute unit responsibilities  $r_{ug} \propto \pi_g \exp(\ell_{ug})$  and normalize over  $g$  for each  $u$  (cf. (11.4));
```

Algorithm 20: PredictMAPSignal($\mathcal{M}_g, X^*, \text{mode}$)

Input: A group model $\mathcal{M}_g$ with exported MAP segmentation $\hat{t}^{(g)}$ and/or Bayes curve; query points $X^* = \{x_1^*, \dots, x_M^*\}$; $\text{mode} \in \{\text{MAP}, \text{Bayes}\}$

Output: Predicted signal values at X^* (and optionally uncertainty)

```

1 if  $\text{mode} = \text{MAP}$  then
2   foreach  $x^*$  in  $X^*$  do
3     Find containing MAP segment  $B(x^*; g)$  (binary search on breakpoint coordinates);
4     Return  $\hat{f}_g^{\text{MAP}}(x^*) = \hat{\mu}_{B(x^*; g)}^{(g)}$ ;
5 else
6   Evaluate/interpolate the exported Bayes regression curve at  $X^*$ ; optionally return posterior variances;
```

11.1.9 Complexity

For Case A, prediction under a fixed group is $\mathcal{O}(m)$ after segment assignment ($\mathcal{O}(m \log k)$ worst case via binary search; often $\mathcal{O}(m)$ with a streaming pointer). For Case B, it is $\mathcal{O}(\sum_u R_u)$ per group, plus segment assignment for internal points. Vector-valued responses add a factor of d in the factorized EF construction. Importantly, prediction does *not* require rerunning the DP unless the optional resegmentation scoring mode is enabled. Membership posterior normalization is performed in log space by subtracting $\text{logsumexp}_g(\log \pi_g + \ell_g)$; unit responsibilities use the same normalization independently for each unit. Segment assignment

is a numerical convention, not an inferential identity: implementations must record whether binary search, streaming pointers, nearest-neighbor assignment, or a declared extrapolation rule was used for out-of-range query points.

11.1.10 Prediction diagnostics

Because the predictive layer reuses segment-level posteriors, the group posterior $P(g \mid \text{new})$ derived from (11.2) and the pointwise predictive summaries inherit their calibration quality from the segmentation. Use two diagnostics for real-data deployments:

- **Held-out log-likelihood (HLL) trace.** Split each test sequence into blocks and compute the per-block posterior-predictive log-likelihood under (11.1). The sum over blocks is the HLL for that sequence. Compare HLL trajectories across candidate segmentations (exported MAP vs. Bayes curve) and across groups; a well-calibrated model yields an HLL ranking consistent with the k -posterior ranking on held-out data.
- **PIT residuals.** For point-valued new observations with continuous predictive CDF, evaluate the probability-integral transform $u_i = F_g(y_i^{\text{new}} \mid \mathcal{M}_g)$. Under correct calibration the $\{u_i\}$ are $\text{Uniform}(0, 1)$; histograms or Kolmogorov–Smirnov summaries flag miscalibration (overconfident predictive widths show up as U-shaped histograms).

These diagnostics are cheap post-hoc checks. In this book they are planned outputs for the real-data analyses in Section 13; the planned per-case-study results tables identify the HLL and boundary-recovery fields that the completed analyses report.

Software organization, data schemas, and reproducibility records

CHAPTER PATH

How the implementation separates observation-model calculations from partition recursions, what the test runs established, and which implementation corrections remain incomplete.

12.1 Implemented organization

The inspected package contains 44 modules and 9,143 source lines. Its principal components are observation-family classes, partition dynamic programming, shared-changepoint and grouped-model wrappers, nonconjugate numerical methods, posterior prediction, changepoint metrics, dataset loaders, experiment scripts, and plotting functions. Observation-family modules populate arrays of segment marginal likelihoods and posterior moments. The partition modules combine those arrays with prior factors. Analysis modules return posterior summaries, a joint MAP partition, predictive quantities, and result metadata.

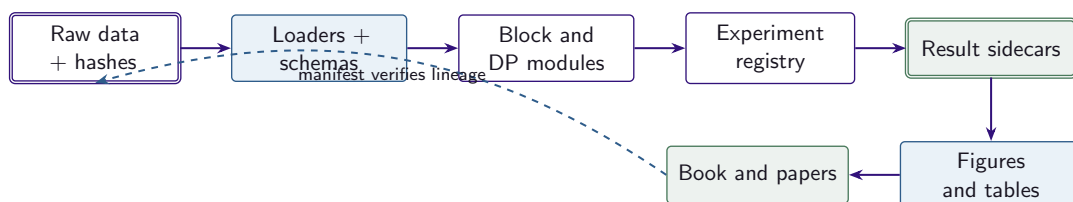


Figure 12.1. Relation among input data, model configuration, numerical calculation, returned results, and generated figures or tables. A portable release should replace absolute author paths with data identifiers and record data, configuration, environment, and output hashes.

12.2 Observed test outcomes

A clean Phase 0 run reported 174 passing tests and five skips caused by optional dependencies. The historical Phase 1 bounded verification collected 179 tests: 157 passed, five were skipped because of optional dependencies, and 17 nonconjugate or grouped-model cases did not complete. An independent Phase 6 rerun collected the same 179 tests: 173 passed, five were skipped, one EP logistic-normal test exceeded a 20-second per-test cap, and none failed. The remaining timeout is neither a pass nor a failure, and no package code was modified for this

verification. CODE-BB-012 specifies termination analysis and profiling for that remaining case. A source comparison found 110 scoped scientific files byte-identical to the fresh extraction.

Verification context	Collected	Passed	Skipped	Unresolved	Failed
Phase 0 package baseline	179	174	5	0	0
Phase 1 bounded profile	179	157	5	17	0
Phase 6 bounded rerun	179	173	5	1	0

Table 12.1. Observed test outcomes. The rows use different execution limits and are reported separately.

12.3 Mathematical invariants checked in software

1. Sum-product and max-sum recursions use the same observation-model and prior contributions but different algebraic operations.
2. A boundary hazard is applied only at interior endpoints; for small n , the partition normalizer is compared with exhaustive enumeration.
3. A latent-group optimization result should return the final value in its recorded objective sequence, and restart ranking should use that value.
4. Comparisons among nonconjugate numerical methods must use the same admissible segments and the same coordinate indexing.
5. Posterior prediction dispatches according to the fitted observation family and rejects unsupported coordinates unless an extrapolation model is specified.
6. Boundary matching first maximizes the number of matches and then minimizes total distance; matched-boundary MAE is NA when no pair is matched.
7. Result metadata distinguish executed results, results valid for a stated analysis, results excluded from a stated analysis, and planned calculations.

12.4 Reproduction commands and portable metadata

The archived report includes editable-installation commands and synthetic or real-data entry points. They identify intended calculations but do not establish that every external data source is currently available. A reproducible run should record the package version, environment, random seeds, model configuration, input hashes, cache history, and output metadata. Absolute paths in archived result files are historical information and should be replaced by portable identifiers in a release.

- `python-mvenv.venv`
- `source.venv/bin/activate`
- `pipinstall-e".[all]"`
- `pytest-q--durations=25`
- `python-mbayesbreak.experiments.synthetic--all--outartifacts/synthetic`
- `python-mbayesbreak.experiments.realdata--dataset<name>--outartifacts/realdata`

12.5 Additional executed diagnostic outputs

The archive also contains mixture-discovery, signal-to-noise sensitivity, multivariate shared-changepoint, segment-count selection, and missing-data diagnostics. These outputs were executed, but they are not required to establish the main results in this volume. Chapter 13 uses the subset needed to describe boundary calibration, generality across observation models, latent-group behavior, and finite-range timing.

Synthetic validation and executed diagnostics

CHAPTER PATH

Real recovery, calibration, portability, approximation, and scaling outputs interpreted at the exact scope supported by their archived protocols.

13.0.1 Single-sequence Gaussian illustration

We begin with a Gaussian sequence containing two true changepoints. Figure 13.1 shows two complementary outputs: the marginal posterior probability that each interior index is a boundary, and the exported joint MAP segmentation together with segment-level posterior intervals. The two dominant posterior spikes align with the true changepoints, while a smaller shoulder of posterior mass just before the first jump reflects local ambiguity caused by a short transitional run of noisy observations. That ambiguity is visible again in the bottom panel, where the exported fit inserts a short intermediate segment before settling into the large middle plateau.

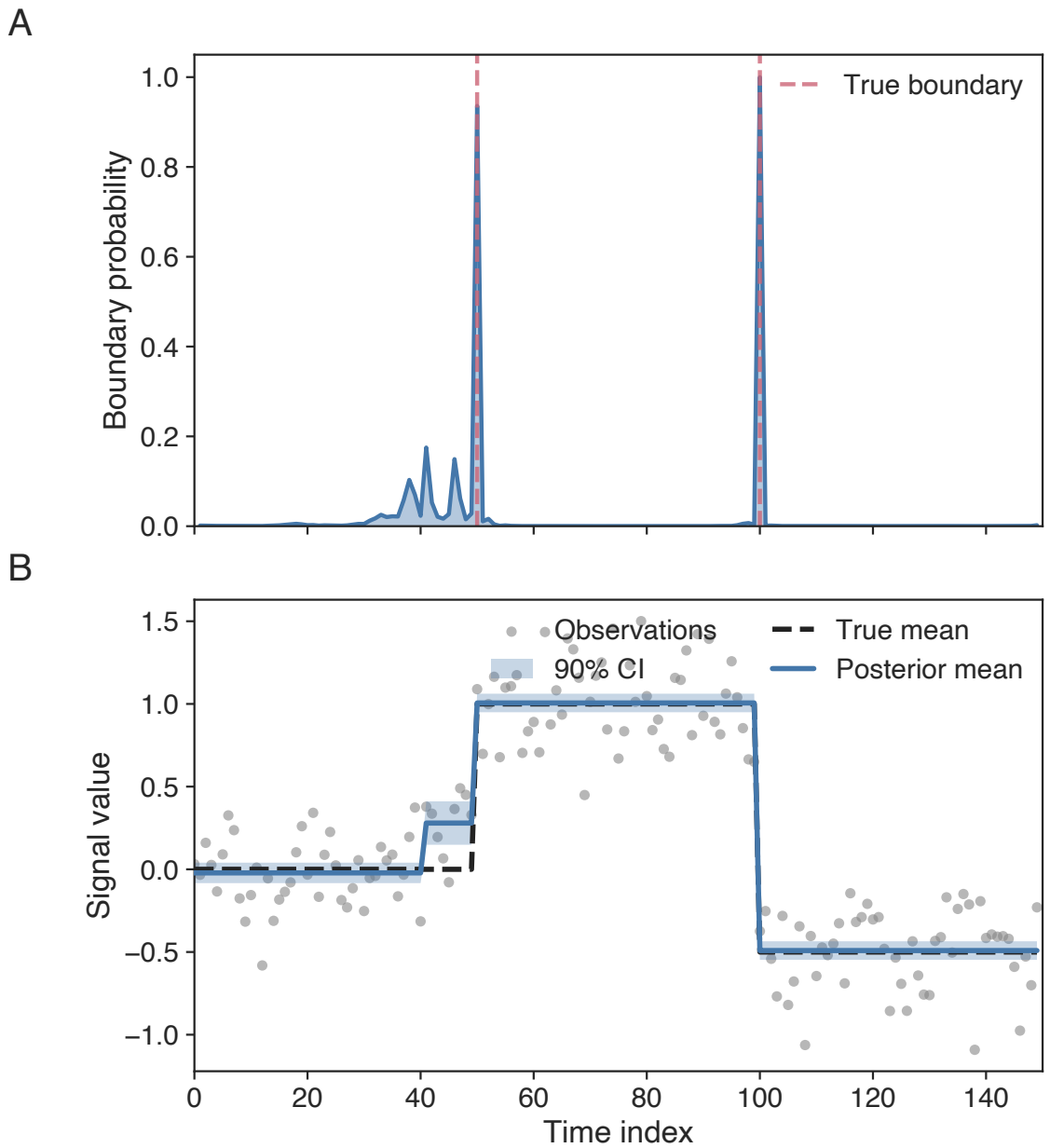


Figure 13.1. Single-sequence Gaussian example. Top: marginal posterior probability that each interior index $i \in \{1, \dots, n - 1\}$ is a changepoint, with dashed vertical lines marking the true changepoints. Bottom: observed data, true latent mean, exported joint MAP segmentation, and 90% segment-level posterior intervals.

Quantity	Value
Selected k (<code>k.ml</code>)	4
Posterior mean $\mathbb{E}[k]$	4.742
MAP k	4
$\log p(y)$	-18.031

Table 13.1. Posterior summary for the single-sequence Gaussian example in Figure 13.1. The table separates the displayed exported segmentation from posterior summaries of the segment count.

Table 13.1 makes an important methodological point explicit: the displayed segmentation and

the posterior over the segment count are related but not identical objects. In the archived example, the exported joint MAP segmentation uses four segments, while the posterior mean $\mathbb{E}[k \mid y]$ is larger, indicating residual posterior mass on more highly segmented configurations. This is precisely why the book distinguishes sum-product posterior summaries from max-sum exported segmentations.

13.0.2 Likelihood-family showcase

BayesBreak is designed so that the global DP layer does not depend on the observation model once block evidences are available. Figure 13.2 visualizes this portability across three conjugate families (Gaussian, Poisson, Binomial) and one low-dimensional quadrature example for $(0,1)$ -valued responses. Across all four panels, the fitted piecewise-constant signal tracks the true latent parameter closely and preserves the correct qualitative segmentation pattern.

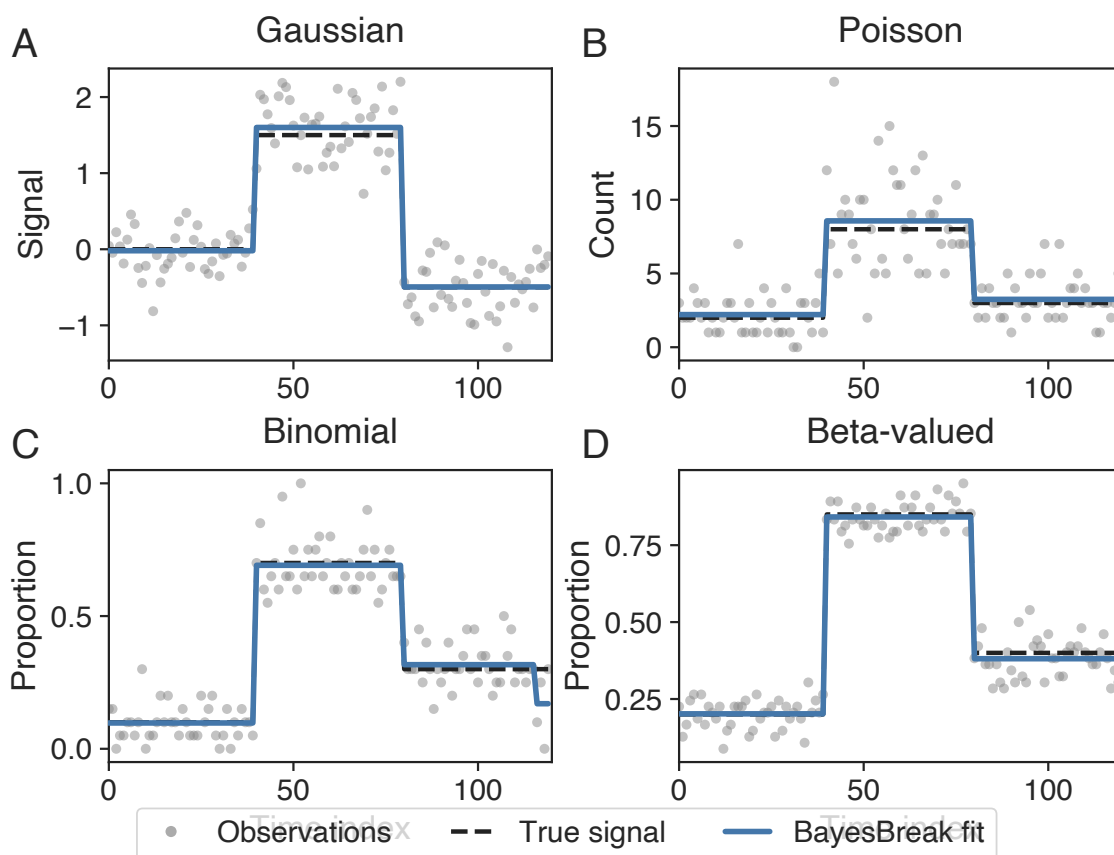


Figure 13.2. Likelihood-family showcase. The first three panels use closed-form conjugate block integrals (Gaussian, Poisson, and Binomial), while the final panel uses the one-dimensional Beta-response quadrature routine. In each case the fitted BayesBreak signal is close to the true piecewise-constant latent parameter.

13.0.3 Quantitative summary across families

Table 13.2 aggregates the small synthetic benchmark packaged with the archive. The main takeaway is not that one family universally dominates another, but that the same DP backend remains usable across very different block models. The Poisson setting is visibly the most challenging among the archived examples: it exhibits the weakest boundary F1 and the largest reconstruction error, which is consistent with the higher count variability shown in Figure 13.2. By contrast, the Beta-response quadrature example achieves the strongest boundary recovery

in this small benchmark.

Family	n	k^*	$\hat{k}$	F1@ τ	MAE	MSE	$-\log p(y)/n$
Gaussian	120	3	3	0.968	0.00	0.0023	0.191
Poisson	120	3	4	0.777	0.20	0.4501	2.162
Binomial	120	3	3	0.960	0.06	0.0004	2.108
Beta-valued	120	3	3	1.000	0.00	0.0001	2.620

Table 13.2. Available synthetic summary across the bundled family-specific examples. The final row corresponds to the Beta-response quadrature example rather than to a conjugate closed-form model.

Because the table reflects one archived benchmark configuration rather than a large Monte Carlo study, the numbers serve as a compact portability and relative-difficulty summary, not as a definitive cross-family ranking.

13.0.4 Calibration of boundary posterior probabilities

A central advantage of the Bayesian formulation is that it returns marginal posterior probabilities for boundary events. Figure 13.3 checks whether those probabilities are calibrated under repeated Gaussian simulations. The reliability curve tracks the diagonal closely across the full probability range, with an empirical calibration error reported in the figure inset of approximately $\text{ECE} \approx 0.010$ and Brier score ≈ 0.011 . The mid-probability bins remain the noisiest because boundary events are intrinsically rare per sequence, but the ordering of confidence levels is sensible: bins assigned larger posterior boundary probability correspond to larger empirical boundary frequency.

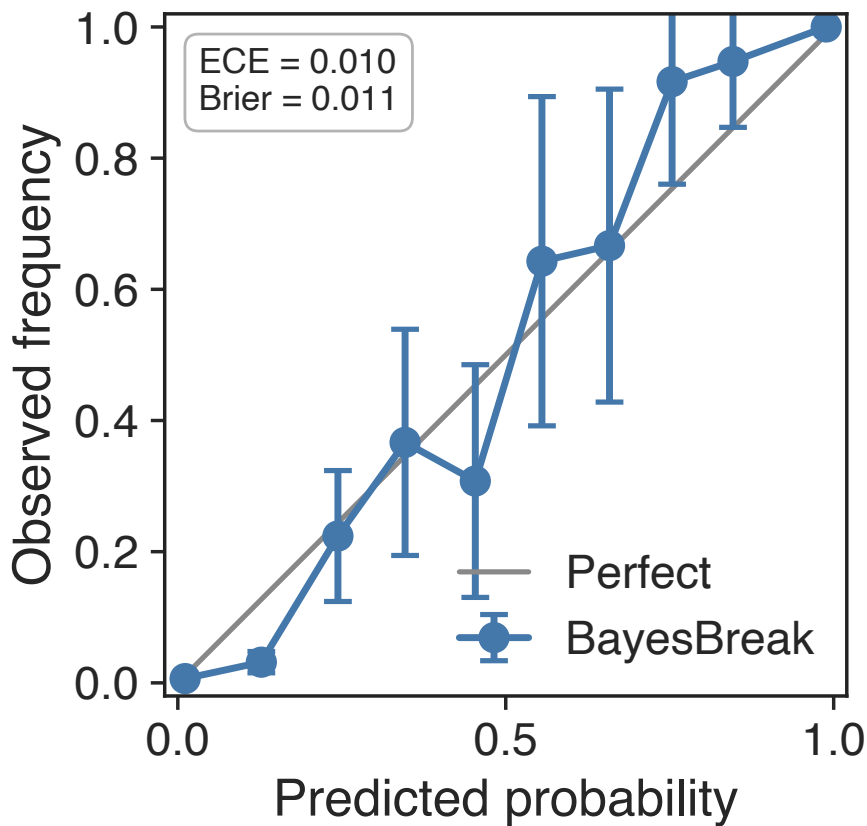


Figure 13.3. Calibration of marginal boundary posterior probabilities under synthetic Gaussian data. The dashed diagonal denotes perfect calibration; the orange curve is the empirical frequency within bins of predicted probability.

This experiment is valuable because segmentation methods are often judged only by recovered changepoints. For BayesBreak, well-behaved posterior probabilities are equally important: they control downstream thresholding, boundary ranking, and any decision rule that is sensitive to the strength of evidence rather than just to the best segmentation.

13.0.5 Latent-group pooling

Figure 13.4 evaluates the finite latent-group template model on $n_{\text{seq}} = 24$ synthetic Gaussian sequences ($n = 80$, $\sigma = 1.0$) drawn from two groups with different changepoint patterns. The left panel shows the posterior responsibilities sorted by truth: the two groups are almost separable, with a small ambiguous band near the class boundary corresponding to a single mis-assignment (hard accuracy 96%). The right panel reports the group-specific boundary marginals induced by the fitted templates; both groups concentrate their posterior mass at the correct latent changepoints, with the cluster-1 marginal locating both boundaries near probability one and the cluster-0 marginal recovering its second boundary at $t \approx 53$ and a softer mode near $t \approx 26$ for the first.

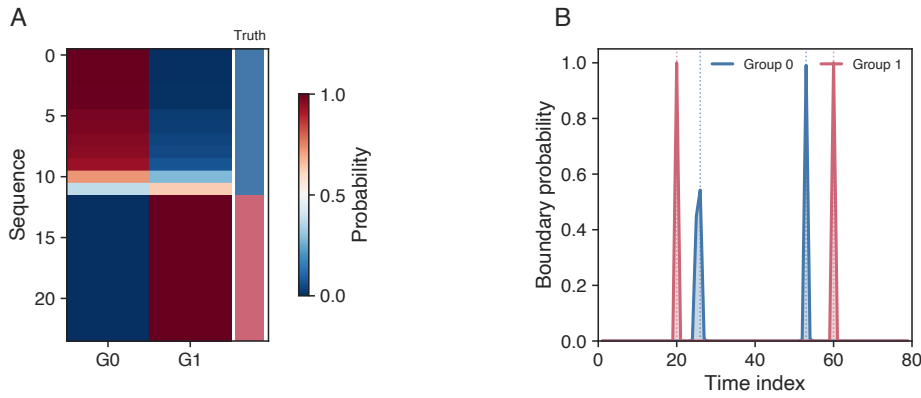


Figure 13.4. Latent-group pooling on synthetic Gaussian sequences ($n = 80$, $\sigma = 1.0$, $n_{\text{seq}} = 24$). Left: posterior responsibilities r_{sg} for each sequence and score-template group, sorted by truth; the right-hand stripe encodes the true label. Right: group-specific marginal boundary probabilities under the fitted latent templates, with dotted vertical lines marking the true per-group changepoints.

A six-panel companion diagnostic (uncropped figure `shared/figures/results/fig4_latent_groups.pdf`) unpacks the same fit further: per-group example sequences with the inferred piecewise mean, the boundary marginals, the responsibility heatmap, the group-averaged signal, and a per-sequence assignment-confidence diagnostic that plots r_{s,y_s} (responsibility on the true group) sorted along the x -axis, with the 0.5 decision threshold shaded and any mis-assigned sequences circled. Even when overall accuracy is high, this view exposes which sequences sit close to the decision boundary and is informative regardless of whether any hard mistakes are made.

13.0.6 Non-conjugate approximation trade-offs

For non-conjugate likelihoods, BayesBreak relies on approximate block evidences while reusing the same DP layer. Table 13.3 compares several such approximations in a bundled logistic-normal synthetic example, using numerical quadrature as a reference. The table shows the observed speed-accuracy trade-off in this archived example: Laplace is the fastest among the listed approximations, while exact quadrature is the slowest. At the same time, block-level approximation error is not perfectly monotone with segmentation quality, because the DP aggregates many local scores nonlinearly.

Method	$\max \Delta \log A^0 $	$q_{95} \Delta \log A^0 $	median $ \Delta \log A^0 $	time (s)	F1@ τ	$\hat{k}$
quadrature	0.000	0.000	0.000	0.093	0.286	6
laplace	1.235	1.188	0.980	0.049	0.286	6
jj	1.265	1.216	1.002	0.071	0.500	3
ep	14.502	14.141	12.680	0.059	0.500	7
pg-vb	1.265	1.216	1.002	0.072	0.500	3

Table 13.3. Approximation trade-offs for a non-conjugate Bernoulli/logistic block model. Quadrature is used as the local reference block integral; the remaining methods trade block-level fidelity against computational cost.

Connection to Proposition 10.7. The large reported values of $\max |\Delta \log A|$ in Table 13.3 (several to tens of log units on the worst reachable block for some approximations) place this

illustration outside the small- ε regime where the uniform-error bound in Proposition 10.7 is tight. The boundary F1 values are therefore descriptive diagnostics, not a confirmation of the small- ε regime. The table still usefully shows that block-level error, runtime, and segmentation quality need not move monotonically together, but it is not a universal ranking of non-conjugate approximation methods.

13.0.7 Runtime scaling

Finally, Figure 13.5 and Table 13.4 report runtime behavior for the Gaussian implementation. Panel A sweeps the series length $n \in \{50, 100, 200, 400, 800\}$ at two fixed segment budgets $k_{\max} \in \{10, 20\}$; panel B sweeps the segment budget $k_{\max} \in \{5, 10, 20, 40, 80\}$ at fixed $n = 200$. Over the archived range, the empirical log-log slopes annotated on Figure 13.5 are ≈ 1.07 in n at $k_{\max} = 10$, ≈ 1.14 in n at $k_{\max} = 20$, and ≈ 0.88 in $k_{\max}$ at fixed $n = 200$. Both sweeps therefore sit well below the $\mathcal{O}(k_{\max}n^2)$ worst case predicted by the exact sum-product DP. The discrepancy reflects two practical effects: (i) cumulative sufficient-statistics caching makes each candidate block evaluation effectively constant time, so the DP cost reduces to traversing the $\mathcal{O}(k_{\max}n)$ reachable $(k, \text{end-point})$ pairs over the small archived range; and (ii) constant-factor overhead from prior evaluation and DP setup dominates at the smallest n , flattening the small- n end of the curve. The $\mathcal{O}(k_{\max}n^2)$ worst case still controls the asymptotic profile and is the relevant bound for the much larger n regimes flagged for the next-pass scaling study in Remark 5.8.

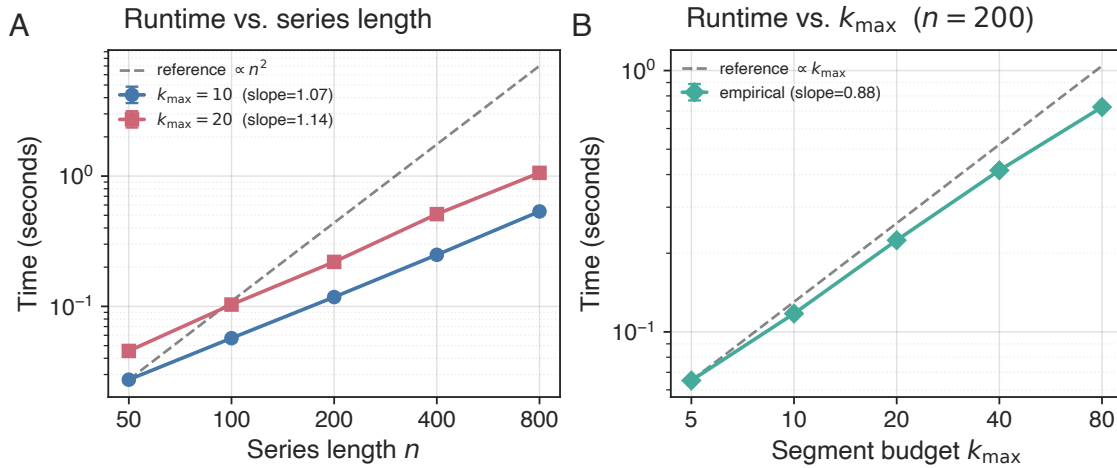


Figure 13.5. Runtime scaling for the Gaussian implementation. Panel A: log-log wall-clock time vs. series length $n \in \{50, 100, 200, 400, 800\}$ at two segment budgets $k_{\max} \in \{10, 20\}$; annotated slopes give the empirical $\log t$ vs. $\log n$ exponent for each curve, and a dashed reference line shows the theoretical $\mathcal{O}(n^2)$ worst case anchored at the smallest measured point. Panel B: log-log time vs. $k_{\max} \in \{5, 10, 20, 40, 80\}$ at fixed $n = 200$ with a dashed linear-in- $k_{\max}$ reference. Points are means over 5 repetitions; error bars denote one standard deviation.

n	$k_{\max}$	mean (s)	std (s)
50	20	0.0455	0.0008
100	20	0.1033	0.0011
200	20	0.2195	0.0011
400	20	0.5106	0.0307
800	20	1.0585	0.0345

Table 13.4. Numerical companion for the same archived Gaussian runtime configuration as Figure 13.5. The bundled table reports the $k_{\max} = 20$ configuration of panel A rather than every curve displayed in the figure; the $k_{\max}$ sweep of panel B is recorded in the figure’s CSV sidebar.

13.1 Evidence interpretation

All populated values and panels in this chapter are real archived computations. Their role is method validation on the bundled synthetic settings, not a comprehensive state-of-the-art claim. Boundary calibration reports ECE approximately 0.010 and Brier approximately 0.011. The latent-group template experiment reports 96% hard recovery at $\sigma = 1.0$ for 24 sequences. The runtime fits report log–log slopes about 1.07 at $k_{\max} = 10$ and 1.14 at $k_{\max} = 20$ for $n = 50, \dots, 800$; these are local empirical summaries, not an asymptotic complexity result.

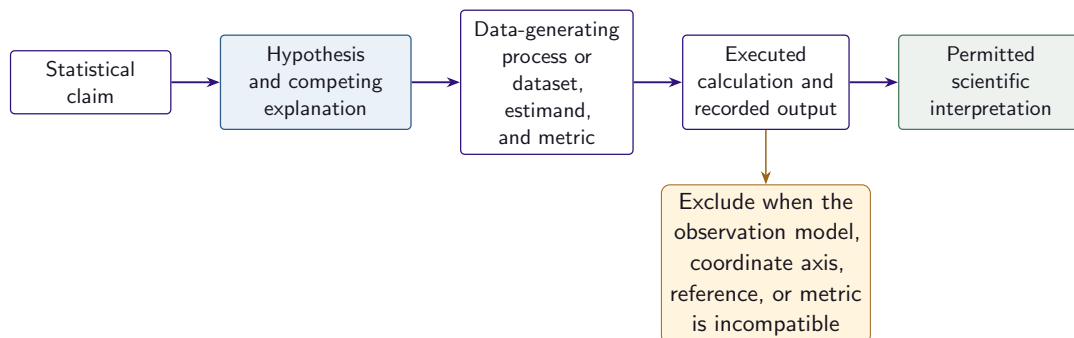


Figure 13.6. Relation among claims, experiments, and interpretation. An executed computation is excluded from a stated conclusion when its observation model, coordinate axis, reference quantity, or metric is incompatible with that conclusion.

13.2 Coverage still required

The archive does not yet provide full repeated uncertainty intervals, fairness-controlled external comparators across every family, large- n scaling, or complete misspecification grids. These are specified as protocols in Appendix D; no planned cell is populated in this chapter.

Real-data studies and interpretation

CHAPTER PATH

Four real executed fits, their exact data and model semantics, comparator limits, and two immutable computations excluded from incompatible scientific interpretations.

14.1 Interpretation policy

The four case studies are real executed BayesBreak fits with archived data hashes and four-of-four DP diagnostic checks. Red vertical lines in the current figures are fitted MAP boundaries unless an independent annotation source is explicitly identified; they are not treated as geology, macro-event, copy-number, or atlas ground truth. Panel D in each generic case-study figure is an in-sample cumulative diagnostic, not held-out evaluation.

14.1.1 Well-log NMR response

The archived source describes the O Ruanaidh well-log NMR response with 4,050 original values and an executed downsampled sequence of $n = 507$. The Gaussian fit used $k_{\max} = 40$, returned $\hat{k} = 23$, and recorded log evidence -4989.283473023698 . A separately executed length-cohesion configuration returned $\hat{k} = 25$ and log evidence -4997.147814570523 ; these values describe different priors and are not a universal ranking of irregular-design strategies.

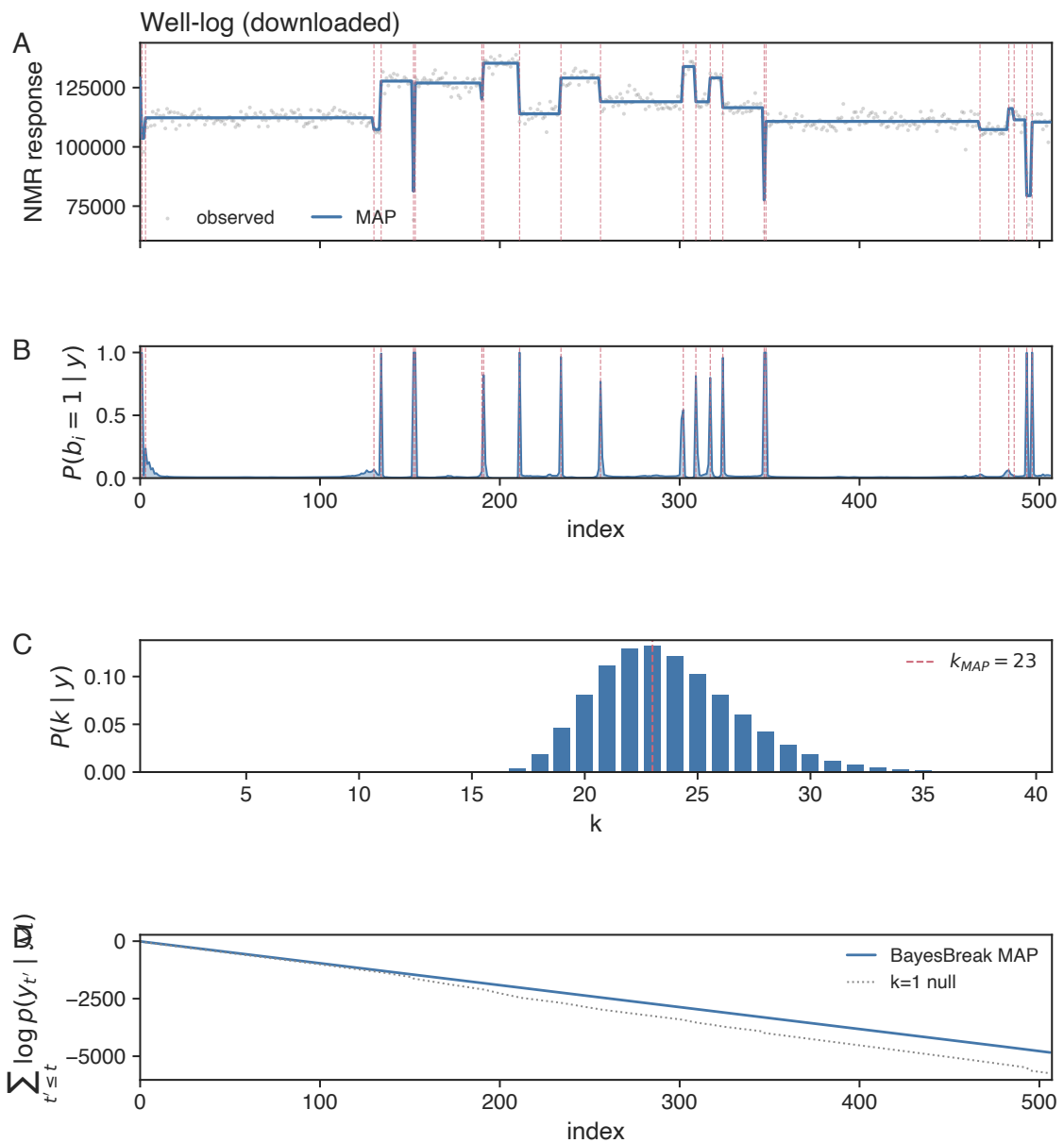


Figure 14.1. Executed well-log fit. The solid segmentation and red vertical markers are fitted BayesBreak summaries. Boundary probabilities and $P(k | y)$ are posterior outputs; the cumulative panel is in-sample.

Configuration	$\hat{k}$	Log evidence	Boundary entropy (nats)
Index-uniform, $c = h = 1$	23	-4989.28	36.46
Archived length cohesion $c(i, j) \propto u_j - u_i$	25	-4997.15	45.46

Table 14.1. Real archived well-log configurations. External boundary accuracy is unavailable.

The comparator cache also contains agreement-to-BayesBreak diagnostics. At tolerance three, optimal partitioning and the Fearnhead-configured reference DP obtained agreement F_1 values 0.91 and 0.93, respectively; binary segmentation obtained 0.86, PELT 0.18, and the archived wild-binary run 0.05. These values measure resemblance to BayesBreak's MAP boundary set, not recovery of independently verified geology.

14.1.2 Array-CGH shared boundaries

The array-CGH archive contains 43 subjects on 2,215 probes. The shared-boundary Gaussian model used $k_{\max} = 15$, returned $\hat{k} = 15$, and recorded pooled log evidence 76359.79958695151. The sum of 43 independently fitted subject evidences is 109617.69544504143; because the two quantities arise from different factorizations, their difference is descriptive and is not a Bayes factor for pooling.

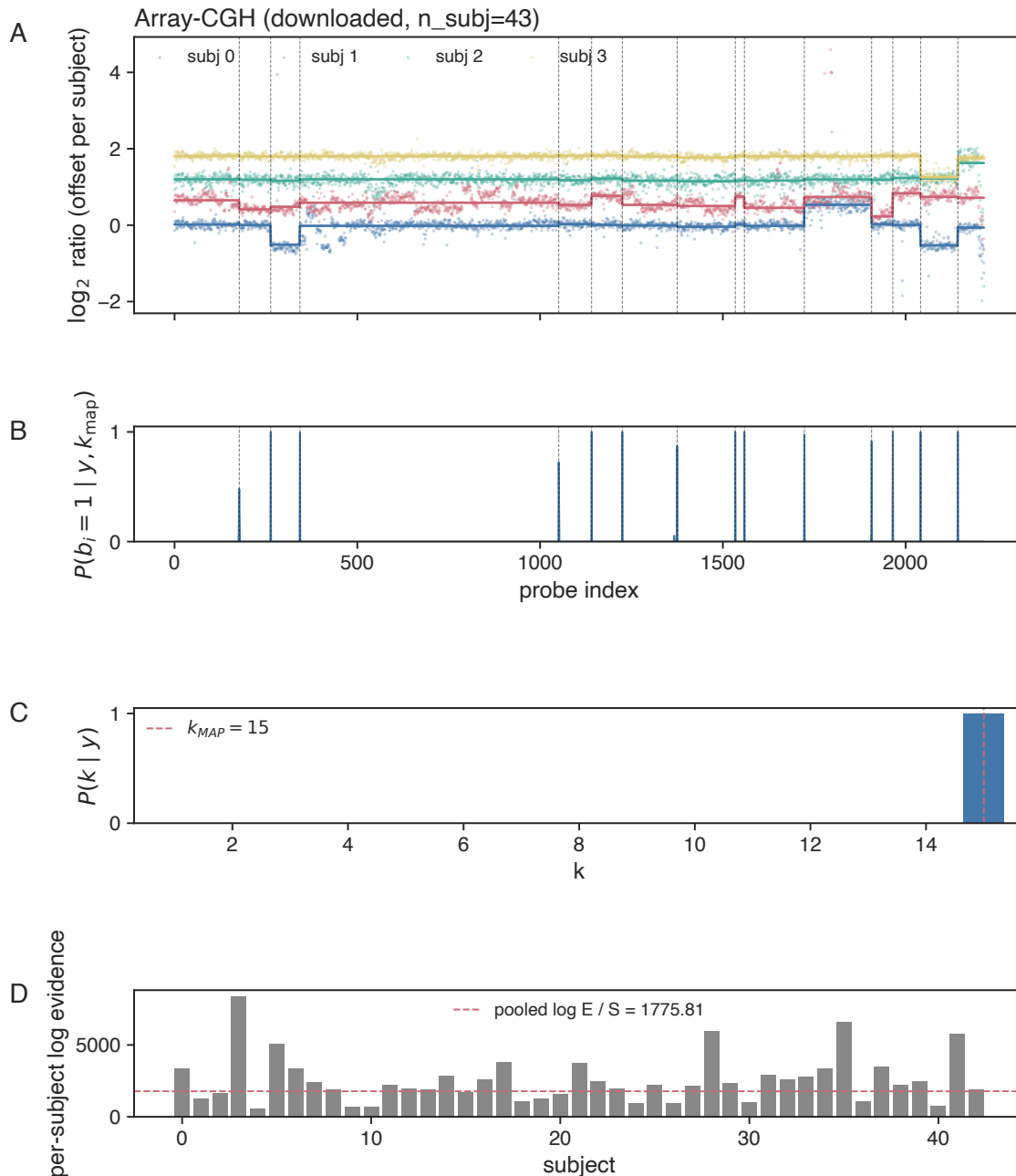


Figure 14.2. Executed shared-boundary array-CGH fit. The common MAP segmentation is fitted across the 43-subject matrix; no external boundary annotation is asserted.

RES-BB-CMP-002: INCOMPATIBLE COMPARATOR AXIS

The comparator cache flattened a 43×2215 matrix into 95,245 values while retaining BayesBreak reference boundaries on the 2,215-probe axis. The resulting zero agreement

values and large distances are real executions of an invalid comparison. They are excluded from all scientific comparison tables. A corrected run must consume the raw subject-by-probe matrix and a validated probe coordinate axis under a new result identifier.

14.1.3 S&P 500 volatility regimes

The source covers January 1, 2015 through December 31, 2023, with approximately 2,263 raw daily observations and a stride-four executed series of $n = 566$. The primary Gaussian model on log-squared returns returned $\hat{k} = 29$ and log evidence -1296.6537252802884 . The secondary series is a Bernoulli indicator of crossing the empirical 95th-percentile threshold, not a Poisson count; it returned $\hat{k} = 50$ and log evidence -114.41004640965144 under a different data model, so the raw evidences are not directly comparable.

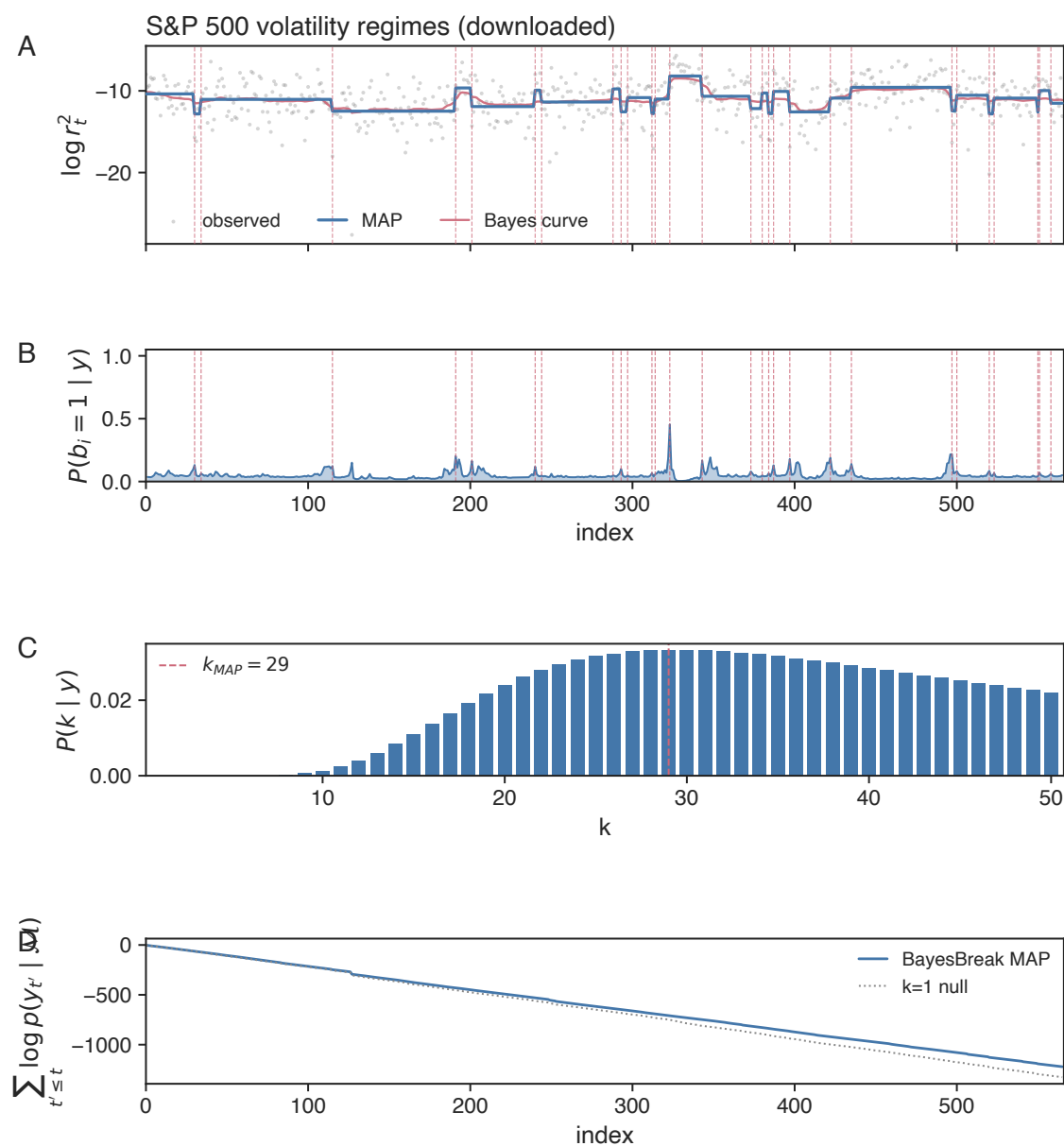


Figure 14.3. Executed S&P 500 volatility fit for 2015–2023 after the archived stride-four preprocessing. Red markers are fitted MAP boundaries, not independently annotated macro events.

Method or signal	$\hat{k}$	Log evidence	Agreement $F_1@3$ vs BB Gaussian
BayesBreak Gaussian, $\log r_t^2$	29	-1296.65	reference
BayesBreak Bernoulli, threshold indicator	50	-114.41	different signal
Optimal partitioning, matched k	29	—	0.82
PELT	79	—	0.53
Binary segmentation, matched k	29	—	0.43
Wild binary segmentation, matched k	29	—	0.07
Fearnhead-configured reference DP	13	—	0.55

Table 14.2. Executed SPX outputs. Agreement columns use BayesBreak MAP boundaries as the reference and are not external accuracy.

14.1.4 Methylation on the methylKit chromosome-21 example

The executed data are the methylKit test1.myCpG chromosome-21 example with $n = 1904$ CpGs and per-CpG read coverage used as known precision. This is not the described Loyfer methylation-atlas analysis. The Beta-observation quadrature fit used $k_{\max} = 15$, returned $\hat{k} = 15$, and recorded log evidence -9518.667508691196 .

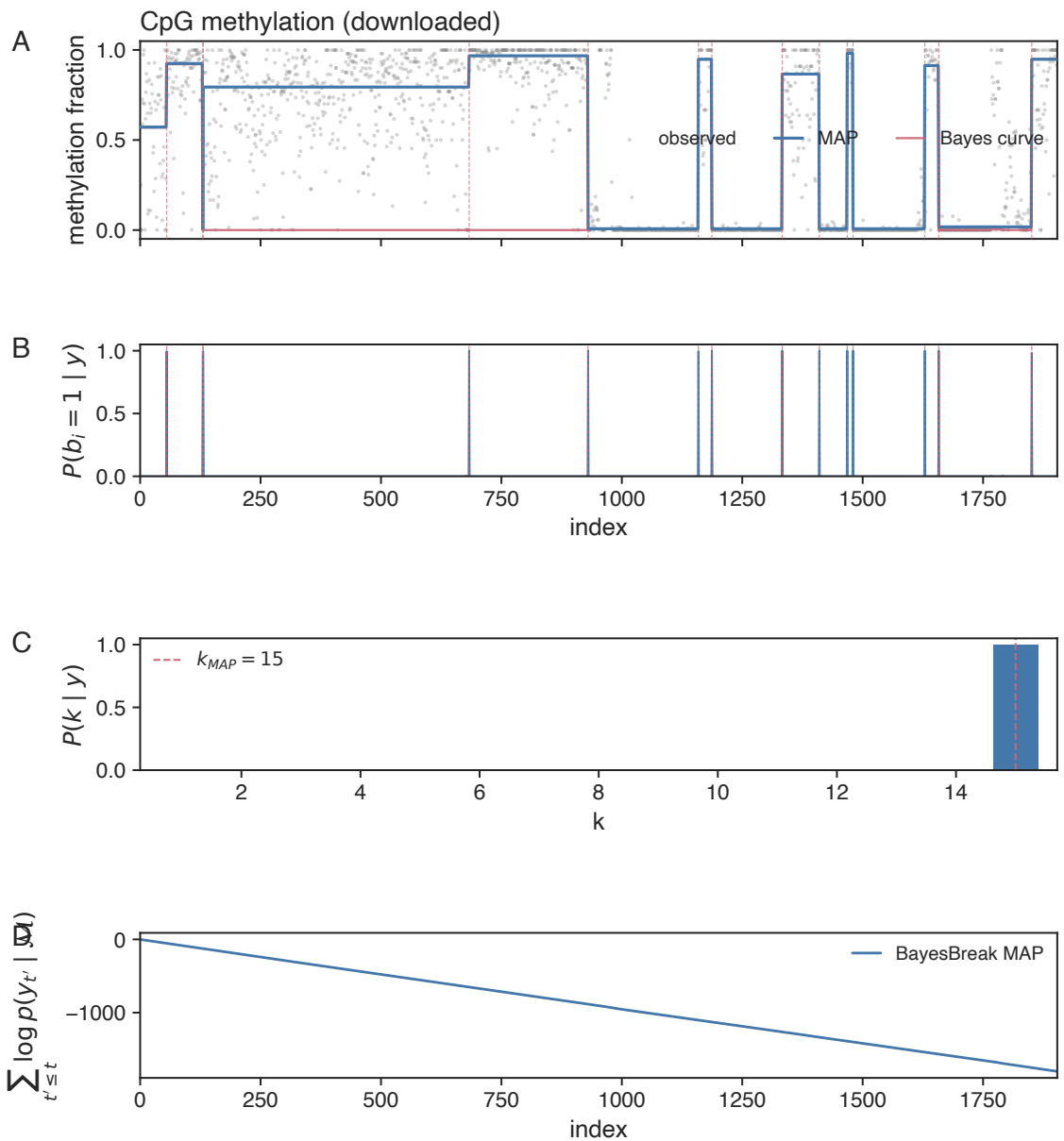


Figure 14.4. Executed methylKit chromosome-21 Beta-observation fit. Red markers are fitted MAP boundaries; no atlas transition annotation is present in the archived source.

Method	$\hat{k}$	Runtime (s)	Agreement $F_1@3$ vs BB
BayesBreak Beta-observation	15	—	reference
Optimal partitioning, matched k	15	46.89	0.79
Binary segmentation, matched k	15	0.10	0.57
PELT	5	3.32	0.44
Wild binary segmentation, matched k	15	0.76	0.14

Table 14.3. Executed methylation comparator agreement. No external atlas accuracy is claimed.

RES-BB-RD-007Q: INVALID FAMILY/SUPPORT PREDICTIVE PATH

The archived held-out value -387.50040013308154 for $m = 381$ remains in the result ledger. It is not shown as valid held-out performance because the implementation used a Gaussian fallback for a Beta-observation estimator and silently clipped out-of-range coordinates to the last MAP block. The segmentation fit and its log evidence remain admissible descriptive real results.

14.2 What these case studies establish

Together, the fits demonstrate that the implemented interface executed on Gaussian single-series, Gaussian shared-boundary, Gaussian volatility, Bernoulli threshold, and Beta-observation settings, and that their archived DP invariants passed. They do not establish external boundary accuracy, causal correspondence to domain events, or superiority to comparators. Those claims require the predeclared protocols in Appendix D.

Limitations and failure modes

CHAPTER PATH

Conditions under which the factorized partition model, segment distribution, numerical approximation, hierarchical assumptions, prediction rule, or empirical comparison is not valid.

The exact results in this volume concern offline Bayesian segmentation on a finite ordered candidate set. They require finite segment marginal likelihoods and a prior whose contribution can be represented by segment or transition factors. Nonconjugate calculations require additional numerical assumptions. This chapter states the corresponding limitations.

15.1 Computational and model support

Sequence length and segment count. The dynamic program has worst-case time complexity $\Theta(k_{\max}n^2)$ and working memory at least $\Theta(k_{\max}n)$ for the principal posterior quantities (Proposition 6.2). It is an offline method. Large sequences may require a restricted changepoint set, pruning, or an approximate decomposition. Such modifications change the set of partitions or the summation unless their exactness is established separately; they do not automatically inherit Theorem 5.5.

Candidate changepoints. A changepoint outside the candidate set cannot receive posterior probability. This is a support restriction of the model, not a numerical error. Scientific locations that must remain possible should be included before fitting.

Nonlocal dependence among segments. The recursions assume that, after any necessary state variable is included, the probability of a partition can be written as products of local contributions. A prior depending on a global pattern of segment lengths, unrestricted interactions among nonadjacent segments, or labels with non-Markov dependence may violate this assumption. Some dependence structures can be handled by an enlarged state space, but the recurrence and complexity must then be derived for that model.

15.2 Observation-model misspecification and prior sensitivity

Misspecified segment distribution. Posterior probabilities are conditional on the chosen within-segment distribution, its descriptors, and its hyperparameters. Heavy-tailed observations

under a Gaussian segment model, zero inflation under a Poisson model, overdispersion under a Binomial model, or unmodeled serial dependence can distort segment marginal likelihoods and therefore the posterior over partitions. Posterior predictive diagnostics can reveal discrepancies, but the method does not automatically select the correct observation family.

Prior sensitivity. The posterior for k and the changepoint probabilities can depend materially on the segment-count prior, segment cohesion $c_u(i, j)$, and boundary hazard $h_u(j)$. This dependence can be stronger on an irregular grid because different priors assign different probabilities to long physical gaps. An analysis should report sensitivity to scientifically plausible prior choices rather than treat a single prior as neutral.

15.3 Numerical approximation

Nonconjugate segment marginal likelihoods. Proposition 10.7 assumes a verified bound on the largest log marginal-likelihood error over every admissible segment. The bound increases with the number of segments, so a fixed segmentwise error can have a larger effect for large k . If the numerical method does not establish this condition, the dynamic program returns the exact posterior for the approximate score array but no error bound relative to the intended posterior.

In the archived comparison, the maximum absolute discrepancy for expectation propagation was approximately 14.5 relative to the quadrature reference on the examined segment set. This is not a small-error setting. The associated boundary score is descriptive and does not verify the stability theorem. A scientifically interpretable nonconjugate analysis should compare numerical methods on the same admissible segments, examine convergence and tail behavior, and use a more accurate calculation when the discrepancy is too large for the desired posterior comparison.

15.4 Hierarchical assumptions

Common changepoints across sequences. The shared-changepoint model assumes aligned candidate coordinates and conditional independence across sequences given a common partition and sequence-specific segment parameters. Unmodeled cross-sequence dependence or genuine sequence-specific changepoints can make the common partition misleading. The finite-candidate consistency theorem also requires a positive average expected log-score gap between the true partition and every alternative; one informative sequence does not satisfy that condition as the number of sequences increases.

Latent-group non-uniqueness. The finite latent-group criterion is invariant under permutation of the group labels and may have duplicate or collapsed templates. Multiple restarts do not prove that a global maximum has been found. Because the sequence-template scores are not assumed to be normalized component densities, finite-mixture identifiability theorems do not apply. The archived implementation must also be corrected so that restart ranking uses the final value in each objective sequence.

Posterior summaries versus joint optimization. Marginal changepoint probabilities and the joint MAP partition are different summaries. Selecting the marginal mode for each ordered

changepoint coordinate need not yield a valid or jointly most probable partition. Applications should report the posterior quantities relevant to the decision rather than interpreting all returned summaries as interchangeable.

15.5 Prediction and evaluation

Coordinate support and extrapolation. Posterior prediction must use the fitted observation family. A coordinate outside the fitted range requires a pre-specified extrapolation model or rejection. Silent endpoint assignment changes the predictive model and can yield a finite but invalid score, as demonstrated by the archived methylation calculation excluded from predictive conclusions.

Changepoint metrics. Tolerance-based boundary F_1 depends on the tolerance and matching rule. When BAYESBREAK MAP boundaries are used as the reference, a comparator score measures agreement with BAYESBREAK, not accuracy against independent truth. A location error is undefined when no pair is matched and is reported as NA. The four real-data analyses do not contain independent changepoint annotations, so their fitted red lines cannot be interpreted as verified geological, genomic, or financial events.

Incompatible comparator coordinates. The array-CGH comparator flattened a 43×2215 fitted matrix to 95,245 values while the reference changepoints remained on the 2,215-probe axis. Those objects do not define a common matching problem. The executed values remain recorded, but they are excluded from comparator conclusions.

15.6 Choice of analysis

- (1) For a single sequence with a conjugate segment model, use the exact segment integration and sum-product or max-sum recursions of Chapters 4–6.
- (2) For a nonconjugate segment model, use a stated numerical method and assess its error on the admissible segment set. Use Proposition 10.7 only when Assumption 10.6 has been established.
- (3) For aligned sequences expected to share changepoints, use the common-partition model of Chapter 8; examine sensitivity to sequence dependence and violations of common structure.
- (4) For observed groups, fit the shared-changepoint model within each group. For unobserved groups, use the finite latent-group criterion of Chapter 9 and interpret its allocations and templates according to that criterion, without importing normalized-mixture claims.
- (5) For prediction, use the posterior predictive distribution of the fitted observation family and state whether conditioning is on a selected partition or averaged over partition uncertainty.

15.7 Empirical coverage

The synthetic studies and four real-data fits demonstrate the calculations under their archived settings. They do not provide a complete fairness-controlled comparison with all relevant

frequentist and Bayesian methods, broad repeated uncertainty intervals, comprehensive misspecification studies, or large- n scaling. Those studies remain necessary before making broad comparative or robustness claims.

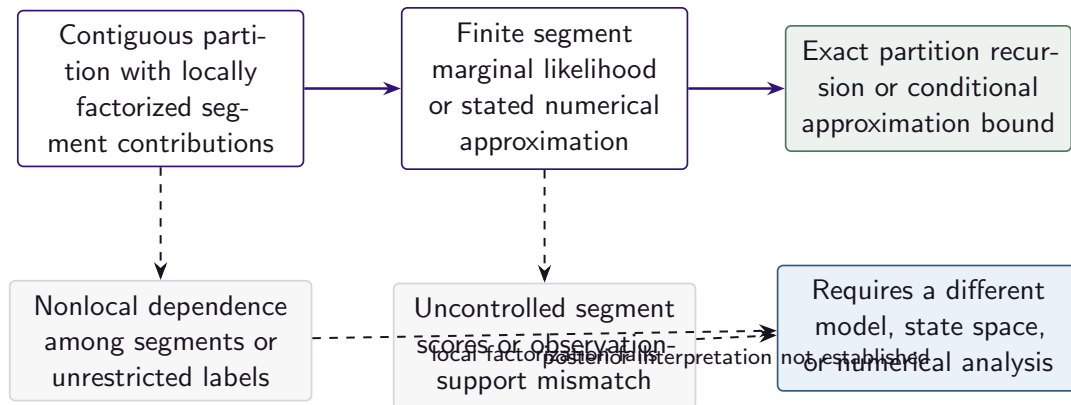


Figure 15.1. Scope of the exact partition calculation. Unsupported candidate sets, nonlocal dependence, misspecified segment distributions, uncontrolled numerical scores, undefined extrapolation, and incompatible comparator axes require a revised model or analysis.

Reproducibility, remaining validation, and conclusion

CHAPTER PATH

A final account of the established statistical results, the executed empirical results, the unresolved implementation work, and the scope of the generalized method.

16.1 Established statistical results

BAYESBREAK is a finite Bayesian calculation over ordered partitions. For regular exponential-family segment models with proper conjugate priors, exact segment marginal likelihoods and posterior moments are obtained from sufficient statistics. Under a factorized partition prior, sum-product dynamic programming yields fixed-count marginal likelihoods, $p(k \mid y)$, changepoint marginals, changepoint-event probabilities, and Bayes curves. A separate max-sum recursion yields the joint MAP partition. For aligned conditionally independent sequences with a common partition, sequence-specific segment marginal likelihoods multiply. The finite latent-group criterion has monotone alternating updates under exact maximization of its Jensen minorizer. For nonconjugate segment models, the posterior-odds bounds are conditional on a uniform error bound for every admissible segment score.

16.2 Interpretation of the executed results

The archive contains executed synthetic recovery, boundary-calibration, observation-model, latent-group, nonconjugate-approximation, and finite-range timing studies, together with four real-data fits. The real-data analyses are descriptive because independent changepoint labels are not archived. Comparator scores that use the BAYESBREAK joint-MAP changepoints as the reference measure agreement rather than external accuracy. The two incompatible computations remain in the result history but are excluded from comparator or predictive conclusions.

16.3 Implementation and validation still required

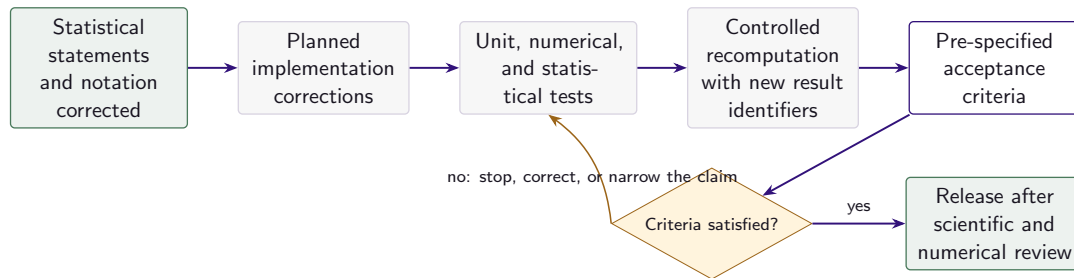


Figure 16.1. Remaining implementation and validation stages. Corrections to source code and controlled recomputations are distinct from the mathematical and editorial revision of this volume.

The highest-priority implementation tasks are: explicit representation of segment cohesion and boundary hazard; correction of the final objective returned by the latent-group optimizer; family-correct posterior prediction for Beta-valued observations; rejection or explicit modeling of out-of-range prediction coordinates; the one-to-one changepoint matching rule; validation of comparator coordinate axes; complete result metadata; and resolution of the 17 bounded-run tests that neither passed nor failed within the previous time limit. Corrected comparator or predictive calculations must receive new result identifiers and must not overwrite the archived outputs.

16.4 Conclusion

The main contribution of `BAYESBREAK` is a generalized hierarchical Bayesian segmentation method that combines exponential-family segment integration with exact posterior computation over ordered partitions and extends that construction to irregular designs, multiple aligned sequences, known groups, and finite latent groups. The generality lies in the observation-model independence of the partition recursion once mathematically valid segment marginal likelihoods are supplied. The hierarchical structure lies in explicit probability models for shared or group-specific changepoints and in the separately stated latent-group optimization criterion.

This formulation is grounded in conjugate analysis, product-partition models, Bayesian multiple-changepoint recursions, hierarchical modeling, and dynamic programming. Its limitations are equally mathematical: exactness depends on finite segment integrals and local factorization; shared-changepoint pooling depends on alignment and conditional independence; nonconjugate results depend on numerical error control; and empirical conclusions are limited by the available references, replications, and comparator definitions. Stating those conditions precisely preserves the breadth of the generalized method without replacing statistical claims by branding language.

Complete annotated literature review

CHAPTER PATH

Every bibliography entry receives one substantive project-specific annotation, with closest-work deltas, transfer limits, and verification status.

BIBLIOGRAPHY-ANNOTATION COMPLETENESS

The final book bibliography contains 38 keys and this appendix contains exactly 38 annotation sources. The machine-readable manifest at `shared/bibliography/annotation_manifest.json` records a one-to-one key/path/hash mapping. An annotation explains how a source is used; it does not upgrade an unverified claim or transfer a theorem beyond its assumptions.

A.1 Recommended reading sequence

A reader focused on the BayesBreak statistical formulation should begin with product partitions and exact Bayesian recursion (Barry–Hartigan, Fearnhead, Hutter), then read design-aware partitioning (Scargle, PPMx, generalized PPM), multiple-sequence work (BASIC and JRPM), and finally the block-local approximation sources. The clustering extension should be read beside Corradin et al.; the finite latent-group criterion in BayesBreak is intentionally narrower than a normalized latent-mixture model.

A.2 Bayesian segmentation and exact-recursion lineage

`adamsmackay2007bocpd`: **Adams and MacKay [1]**. Adams and MacKay formulate online Bayesian changepoint detection through a recursive posterior over run length. BayesBreak uses this work to delimit its scope: BayesBreak is an offline contiguous-partition method that returns a posterior over segment count, fixed-count boundary marginals, Bayes curves, and a separate joint MAP partition, rather than an online filtering distribution. The source therefore informs the online/offline distinction and potential future streaming comparisons, but it is not treated as evidence for BayesBreak’s offline exact-DP claims. *Verification status: project bibliography metadata retained; role checked in the Phase 1 landscape review.*

`auger1989segment`: **Auger and Lawrence [2]**. Auger and Lawrence develop dynamic-programming algorithms for identifying segment neighborhoods under broad segment-fit

criteria, including least squares and maximum likelihood. The paper is part of the classical algorithmic lineage for Bellman-style segmentation and shows that an optimal-partition max-sum recurrence is not itself a BayesBreak contribution. BayesBreak differs by using integrated segment likelihoods in Bayesian recursions for normalization and posterior marginals, while retaining a separate max-sum recursion for the joint MAP partition. The paper is used as algorithmic background rather than as a source for Bayesian posterior calculations. *Verification status: title, venue, volume, pages, and DOI independently checked against the publisher record in Phase 6.*

barry1992ppm: **Barry and Hartigan [3]**. Barry and Hartigan introduce product partition models for changepoint problems, including the cohesion factorization that underlies segment-local prior contributions. This is direct prior art for BayesBreak's factorized partition prior. BayesBreak narrows the setting to ordered contiguous partitions and makes the local prior factorization explicit by separating segment cohesion $c_x(i, j)$ from an interior-boundary hazard $h_x(j)$, both of which preserve dynamic-programming factorization. The source prevents any claim that cohesion priors originate here. *Verification status: central primary source checked in Phase 1.*

barry1993bayesCP: **Barry and Hartigan [4]**. Barry and Hartigan develop a Bayesian analysis for changepoint problems using the product-partition framework. BayesBreak cites this paper for the Bayesian changepoint and cohesion lineage, including posterior uncertainty about partitions. The exact BayesBreak delta is not generic Bayesian changepoint inference; it is the generalized exponential-family segment formulation, the collection of posterior quantities computed by the sum-product recursion, the hierarchical extensions, and the separately computed joint MAP partition. The comparison also motivates calibrated novelty language. *Verification status: central primary source checked in Phase 1.*

carlin1992hierBayesCP: **Carlin et al. [8]**. Carlin, Gelfand, and Smith present hierarchical Bayesian changepoint analysis and show how uncertainty and pooling can be represented beyond a single flat partition model. BayesBreak uses this paper to locate its shared and grouped extensions within a broader hierarchical Bayesian lineage. The current exact BayesBreak results apply only where subject-level contributions factor through a common boundary vector or known grouping; richer hierarchical dependence generally requires additional computation. The source therefore informs both motivation and scope limits. *Verification status: project metadata retained; supporting lineage.*

denison1998bayesian: **Denison et al. [11]**. Denison, Mallick, and Smith develop automatic Bayesian curve fitting with a joint model for the number and positions of knots in piecewise-polynomial representations and reversible-jump Markov chain Monte Carlo for posterior computation. BayesBreak uses the paper as a contrasting Bayesian model-space and computational lineage rather than as an exact changepoint dynamic-programming reference. The comparison makes BayesBreak's scope precise: its exact recursions require a finite ordered candidate set and segment-factorized marginal likelihoods, and do not subsume arbitrary trans-dimensional curve models. *Verification status: title, journal, volume, pages, DOI, and methodological description independently checked against the Royal Statistical Society publisher record in Phase 6; the previously supplied JASA changepoint citation was incorrect.*

fearnhead2006exact: **Fearnhead [14]**. Fearnhead gives exact and efficient Bayesian inference for a class of multiple-changepoint models with unknown changepoint count, conditionally independent segments, product-partition structure, and quadratic exact computation. This is the closest foundational work for BayesBreak's sum-product recursion and materially constrains novelty. BayesBreak's contribution is the generalized exponential-family segment formulation, posterior moments and changepoint summaries computed by common recursions, the separate max-sum calculation, irregular-design and multiple-sequence extensions, and the finite latent-group criterion. *Verification status: central primary source checked in Phase 1.*

green1995rjMCMC: **Green [18]**. Green introduces reversible-jump Markov chain Monte Carlo for Bayesian model determination across spaces of differing dimension. Changepoint models can use this machinery to sample segment count and boundaries without the finite factorization required by exact dynamic programming. BayesBreak cites the work as the principal trans-dimensional Bayesian alternative and uses it to state its own restriction: exactness comes from finite ordered support and local factorization, not from universal Bayesian computation. *Verification status: metadata retained; foundational computational role.*

hutter2006bpcr: **Hutter [19]**. Hutter derives exact Bayesian regression for piecewise-constant functions, including unknown segment count and boundaries, evidence, break probabilities, error estimates, and a Bayesian regression curve. This is direct closest work for several BayesBreak headline outputs. The revised book therefore does not claim that evidence, boundary probabilities, or a Bayes curve originate with BayesBreak. The remaining contribution is the generalized use of exposure-aware segment sufficient statistics, shared and grouped extensions, separate posterior and joint-MAP calculations, and one implementation of these statistical quantities. *Verification status: title, 2007 journal metadata, and DOI corrected in the project bibliography during Phase 1/3.*

punskaya2002bayesian: **Punskaya et al. [28]**. Punskaya and coauthors develop Bayesian curve fitting by MCMC with signal-segmentation applications. The paper supplies a sampling-based Bayesian alternative that can accommodate models outside BayesBreak's exact factorized regime. BayesBreak uses it to motivate posterior uncertainty while explaining why its own exact recursions require segment-local integrated scores. It is not used to justify the archived BayesBreak numerical values. *Verification status: canonical IEEE journal metadata retained in the supplied bibliography.*

yao1988biometrika: **Yao [38]**. Yao studies estimation of the number of changepoints in an independent normal sequence using Schwarz's criterion and establishes weak consistency of the proposed estimator. This paper represents information-criterion selection of segment count. BayesBreak uses it as a contrast to assigning a prior to the number of segments, integrating over that number, and reporting its posterior distribution together with posterior changepoint summaries and a separate joint MAP partition. These procedures address related but nonidentical inferential targets. *Verification status: author name, title, journal, volume, pages, DOI, and stated result independently checked against the publisher record in Phase 6.*

A.3 Conjugacy, approximate inference, and numerical computation

bernardo1994bayesian: **Bernardo and Smith [5]**. Bernardo and Smith provide the general Bayesian foundations used to interpret marginal likelihoods, posterior model probabilities, posterior predictive quantities, and decision-relevant uncertainty. In BayesBreak, the book supports the distinction between evidence, posterior summaries, and exported point decisions. It does not supply a changepoint algorithm or validate any archived numerical result. Its role is foundational vocabulary and interpretation, especially where the manuscript separates a posterior over partitions from a joint MAP action. *Verification status: metadata retained from the supplied bibliography; foundational role only.*

diaconis1979conjugate: **Diaconis and Ylvisaker [12]**. Diaconis and Ylvisaker characterize conjugate priors for exponential families. Their normalizer convention supplies the algebra behind BayesBreak's integrated block evidence: cumulative statistic contributions and cumulative log-partition multipliers update the conjugate hyperparameters, and the evidence is a ratio of normalizers times the powered base measure. BayesBreak extends the presentation with an exposure-aware descriptor d_t and a separately declared power/reliability weight a_t so Poisson exposures and binomial trial counts are not conflated with likelihood tempering. *Verification status: bibliography metadata retained; mathematical use is explicit in Chapter 4.*

tierney1986laplace: **Tierney and Kadane [36]**. Tierney and Kadane develop Laplace approximations for posterior moments and marginal posterior densities under regularity conditions that include a suitably regular, unimodal likelihood-times-prior integrand. BayesBreak cites this paper as a foundation for Laplace evaluation of a nonconjugate segment marginal likelihood. The source does not imply a uniform error bound over all candidate segments, nor does it establish a BayesBreak-specific convergence rate. Any use in BayesBreak therefore requires the segment objective, mode, curvature, and approximation error to be assessed over the admissible segment set; the partition-level error statements apply only after such a segment-level bound has been established. *Verification status: publisher metadata and article scope checked for the corrective revision.*

jaakkola2000logisticvb: **Jaakkola and Jordan [20]**. Jaakkola and Jordan develop a variational bound for Bayesian logistic models. BayesBreak uses the construction only as an established segment-local approximation option for nonconjugate logistic marginal likelihoods or moments. The approximation is not a BayesBreak contribution, and the source does not supply a uniform segment-error rate for the present changepoint setting. Application of the BayesBreak posterior perturbation result therefore requires a separate, model-specific uniform bound over all reachable segments under explicit parameter, initialization, convergence, and tail conditions. *Verification status: bibliographic metadata and DOI independently checked in Phase 6; the limitation on transferred error guarantees is retained.*

minka2001ep: **Minka [23]**. Minka develops expectation propagation as an approximate Bayesian inference method based on iterative moment matching. BayesBreak treats EP as one possible block-local evidence routine for a nonconjugate family. The revised method makes no routine-wide convergence or accuracy claim: EP must return diagnostics and, where global perturbation claims are desired, a certified error against a trustworthy reference on

every reachable block. The source contributes the approximation mechanism, not evidence for BayesBreak-specific rates. *Verification status: central source checked in Phase 1.*

polson2013pg: **Polson et al. [27]**. Polson, Scott, and Windle introduce Pólya–Gamma data augmentation for Bayesian logistic models, yielding conditional Gaussian structure. BayesBreak uses this identity as an established route for block-local logistic computation and diagnostics. The source does not imply that a mean-field or approximate BayesBreak block score has a uniform rate over all reachable blocks. Such a rate would require additional assumptions and proof; in this book the global theorem is explicitly conditional on a supplied local error certificate. *Verification status: primary source checked in Phase 1.*

dempster1977em: **Dempster et al. [10]**. Dempster, Laird, and Rubin establish the expectation–maximization algorithm for maximum-likelihood estimation with incomplete data. The BayesBreak latent-template routine uses a responsibility-weighted alternating optimization, but its subject–template quantities are prior-weighted partition scores rather than normalized component densities. The book therefore cites EM as optimization background and describes the routine through its displayed finite objective and Jensen minorization; it does not claim an observed-data finite-mixture likelihood or import identifiability conclusions from mixture theory. *Verification status: journal metadata and DOI independently checked against the publisher record in Phase 6; the source is used only for the established EM optimization lineage.*

blanchard2021lse: **Blanchard et al. [6]**. Blanchard, Higham, and Higham analyze stable computation of log-sum-exp and softmax functions. BayesBreak uses these numerical primitives throughout log-domain forward/backward recursions, pooled evidence aggregation, template responsibilities, and normalization checks. The source informs implementation and numerical QA rather than the statistical model. It is particularly relevant where underflow would otherwise turn very small but nonzero partition mass into an apparent hard exclusion. *Verification status: metadata retained; numerical role linked to the software chapter.*

teicher1963identifiability: **Teicher [34]**. Teicher establishes classical identifiability results for finite mixtures under conditions on normalized component distributions. The reference is retained to mark a boundary of applicability. BayesBreak’s current subject–template scores are positive partition scores rather than normalized subject-level component densities, so finite-mixture identifiability does not follow from Teicher’s result. The book limits the established latent-template theory to monotonicity of the stated objective and records label permutations, duplicate templates, and collapsed solutions as failure modes. *Verification status: journal metadata and DOI independently checked against the Project Euclid record in Phase 6; earlier overextension of the identifiability result remains removed.*

A.4 Optimization and frequentist comparator lineage

frick2014smuce: **Frick et al. [16]**. Frick, Munk, and Sieling develop multiscale changepoint inference with simultaneous error control and confidence statements. SMUCE is a relevant baseline because it emphasizes inferential reliability rather than only point recovery. BayesBreak uses it to design task-matched comparisons for boundary recovery and uncertainty, while recognizing that its posterior probabilities and Bayes curves are not directly equivalent to

multiscale confidence sets. Any benchmark must therefore specify the inferential target and calibration criterion. *Verification status: publisher metadata independently checked and corrected in Phase 6; baseline role retained.*

fryzlewicz2014wbs: **Fryzlewicz [17]**. Fryzlewicz introduces wild binary segmentation for efficient multiple-changepoint detection, especially when many changes may occur. It is a computationally attractive point-estimation baseline in the BayesBreak experiments. The source informs scaling and recovery comparisons, but it does not return the same posterior objects as BayesBreak. Archived F1 values computed against BayesBreak MAP boundaries are consequently labeled agreement diagnostics, not independent accuracy or evidence of superiority. *Verification status: metadata retained; baseline use is explicit in the evidence ledger.*

jackson2005optpart: **Jackson et al. [21]**. Jackson and coauthors give an algorithm for optimal partitioning of interval data, reinforcing the dynamic-programming lineage for globally optimal segmentations. BayesBreak uses this paper to distinguish an optimal partition computed by max-sum recursion from posterior quantities computed by sum-product recursion. Both routes consume the same local scores, but they answer different questions; marginal boundary modes cannot be substituted for the joint MAP partition. *Verification status: metadata retained; algorithmic background.*

killick2012pelt: **Killick et al. [22]**. Killick, Fearnhead, and Eckley introduce PELT, a pruned optimization method with linear expected computational cost under stated conditions. PELT is a central efficiency comparator for BayesBreak's worst-case $O(k_{\max}n^2)$ exact posterior engine. The comparison is intentionally asymmetric in output richness: PELT targets a penalized optimal segmentation, whereas BayesBreak also returns evidence, segment-count uncertainty, boundary marginals, and model-averaged curves. Fair evaluation must match objectives and tuning budgets. *Verification status: central primary source checked in Phase 1.*

olshen2004cbs: **Olshen et al. [25]**. Olshen and coauthors introduce circular binary segmentation for array-based DNA copy-number analysis. CBS is a domain-standard comparator for the array-CGH case study. It operates on point-estimate segmentation rather than BayesBreak's posterior partition distribution. The archived CGH comparator computation is excluded from scientific comparison because a 43×2215 fitted array was flattened to 95,245 values while the BayesBreak boundaries remained on the 2,215-probe axis; that execution cannot support a scientific comparison. *Verification status: metadata retained; domain role and exclusion reason documented from the archive.*

rigaill2010pruned: **Rigaill [30]**. Rigaill develops pruned dynamic programming for recovering best segmentations over a range of segment counts. The work belongs to the optimization/pruning lineage and informs possible efficiency comparisons. BayesBreak's current core deliberately preserves the unpruned exact posterior recursion, so the empirical slopes near 1.07 and 1.14 are reported only as finite-range observations and never as a change to the $O(k_{\max}n^2)$ worst-case complexity. *Verification status: published 2015 version recorded in the project bibliography.*

rudin1992rof: **Rudin et al. [31]**. Rudin, Osher, and Fatemi introduce total-variation regularization for edge-preserving reconstruction. The work is a mathematical ancestor of piecewise-constant penalization, not a Bayesian changepoint posterior method. BayesBreak uses this lineage to contextualize fused and total-variation baselines and to distinguish a penalized point estimator from a posterior distribution over partitions. *Verification status: bibliographic metadata and DOI independently checked against the publisher record in Phase 6; relationship is background only.*

tibshirani2005fused: **Tibshirani et al. [35]**. Tibshirani and coauthors formulate the fused lasso, combining sparsity and adjacent-difference penalties to estimate piecewise-constant structure. It is a natural point-estimation baseline for offline segmentation. BayesBreak differs through probabilistic integration over segment parameters and posterior inference over changepoint configurations; a fused-lasso tuning path is not equivalent to a prior on the number of segments. The source informs baseline selection and misspecification analysis. *Verification status: journal title, volume, pages, and DOI independently checked against the publisher record in Phase 6.*

truong2020review: **Truong et al. [37]**. Truong, Oudre, and Vayatis provide a selective review and taxonomy of offline changepoint methods, including costs, search methods, and evaluation considerations. BayesBreak uses the review to organize baseline families and to prevent category errors between detection, segmentation, online monitoring, and posterior inference. It is a secondary synthesis rather than evidence for a BayesBreak theorem or result. *Verification status: central review source checked in Phase 1.*

A.5 Design-aware and generalized partition priors

scargle2013bayesianblocks: **Scargle et al. [33]**. Scargle and coauthors develop Bayesian Blocks for optimal representations under arbitrary sampling modes, gaps, exposures, and several data modes. This is important prior art for irregularly sampled segmentation. BayesBreak therefore narrows its claim: likelihood exposure, segment cohesion, and boundary hazard are distinct modeling components, and ordered cohesion-plus-hazard factors can be consumed by the same DP. The source is closest work for irregular-design handling, not support for the earlier conflated Poisson-gap derivation. *Verification status: primary source checked in Phase 1.*

muller2011ppmx: **Müller et al. [24]**. Müller, Quintana, and Rosner introduce PPMx, adding covariate similarity to product-partition cohesion. This is direct lineage for data- or design-dependent partition priors. BayesBreak's contribution is a restricted ordered-contiguous factorization that makes segment cohesion and boundary hazard explicit and preserves exact normalization by dynamic programming. The source prevents claims that covariate-dependent or design-aware cohesion is new in general. *Verification status: primary source checked in Phase 1.*

park2010gppm: **Park and Dunson [26]**. Park and Dunson develop generalized product partition models, broadening the ways local partition factors can encode prior structure. BayesBreak uses the work to situate its ordered prior specification: only factors that can be

assigned locally to admissible contiguous blocks and interior endpoints can enter the unchanged finite DP. The book's result is therefore a transparent specialization and implementation rule, not a general invention of product-partition priors. *Verification status: primary source checked in Phase 1.*

A.6 Multiple sequences, dependence, and shared boundaries

bleakley2011groupfused: **Bleakley and Vert [7]**. Bleakley and Vert propose the group fused lasso for detecting changepoints shared across multiple one-dimensional signals by penalizing multidimensional total variation. It is a task-relevant frequentist comparator for common structural changes across sequences. BayesBreak differs by integrating sequence-specific segment parameters and, in its strict common-boundary model, multiplying sequence-level segment marginal likelihoods before Bayesian dynamic programming. The methods return different inferential objects, so comparisons must use comparable tuning budgets and distinguish point-estimate agreement from posterior uncertainty. *Verification status: the cited item is the 2011 arXiv preprint 1106.4199; the previously supplied Annals of Applied Statistics metadata was not supported and was corrected in Phase 6.*

fearnhead2011dependence: **Fearnhead and Liu [15]**. Fearnhead and Liu develop efficient Bayesian multiple-changepoint analysis when parameters can depend across adjacent segments. BayesBreak's protected exact regime instead assumes that, after integration and declared prior factors, a partition score decomposes into local block contributions. This source marks a clear boundary: dependence across segments is scientifically useful but is not covered by the current exact block API without state augmentation or a different recursion. *Verification status: primary source checked in Phase 1.*

ruggieri2014bcpvs: **Ruggieri and Lawrence [32]**. Ruggieri and Lawrence combine Bayesian changepoint inference with variable selection and apply the method to a paleoclimate proxy record. The work demonstrates that scientifically meaningful segmentation often couples regime changes with additional structural choices. BayesBreak cites it as an application and model-extension lineage while keeping variable selection outside the protected core. It informs future-work boundaries rather than a current contribution claim. *Verification status: metadata retained; supporting application role.*

fan2017basic: **Fan and Mackey [13]**. Fan and Mackey's BASIC model addresses simultaneous changepoints across many sequences and includes copy-number and S&P volatility applications. It is the closest multisequence comparator and overlaps the domains of two BayesBreak case studies. BayesBreak's narrower exact result assumes one common boundary vector and conditionally independent subject contributions whose integrated block evidences multiply. The revised book presents this as a restricted exact pooling regime, not the first Bayesian multisequence changepoint method. *Verification status: central primary source checked in Phase 1.*

quinlan2024jrpm: **Quinlan et al. [29]**. Quinlan, Page, and Castro develop joint random partition models that permit dependence among multiple partitions. This is richer than BayesBreak's exact common-boundary specialization. The source is used to state the tradeoff

precisely: BayesBreak gains a simple exact DP by imposing a shared partition (or known groups), while JRPM targets flexible dependence across series. Broad claims about general multivariate partition inference are therefore excluded. *Verification status: central primary source checked in Phase 1.*

A.7 Clustering by structural changes

corradin2026commonstructural: **Corradin et al. [9]**. Corradin, Danese, KhudaBukhsh, and Ongaro cluster time-dependent observations according to common structural-change locations using a latent random-order representation and Bayesian computation. This is the closest verified work for the broad latent-group narrative. BayesBreak retains only a narrower computational object: a finite collection of deterministic changepoint templates optimized under the explicitly stated criterion in Chapter 9, using auxiliary allocation weights and exact responsibility-weighted max-sum updates. The two models should not be conflated, and no claim of first clustering by shared changepoints remains. *Verification status: publisher page checked; online publication 28 October 2025, Statistics and Computing 36, article 7 (2026), DOI 10.1007/s11222-025-10756-x.*

A.8 Bibliography-key completeness table

Table A.1. One-to-one correspondence between bibliography entries and annotations.

Annotation	Bibliography key	Canonical annotation source
ANN-BB-001	adamsmackay2007bocpd	shared/bibliography/annotated_entries/adamsmackay2007bocpd.tex
ANN-BB-002	auger1989segment	shared/bibliography/annotated_entries/auger1989segment.tex
ANN-BB-003	barry1992ppm	shared/bibliography/annotated_entries/barry1992ppm.tex
ANN-BB-004	barry1993bayesCP	shared/bibliography/annotated_entries/barry1993bayesCP.tex
ANN-BB-005	bernardo1994bayesian	shared/bibliography/annotated_entries/bernardo1994bayesian.tex
ANN-BB-006	bleakley2011groupfused	shared/bibliography/annotated_entries/bleakley2011groupfused.tex
ANN-BB-007	carlin1992hierBayesCP	shared/bibliography/annotated_entries/carlin1992hierBayesCP.tex
ANN-BB-008	denison1998bayesian	shared/bibliography/annotated_entries/denison1998bayesian.tex
ANN-BB-009	diaconis1979conjugate	shared/bibliography/annotated_entries/diaconis1979conjugate.tex

Annotation	Bibliography key	Canonical annotation source
ANN-BB-010	fearnhead2006exact	shared/bibliography/annotated_entries/fearnhead2006exact.tex
ANN-BB-011	frick2014smuce	shared/bibliography/annotated_entries/frick2014smuce.tex
ANN-BB-012	fryzlewicz2014wbs	shared/bibliography/annotated_entries/fryzlewicz2014wbs.tex
ANN-BB-013	green1995rjMCMC	shared/bibliography/annotated_entries/green1995rjMCMC.tex
ANN-BB-014	hutter2006bpcr	shared/bibliography/annotated_entries/hutter2006bpcr.tex
ANN-BB-015	jaakkola2000logisticvb	shared/bibliography/annotated_entries/jaakkola2000logisticvb.tex
ANN-BB-016	jackson2005optpart	shared/bibliography/annotated_entries/jackson2005optpart.tex
ANN-BB-017	killick2012pelt	shared/bibliography/annotated_entries/killick2012pelt.tex
ANN-BB-018	minka2001ep	shared/bibliography/annotated_entries/minka2001ep.tex
ANN-BB-019	olshen2004cbs	shared/bibliography/annotated_entries/olshen2004cbs.tex
ANN-BB-020	polson2013pg	shared/bibliography/annotated_entries/polson2013pg.tex
ANN-BB-021	punskaya2002bayesian	shared/bibliography/annotated_entries/punskaya2002bayesian.tex
ANN-BB-022	rigaill2010pruned	shared/bibliography/annotated_entries/rigaill2010pruned.tex
ANN-BB-023	rudin1992rof	shared/bibliography/annotated_entries/rudin1992rof.tex
ANN-BB-024	tibshirani2005fused	shared/bibliography/annotated_entries/tibshirani2005fused.tex
ANN-BB-025	yao1988biometrika	shared/bibliography/annotated_entries/yao1988biometrika.tex
ANN-BB-026	truong2020review	shared/bibliography/annotated_entries/truong2020review.tex
ANN-BB-027	scargle2013bayesianblocks	shared/bibliography/annotated_entries/scargle2013bayesianblocks.tex
ANN-BB-028	fearnhead2011dependence	shared/bibliography/annotated_entries/fearnhead2011dependence.tex
ANN-BB-029	ruggieri2014bcpvs	shared/bibliography/annotated_entries/ruggieri2014bcpvs.tex
ANN-BB-030	fan2017basic	shared/bibliography/annotated_entries/fan2017basic.tex

Annotation	Bibliography key	Canonical annotation source
ANN-BB-031	quinlan2024jrpm	shared/bibliography/annotated_entries/quinlan2024jrpm.tex
ANN-BB-032	muller2011ppmx	shared/bibliography/annotated_entries/muller2011ppmx.tex
ANN-BB-033	park2010gppm	shared/bibliography/annotated_entries/park2010gppm.tex
ANN-BB-034	dempster1977em	shared/bibliography/annotated_entries/dempster1977em.tex
ANN-BB-035	blanchard2021lse	shared/bibliography/annotated_entries/blanchard2021lse.tex
ANN-BB-036	teicher1963 identifiability	shared/bibliography/ annotated_entries/ teicher1963identifiability.tex
ANN-BB-037	corradin2026 commonstructural	shared/bibliography/ annotated_entries/ corradin2026commonstructural. tex
ANN-BB-038	tierney1986laplace	shared/bibliography/annotated_entries/tierney1986laplace.tex

Detailed implementation plan

CHAPTER PATH

A module-by-module plan for implementing the stated probability models, numerical methods, experiments, data schemas, tests, and reproducibility records.

B.1 Scope and non-goals

The implementation work preserves the author-approved method: generalized hierarchical Bayesian segmentation with exponential-family segment models, irregular-design priors, multiple-sequence hierarchies, and known or latent groups. The core software separates observation-model calculations from sum-product and max-sum recursions over ordered partitions. The plan does not introduce an online detector, reinterpret the finite latent-group criterion as a normalized mixture likelihood, change the scientific endpoint, or overwrite archived numerical results. A corrected computation receives a new result identifier and records its relationship to the earlier output.

B.2 Repository organization

`src/bayesbreak/` Observation families, segment priors, sum-product recursion, max-sum MAP calculation, multiple-sequence models, grouped models, and prediction.

`configs/` Versioned model, comparator, dataset, and execution specifications.

`experiments/` Drivers for EPR-BB-001 through EPR-BB-015.

`scripts/` Data-schema, result-metadata, bibliography, build, and release validators.

`tests/` Unit, property-based, numerical, statistical, regression, and reproducibility tests.

`data/` Immutable source records and versioned processed datasets.

`artifacts/` Result files, figures, tables, numerical error records, and records of excluded computations.

`docs/schemas/` Schemas for results, numerical-error summaries, metrics, and reproducibility metadata.

Responsibility is organized by component: observation models, partition inference, posterior prediction and metrics, experiments and result metadata, and release verification. A task is complete only when the corresponding mathematical invariant and regression test pass together.

B.3 Public functions and returned objects

Observation-family calculation Returns a log segment marginal likelihood, requested posterior moments or posterior state, support information, and, for a numerical approximation, the available convergence and error information for each admissible segment.

Partition prior Returns log segment cohesion and log interior-boundary hazard separately, including zero-probability cases.

Partition inference Uses the same prior-adjusted segment scores in sum-product and maximum recursions and returns marginal likelihoods, $P(k | y)$, changepoint marginals, Bayes curves, and a joint MAP partition.

Posterior prediction Dispatches according to the fitted observation family and applies the stated coordinate-support or extrapolation rule.

Changepoint metrics Validates coordinate axes and uses maximum-cardinality, minimum-distance one-to-one matching; matched-location MAE is NA when no pair is matched.

Result record Distinguishes execution status from validity for a specified analysis and does not mutate an archived result in place.

B.4 Task registry and acceptance criteria

Table B.1. Implementation task registry. Detailed files, interfaces, and invariants are in Appendix C.

ID	Work item	Acceptance criterion
CODE-BB-001	Implement explicit segment-cohesion and boundary-hazard prior specification	Analytic small-n enumeration and sensitivity fixtures pass.
CODE-BB-002	Integrate repaired prior with exact DP and property tests	EPR-BB-001 randomized invariant suite passes.
CODE-BB-003	Repair the finite latent-group criterion and restart selection	All exit paths, ties, collapsed groups, and restart ordering tests pass.
CODE-BB-004	Harden grouped and replicate log-domain aggregation	Adversarial magnitude tests pass without overflow/underflow-induced ranking changes.
CODE-BB-005	Add a schema for nonconjugate segment-error assessment	Nested tolerances tighten the verified bound or return an explicit failure state; synthetic propagation checks hold.
CODE-BB-006	Implement Beta-observation posterior predictive scoring	Normalization/reference tests and edge-case finite checks pass.

ID	Work item	Acceptance criterion
CODE-BB-007	Make extrapolation policy explicit	Support-boundary and provenance tests pass for every policy.
CODE-BB-008	Canonicalize boundary matching and metrics	Hand-computed, tie, empty, and symmetry fixtures pass.
CODE-BB-009	Validate comparator axes and tuning budgets	CGH cached single-trace artifact is rejected deterministically; valid raw matrix route is covered.
CODE-BB-010	Version result and provenance schemas	Every new sidecar validates and every archived sidecar has a documented migration/read-only path.
CODE-BB-011	Synchronize real-data loaders, cards, captions, and hashes	Well-log, CGH, SPX, and methylKit cards validate against generated sidecars and captions.
CODE-BB-012	Profile and re-run the full test suite	All required tests terminate under pinned profiles or remain explicitly blocked with owner and reason.
CODE-BB-013	Reconcile version, CI, changelog, and preprint lineage	Clean checkout reports one version and citation target; lineage table is complete.
CODE-BB-014	Render canonical handoff and appendix fragments	Book appendix and standalone handoff pass semantic and ID synchronization checks.
CODE-BB-015	Regenerate figures and tables with semantic validation	Every figure/table has source hashes, status metadata, and caption anchor; visual QA passes.
CODE-BB-016	Automate bibliography annotation completeness	Completeness checker returns zero missing, duplicate, orphaned, or relationship-free entries.

B.5 Data, result metadata, and leakage controls

Each dataset receives a source record, immutable hash, license or access note, raw schema, and a versioned sequence of transformations. Splits are fixed before fitting. Simulation truth is stored separately from fitted outputs. Comparator tuning uses only the declared training or validation information. Agreement with a `BAYESBREAK` fit is not relabelled as external truth. The well-log, array-CGH, equity, and methylation datasets retain their executed preprocessing and hashes; a corrected or extended analysis is stored as a new version.

B.6 Execution, caching, and failure states

A run records code, data, model specification, environment, random seed, wall time, memory, convergence state, and output hashes. Cached segment calculations are reused only when the observation family, descriptors, priors, support, and code hash agree. A nonconjugate method may finish with a numerical value but without a verified error bound; a comparator calculation may finish but be invalid for a specified comparison; a test may remain unresolved because it does not terminate under the execution limit. These states are reported separately.

B.7 Tests

Unit tests cover sufficient statistics, conjugate normalizers, support, and segment factorization. Property-based tests compare sum-product and max-sum recursions with exhaustive enumeration on small problems. Numerical tests check log-domain normalization and segmentwise error calculations. Statistical tests cover calibration, null simulations, and repeated-seed behavior. Regression tests preserve archived result identifiers and ensure that excluded values cannot enter incompatible figures or tables. Reproducibility tests rebuild figures and tables and compare declared hashes or numerical tolerances.

B.8 Implementation order and stopping criteria

The dependency order is: segment-cohesion and boundary-hazard semantics; dynamic-programming invariants; latent-group and shared-changepoint numerical corrections; posterior prediction and changepoint metrics; comparator and data schemas; full-suite profiling; controlled recomputation; figure and table regeneration; and source synchronization. Each stage has a pre-specified acceptance criterion. When a theorem assumption, numerical error bound, comparator definition, or test cannot be satisfied, the corresponding claim is narrowed or the calculation remains excluded from that scientific interpretation. An unimplemented item remains explicitly incomplete.

Complete coding-agent handoff

CHAPTER PATH

Implementation specification, module interfaces, test obligations, experiment protocols, and explicit incomplete states rendered from the same source as the standalone handoff.

CANONICAL SYNCHRONIZATION

Canonical handoff ID: HANDOFF-BB-CANONICAL-001. Phase: 6. Canonical JSON SHA-256: 4a778128eb66cd2bce96497e2176ac8e9615d3efbf2e6faf11f97dcdb17c4f97. The book appendix, task registry, standalone Markdown handoff, repository registries, and synchronization manifest are checked mechanically.

C.1 Author-fixed scientific scope

BayesBreak is the method developed in Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs. For supported observation models, family-specific segment marginal likelihoods and posterior moments enter common sum-product recursions for posterior quantities and a separate max-sum recursion with backtracking for the joint MAP partition. The implementation must preserve irregular-design priors, common and group-specific changepoints across multiple sequences, known groups, and the finite latent-group criterion stated in the manuscript.

C.1.1 Do not change

- Do not change the manuscript title or replace generalized hierarchical Bayesian segmentation as the main scientific narrative without explicit author approval.
- Do not overwrite populated archived numerical results; a corrected computation receives a new result ID and an explicit parent-result link.
- Do not interpret the finite latent-group criterion as a normalized finite-mixture likelihood.
- Do not use marginal changepoint modes as the joint MAP partition.
- Do not treat agreement with BayesBreak MAP changepoints as accuracy against independent truth.
- Do not use RES-BB-CMP-002 or RES-BB-RD-007Q in comparator or posterior-predictive conclusions.

- Do not describe the method with vague branding; use the defined probability model, prior, recursion, optimization objective, metric, or numerical criterion.

C.1.2 Non-goals

- No normalized latent-mixture model is implemented in this work program.
- No online changepoint detector, slope-change model, or graph segmentation method is introduced.
- No archived result is silently corrected in place.
- No fixed package version, compute budget, or completion date is invented before version and resource profiling are resolved.
- No slide deck is generated by this manuscript workflow.

C.2 Repository bootstrap and explicit status

REPOSITORY STATUS

Root. coding/repository_skeleton

Status. interface blueprint; scientific implementation intentionally incomplete

Environment. Use a pinned environment selected during CODE-BB-012; the exact package version source is unresolved under CODE-BB-013.

Use restriction. Do not publish or install the skeleton as a completed BayesBreak release. It exists to expose approved modules, schemas, task IDs, and explicit incomplete states.

Required top-level paths.

- src/bayesbreak
- tests
- configs
- schemas
- experiments
- scripts
- data
- docs/data_cards
- artifacts
- provenance

C.2.1 Module ownership

Table C.1. Approved repository modules and task ownership.

Path	Responsibility	Tasks
src/bayesbreak/ families/	Observation-family segment marginal likelihoods, posterior moments, and family-specific posterior prediction.	CODE-BB-005;CODE-BB-006
src/bayesbreak/ priors.py;src/ bayesbreak/design_ prior.py	Segment cohesion, interior-boundary hazard, and partition-prior factors.	CODE-BB-001;CODE-BB-002
src/bayesbreak/ dp.py;src/ bayesbreak/map.py; src/bayesbreak/ posterior.py	Sum-product posterior calculations, max-sum joint MAP recovery, and posterior summaries.	CODE-BB-002
src/bayesbreak/ replicates.py;src/ bayesbreak/groups.py	Common-partition pooling and known-group fits with stable log-domain aggregation.	CODE-BB-004
src/bayesbreak/ mixture.py	Finite latent-group minorization–maximization and restart selection.	CODE-BB-003
src/bayesbreak/ prediction.py	Coordinate assignment, explicit extrapolation, and family-specific posterior prediction.	CODE-BB-006;CODE-BB-007
src/bayesbreak/ metrics.py;src/ bayesbreak/ comparators.py	One-to-one boundary matching, metrics, comparator-axis validation, and tuning-budget records.	CODE-BB-008;CODE-BB-009
src/bayesbreak/ provenance.py; schemas/	Versioned result records, parent links, hashes, interpretation status, and schema validation.	CODE-BB-010
src/bayesbreak/ datasets/;docs/data_ cards/	Data loading, provenance, preprocessing descriptions, and application-specific cards.	CODE-BB-011
tests/;scripts/ ;experiments/	Unit, numerical, statistical, regression, reproducibility, and claim-mapped experiment execution.	CODE-BB-012;CODE-BB-015;CODE-BB-016

C.3 Configuration keys, commands, and schemas

C.3.1 Configuration keys

Observation Model.

- `model.family`
- `model.family_descriptors`
- `model.prior`
- `model.likelihood_power_weights`

Partition.

- `partition.k_max`
- `partition.min_segment_length`
- `partition.segment_cohesion`
- `partition.boundary_hazard`
- `partition.allowed_boundaries`

Inference.

- `inference.posterior_marginals`
- `inference.joint_map`
- `inference.tie_breaking`
- `inference.numeric_dtype`

Hierarchy.

- `hierarchy.common_partition`
- `hierarchy.known_group_labels`
- `hierarchy.n_latent_groups`
- `hierarchy.n_restarts`
- `hierarchy.optimization_tolerance`

Approximation.

- `approximation.method`
- `approximation.reference_method`
- `approximation.tolerance`
- `approximation.tail_control`

Prediction.

- `prediction.extrapolation`
- `prediction.coordinate_support`
- `prediction.return_log_score`

Metrics.

- `metrics.boundary_tolerance`
- `metrics.matching_rule`
- `metrics.reference_type`
- `metrics.metric_version`

Execution.

- `execution.seed`
- `execution.environment_lock_hash`
- `execution.data_hash`
- `execution.config_hash`

- `execution.code_hash`
- `execution.result_id`
- `execution.parent_result_id`

C.3.2 Commands

Skeleton Validation.

- `pythoncoding/repository_skeleton/scripts/check_skeleton.py`
- `pythonscripts/check_sync.py`

Implementation Tests After Code Exists.

- `python-mpytest-q`
- `python-mpytest-qtests/test_dp_properties.pytests/test_dp_priors.py`
- `python-mpytest-qtests/test_prediction_support.pytests/test_boundary_matching.pytests/test_comparator_schemas.py`

Document Builds.

- `makebook`
- `makepaper-all`
- `makeexecutive`
- `makevalidate-phase6`

Result Validation After Runs.

- `pythonscripts/validate_result_schema.py`
- `pythonscripts/check_asset_semantics.py`
- `pythonscripts/check_annotation_completeness.py`

C.3.3 Required schema fields

Result Record.

- `result_id`
- `parent_result_id`
- `execution_status`
- `scientific_interpretation`
- `data_hash`
- `config_hash`
- `code_hash`
- `metrics`
- `artifacts`

Boundary Metric.

- `prediction_axis`

- reference_axis
- tolerance
- matching_rule
- matched_pairs
- precision
- recall
- f1
- mae_or_na

Approximation Error Record.

- family
- block_support_hash
- reference_method
- max_log_score_error
- optimization_residual
- tail_bound
- quadrature_error
- convergence_status

C.4 Exact files, interfaces, invariants, and acceptance tests

The task cards below are generated directly from the canonical handoff. No task is represented as complete merely because its interface appears in the repository skeleton.

CODE-BB-001

Task. Implement explicit segment-cohesion and boundary-hazard prior specification
Files.

- src/bayesbreak/priors.py
- src/bayesbreak/design_prior.py
- tests/test_priors.py

Interfaces.

- log_cohesion(i, j, x, config)
- log_boundary_hazard(j, x, config)
- partition_log_prior(t, x, config)

Theory-to-code invariants.

- hazard applies only to interior boundaries.
- regular-grid special case matches declared formula.
- local factors sum to normalized prior after DP normalizer.

Dependencies. REV-BB-004

Acceptance. Analytic small-n enumeration and sensitivity fixtures pass.

CODE-BB-002

Task. Integrate repaired prior with exact DP and property tests

Files.

- `src/bayesbreak/dp.py`
- `tests/test_dp_priors.py`
- `tests/test_dp_properties.py`

Interfaces.

- `local_partition_score(i, j, is_terminal, prior_config)`

Theory-to-code invariants.

- sum-product and max-sum consume identical local prior factors.
- zero support is preserved.
- normalizers match exhaustive enumeration.

Dependencies. CODE-BB-001;REV-BB-005

Acceptance. EPR-BB-001 randomized invariant suite passes.

CODE-BB-003

Task. Repair the finite latent-group criterion and restart selection

Files.

- `src/bayesbreak/mixture.py`
- `tests/test_mixture.py`
- `tests/test_mixture_restarts.py`

Interfaces.

- `template_log_score`
- `fit_score_templates`
- `objective_trace_`

Theory-to-code invariants.

- returned objective equals `trace[-1]`.
- objective is monotone up to tolerance.
- selection uses final objective.
- API does not call the criterion a likelihood.

Dependencies. REV-BB-007

Acceptance. All exit paths, ties, collapsed groups, and restart ordering tests pass.

CODE-BB-004

Task. Harden grouped and replicate log-domain aggregation

Files.

- `src/bayesbreak/replicates.py`
- `src/bayesbreak/groups.py`
- `tests/test_replicates_stability.py`

Interfaces.

- `aggregate_block_log_evidence`
- `shared_fit`

Theory-to-code invariants.

- no direct exponentiation of unbounded log evidence.
- finite checks and diagnostic exceptions.
- small cases match high precision.

Dependencies. REV-BB-006;REV-BB-012

Acceptance. Adversarial magnitude tests pass without overflow/underflow-induced ranking changes.

CODE-BB-005

Task. Add a schema for nonconjugate segment-error assessment

Files.

- `src/bayesbreak/nonconjugate/error_bounds.py`
- `src/bayesbreak/nonconjugate/integration.py`
- `tests/test_nonconjugate_error_bounds.py`

Interfaces.

- `SegmentErrorRecord`
- `evaluate_reachable_segment_error`
- `propagate_partition_bounds`

Theory-to-code invariants.

- reference and approximate values share block coordinates.
- tail and quadrature error are separate.
- global bound uses maximum reachable error.

Dependencies. REV-BB-008

Acceptance. Nested tolerances tighten the verified bound or return an explicit failure state; synthetic propagation checks hold.

CODE-BB-006

Task. Implement Beta-observation posterior predictive scoring

Files.

- `src/bayesbreak/families/beta_obs.py`
- `tests/test_beta_obs_prediction.py`

Interfaces.

- `posterior_predictive_logpdf_block`

Theory-to-code invariants.

- no Gaussian fallback.
- parameters respect Beta support.
- analytic/numerical reference agreement.

Dependencies. REV-BB-009

Acceptance. Normalization/reference tests and edge-case finite checks pass.

CODE-BB-007

Task. Make extrapolation policy explicit

Files.

- `src/bayesbreak/prediction.py`
- `tests/test_prediction_support.py`

Interfaces.

- `predict(..., extrapolation="error")`
- `assign_to_partition`

Theory-to-code invariants.

- default rejects left/right out-of-range coordinates.
- named policies are recorded in provenance.
- in-range assignments are unchanged.

Dependencies. REV-BB-009

Acceptance. Support-boundary and provenance tests pass for every policy.

CODE-BB-008

Task. Canonicalize boundary matching and metrics

Files.

- `src/bayesbreak/metrics.py`
- `tests/test_boundary_matching.py`
- `scripts/baselines_metrics.py`

Interfaces.

- `match_boundaries_one_to_one`
- `boundary_metrics`

Theory-to-code invariants.

- maximum cardinality before minimum distance.
- no reference/prediction reuse.
- matched MAE is NA with zero matches.
- metric version recorded.

Dependencies. REV-BB-009;REV-BB-010

Acceptance. Hand-computed, tie, empty, and symmetry fixtures pass.

CODE-BB-009

Task. Validate comparator axes and tuning budgets

Files.

- `src/bayesbreak/comparators.py`
- `scripts/run_comparators.py`
- `tests/test_comparator_schemas.py`

Interfaces.

- `ComparatorInputSchema`
- `TuningBudget`
- `validate_axis`

Theory-to-code invariants.

- multisequence methods receive raw matrix plus coordinate axis.
- flattening cannot pass schema validation.
- task type and tuning stratum are explicit.

Dependencies. REV-BB-010

Acceptance. CGH cached single-trace artifact is rejected deterministically; valid raw matrix route is covered.

CODE-BB-010

Task. Version result and provenance schemas

Files.

- src/bayesbreak/provenance.py
- schemas/result_sidecar.schema.json
- tests/test_result_schema.py

Interfaces.

- ResultRecord
- ResultStatus
- write_sidecar

Theory-to-code invariants.

- executed/valid-for-stated-interpretation/excluded/planned are distinct.
- raw data, weights, split, configuration, environment, and output hashes are supported.
- absolute author paths are rejected in release mode.

Dependencies. REV-BB-001;REV-BB-011

Acceptance. Every new sidecar validates and every archived sidecar has a documented migration/read-only path.

CODE-BB-011

Task. Synchronize real-data loaders, cards, captions, and hashes

Files.

- src/bayesbreak/datasets/
- scripts/figures/
- docs/data_cards/
- tests/test_dataset_cards.py

Interfaces.

- DatasetCard
- load_with_provenance

Theory-to-code invariants.

- source/date/n/stride/family/weights agree.
- external annotations have source and matching rule.
- MAP markers are never labeled external events.

Dependencies. REV-BB-011;CODE-BB-010

Acceptance. Well-log, CGH, SPX, and methylKit cards validate against generated

sidecars and captions.

CODE-BB-012

Task. Profile and re-run the full test suite

Files.

- pyproject.toml
- environment-lock.*
- .github/workflows/
- tests/

Interfaces.

- testmanifest
- timeoutmarker
- performanceprofile

Theory-to-code invariants.

- pass/skip/fail/unresolved reported separately.
- no bounded-run timeout is counted as pass.
- environment hash recorded.

Dependencies. REV-BB-012

Acceptance. All required tests terminate under pinned profiles or remain explicitly blocked with owner and reason.

CODE-BB-013

Task. Reconcile version, CI, changelog, and preprint lineage

Files.

- pyproject.toml
- src/bayesbreak/_version.py
- CHANGELOG.md
- README.md
- CITATION.cff
- docs/version_lineage.md
- .github/workflows/

Interfaces.

- singleversionsource
- releaseprovenancerecord

Theory-to-code invariants.

- badge points to existing workflow.
- generated version matches clean tag.
- preprint is lineage, not independent validation.

Dependencies. REV-BB-012

Acceptance. Clean checkout reports one version and citation target; lineage table is complete.

CODE-BB-014

Task. Render canonical handoff and appendix fragments

Files.

- shared/handoffs/
- book/appendices/coding_agent_handoff.tex
- coding/CODING_AGENT_HANDOFF.md
- scripts/check_sync.py

Interfaces.

- canonicalhandoffschema
- Markdownrenderer
- LaTeXrenderer

Theory-to-code invariants.

- stable IDs preserved.
- no manual divergent copies.
- hash of normalized content recorded.

Dependencies. REV-BB-002;REV-BB-012

Acceptance. Book appendix and standalone handoff pass semantic and ID synchronization checks.

CODE-BB-015

Task. Regenerate figures and tables with semantic validation

Files.

- scripts/figures/
- scripts/tables/
- shared/tables/
- tests/test_asset_semantics.py

Interfaces.

- assetsidecar
- markerstyleregistry
- result-interpretationfilter

Theory-to-code invariants.

- excluded archived values cannot enter incompatible scientific panels.
- planned cells remain visibly empty or planned.
- external, MAP, and contextual markers have distinct styles.

Dependencies. REV-BB-001;REV-BB-010;REV-BB-011

Acceptance. Every figure/table has source hashes, status metadata, and caption anchor; visual QA passes.

CODE-BB-016

Task. Automate bibliography annotation completeness

Files.

- shared/bibliography/references.bib

- `shared/bibliography/annotated_entries/`
- `scripts/check_annotations.py`

Interfaces.

- `annotationrecordschema`
- `bibliographykeyscanner`

Theory-to-code invariants.

- exactly one annotation per included key.
- each annotation has explicit project relationship.
- metadata verification status present.

Dependencies. APP-BB-A;REV-BB-012

Acceptance. Completeness checker returns zero missing, duplicate, orphaned, or relationship-free entries.

C.5 Stage sequence

Table C.2. Implementation and validation sequence.

Stage	Purpose	Tasks	Advance criterion
S1	Prior factors and exact-DP integration	CODE-BB-001;CODE-BB-002	Exhaustive finite-case checks and prior sensitivity fixtures pass.
S2	Latent-group objective and stable pooling	CODE-BB-003;CODE-BB-004	Final-objective, monotonicity, restart, and high-precision aggregation tests pass.
S3	Prediction, metrics, comparator schemas, and result records	CODE-BB-006;CODE-BB-007;CODE-BB-008;CODE-BB-009;CODE-BB-010;CODE-BB-011	Family, support, matching, axis, data-card, and result-schema tests pass.
S4	Pinned test completion and release-lineage repair	CODE-BB-012;CODE-BB-013;CODE-BB-014;CODE-BB-016	Every required test has a terminal status and canonical sources are synchronized.
S5	Corrected CGH and methylation computations	CODE-BB-009;CODE-BB-010;CODE-BB-011;CODE-BB-015	New result IDs, parent links, hashes, correct axes or observation family, and independent review are complete.
S6	Expanded calibration, robustness, fairness, and scaling	experiment protocols EPR-BB-001 through EPR-BB-015 as applicable	Pre-specified repetitions, uncertainty, negative controls, and falsifying outcomes are reported.
S7	Independent scientific and software release review	Phase 6 verification	All release checks are evaluated and every incomplete item is disclosed.

C.6 Experiment matrix and output ownership

Each experiment driver owns a machine-readable run registry, configuration set, raw metric records, and figure or table target. Populated archived values remain read-only. Corrected runs receive a new result ID and a parent pointer.

C.6.1 EPR-BB-001 – Randomized exact-DP and semiring invariants

Claims. CLM-BB-001;CLM-BB-003;CLM-BB-004

Status. executed plus expansion planned

Hypothesis. Sum-product and max-sum implementations match exhaustive enumeration on all sampled admissible finite cases.

Competing explanation. Agreement is limited to hand-picked priors or benign support.

Design. Property-based generation over n , $k_{\max}$, local scores, zero-support blocks, priors, and ties; compare evidence, $P(k)$, marginals, and joint MAP.

Metrics. absolute/relative error, normalization error, exact partition equality, failure count with seed.

Repetition and uncertainty. At least 1,000 deterministic seeded cases per configuration family; report zero-failure upper bound.

Dependencies. CODE-BB-002;CODE-BB-012

Output. tests and a machine-readable invariant report.

C.6.2 EPR-BB-002 – Conjugate-family recovery and null/misspecification grid

Claims. CLM-BB-002;CLM-BB-013

Status. executed real values plus expanded protocol planned

Hypothesis. The common interface preserves calibrated segmentation behavior across supported families when their data model is correct.

Competing explanation. Apparent portability is driven by a narrow SNR or tuned prior.

Design. Factorial grid over family, n , segment count, spacing, SNR, weight/exposure patterns, null data, and selected misspecification.

Metrics. boundary precision/recall/F1 under external truth, k error, posterior coverage, log evidence, runtime.

Repetition and uncertainty. Predeclared repetitions with bootstrap or t-intervals by cell and family-level multiplicity control for headline contrasts.

Dependencies. CODE-BB-001;CODE-BB-002;CODE-BB-010

Output. family performance table, calibration plots, and null-control table.

C.6.3 EPR-BB-003 – Boundary posterior calibration

Claims. CLM-BB-003;CLM-BB-013

Status. executed ECE/Brier values retained; expanded uncertainty and calibration checks planned

Hypothesis. Bayes curve probabilities are calibrated for the declared model after marginalizing over k .

Competing explanation. Low aggregate ECE is caused by severe class imbalance or bin choices.

Design. Repeated simulated datasets with fixed- k and k -marginalized posterior variants; preregistered equal-width and equal-mass bins.

Metrics. Brier, log score, ECE, reliability slope/intercept, prevalence, calibration curves.

Repetition and uncertainty. Dataset-level intervals and paired comparison of fixed- k versus marginalized variants.

Dependencies. CODE-BB-002;CODE-BB-010

Output. calibration figure with intervals and protocol sidecar.

C.6.4 EPR-BB-004 – Segment-count and SNR behavior

Claims. CLM-BB-003;CLM-BB-013

Status. executed plus interpretation expansion planned

Hypothesis. Posterior uncertainty over k changes predictably with signal strength and prior concentration.

Competing explanation. Observed k behavior is a prior artifact or oversegmentation bias.

Design. Cross SNR, n , true k , minimum segment length, and prior settings; include no-change and dense-change controls.

Metrics. $P(k_{\text{true}})$, posterior entropy, k bias, boundary loss, false discoveries.

Repetition and uncertainty. Repeated-seed intervals and decomposition by prior regime.

Dependencies. CODE-BB-001;CODE-BB-002

Output. k -posterior panels and falsification table.

C.6.5 EPR-BB-005 – Finite latent-group criterion stress test

Claims. CLM-BB-008;CLM-BB-009;CLM-BB-013

Status. archived 96 percent value real; rerun required after semantic/code repair

Hypothesis. Coordinate updates monotonically improve the stated score and recover shared templates in identifiable finite simulations.

Competing explanation. Recovery is restart luck, label coding, or collapse under overspecified G .

Design. Vary S , $G_{\text{true}}/G_{\text{fit}}$, n , noise, template separation, imbalance, initialization, and collapsed/duplicate templates.

Metrics. objective monotonicity, final objective, label-invariant ARI, template distance, collapse rate, restart selection stability.

Repetition and uncertainty. At least 50 datasets per cell and multiple restarts; paired pre/post repair diagnostic.

Dependencies. CODE-BB-003;CODE-BB-012

Output. stress-test table, objective traces, and regression report.

C.6.6 EPR-BB-006 – Shared-boundary weak-signal pooling

Claims. CLM-BB-006;CLM-BB-007

Status. executed supplementary outputs plus fair comparison planned

Hypothesis. Exact shared-boundary pooling improves boundary recovery when multiple subjects share changes and individual signals are weak.

Competing explanation. Gains are caused by forced-sharing bias or tuned k .

Design. Vary subject count, information heterogeneity, shared fraction, subject-specific deviations, noise, and prior; compare shared, independent, and suitable multisequence methods.

Metrics. external-truth boundary loss, k error, calibration, predictive score under a common joint target, runtime.

Repetition and uncertainty. Repeated datasets with paired contrasts and heterogeneity-stratified intervals.

Dependencies. CODE-BB-004;CODE-BB-009;CODE-BB-010

Output. pooling benefit/failure surface and model-assumption table.

C.6.7 EPR-BB-007 – Nonconjugate local certificate and global sensitivity

Claims. CLM-BB-010;CLM-BB-011

Status. real benchmark values retained; certification expansion planned

Hypothesis. Measured reachable-block log-score tolerances predict the observed perturbation of evidence and posterior summaries.

Competing explanation. Average block error hides rare reachable blocks that dominate the posterior.

Design. Construct high-accuracy references, census or sample all reachable blocks, separate truncation from numerical error, and rerun DP at nested tolerances.

Metrics. max/quantile block error, evidence ratio, posterior-odds distortion, TV distance, boundary/ k changes, time/memory.

Repetition and uncertainty. Multiple datasets and tolerance ladders; deterministic reference checks where possible.

Dependencies. CODE-BB-005;CODE-BB-012

Output. certificate files and segment-to-partition error-bound table.

C.6.8 EPR-BB-008 – Runtime and memory decomposition

Claims. CLM-BB-003;CLM-BB-013

Status. real local slopes retained; extended scaling planned

Hypothesis. Measured cost decomposes into block construction and $O(k_{\max} n^2)$ DP transitions with predictable memory scaling.

Competing explanation. Near-linear local slopes reflect cache/warmup or a narrow n range, not algorithmic structure.

Design. Extend n and $k_{\max}$ until resource limits; separate cold/warm cache, block construction, DP, backtracking, and I/O; use fixed hardware and repeated runs.

Metrics. wall time, CPU time, peak RSS, transition count, slope by range, residuals against $n^2 k$ model.

Repetition and uncertainty. Median and dispersion across repetitions, machines reported separately.

Dependencies. CODE-BB-012

Output. scaling curves, resource table, and profiler artifacts.

C.6.9 EPR-BB-009 – Well-log prior sensitivity case study

Claims. CLM-BB-005;CLM-BB-014

Status. two executed real fits retained

Hypothesis. Cohesion and boundary-hazard choices change segmentation in interpretable ways without being mislabeled as external accuracy.

Competing explanation. Differences are arbitrary prior sensitivity with no physical interpretation.

Design. Reproduce archived index-uniform and length-aware fits, then vary repaired c/h choices on the same 507-point stride-8 series.

Metrics. $k_{\hat{}}$, log evidence within a common model family, boundary stability, posterior uncertainty, runtime.

Repetition and uncertainty. Sensitivity grid rather than sampling uncertainty; external annotations only if independently verified.

Dependencies. CODE-BB-001;CODE-BB-011;CODE-BB-015

Output. case-study figure, prior sensitivity table, provenance card.

C.6.10 EPR-BB-010 – Array-CGH shared-boundary case study

Claims. CLM-BB-006;CLM-BB-014

Status. 43 by 2215 executed fits retained; invalid comparator excluded from comparator conclusions

Hypothesis. The shared model provides a coherent common-boundary summary across the multisequence matrix.

Competing explanation. Forced sharing obscures subject-specific changes or comparator conclusions arise from axis mismatch.

Design. Reproduce shared and independent fits; validate axis before every comparator; use raw matrix for multisequence baselines and common joint targets for model comparison.

Metrics. boundary summaries, subject fit diagnostics, predictive/common-target scores, external annotations when verified, runtime.

Repetition and uncertainty. Subject bootstrap or simulation-based uncertainty where scientifically justified; no pseudo-replication across probes.

Dependencies. CODE-BB-004;CODE-BB-009;CODE-BB-011;CODE-BB-015

Output. valid multiple-sequence comparison table and provenance link to the excluded computation.

C.6.11 EPR-BB-011 – SPX regime case study

Claims. CLM-BB-014

Status. Gaussian and Bernoulli fits are executed real results

Hypothesis. The two declared observation models expose different posterior regime summaries over 2015-2023.

Competing explanation. Event alignment is post hoc and family evidence values are incommensurate.

Design. Reproduce stride-4 $n=566$ fits, define response transformations, and predeclare any out-of-sample or contextual-event analysis.

Metrics. k_{hat} , within-model evidence, posterior uncertainty, predictive score under declared target, qualitative event overlay.

Repetition and uncertainty. Block-aware time split or rolling evaluation for predictive claims; event overlay remains descriptive.

Dependencies. CODE-BB-007;CODE-BB-011;CODE-BB-015

Output. synchronized case-study figure/caption/data card.

C.6.12 EPR-BB-012 – MethyKit chr21 segmentation and predictive repair

Claims. CLM-BB-002;CLM-BB-012;CLM-BB-014

Status. segmentation real; archived held-out value excluded from predictive conclusions; rerun planned

Hypothesis. A family-correct Beta-observation predictive distribution yields a valid held-out score on a scientifically justified in-support split.

Competing explanation. The apparent score was an artifact of Gaussian fallback and endpoint clipping.

Design. Use exact executed methylKit test1.myCpG chr21 source, hash weights/split, implement Beta predictive, keep test coordinates in declared support or use named extrapolation, repeat block-aware splits.

Metrics. total and mean log predictive score with denominator m , calibration where defined, k_{hat} , boundary stability, runtime.

Repetition and uncertainty. Repeated splits with intervals; second region/cell type remains planned until sourced.

Dependencies. CODE-BB-006;CODE-BB-007;CODE-BB-010;CODE-BB-011

Output. new sidecar with lineage to the excluded value and corrected figure/table.

C.6.13 EPR-BB-013 – Comparator agreement, tuning fairness, and external accuracy

Claims. CLM-BB-014

Status. non-CGH archived agreement values real; fair comparison planned

Hypothesis. Conclusions remain stable when matched-k agreement, independently tuned prediction, and external-truth accuracy are reported as separate tasks.

Competing explanation. Reported superiority is induced by BayesBreak-referenced labels or asymmetric tuning.

Design. Three strata: matched-k diagnostic, equal-budget prediction tuning, and external annotation/simulation truth; canonical one-to-one matcher.

Metrics. agreement F1/Jaccard/matched MAE, predictive proper score, external precision/recall/F1, runtime and tuning cost.

Repetition and uncertainty. Repeated splits/datasets and paired intervals; NA when no matched boundaries.

Dependencies. CODE-BB-008;CODE-BB-009;CODE-BB-010

Output. task-separated comparator table and tuning ledger.

C.6.14 EPR-BB-014 – Family-specific posterior-prediction tests

Claims. CLM-BB-012

Status. partial tests; expansion required

Hypothesis. Every family produces finite, normalized or correctly generalized predictive scores on supported inputs and rejects unsupported extrapolation by default.

Competing explanation. Base-class fallbacks hide family-specific errors.

Design. Unit and integration cases for each family, MAP versus model-averaged prediction, left/right out-of-range coordinates, zero weights, edge parameters, and invalid inputs.

Metrics. analytic/reference agreement, finite checks, normalization where applicable, expected exceptions, reproducibility.

Repetition and uncertainty. Deterministic tests plus randomized property cases.

Dependencies. CODE-BB-006;CODE-BB-007;CODE-BB-012

Output. prediction certification report.

C.6.15 EPR-BB-015 – Misspecification, failure-boundary, and negative-control suite

Claims. CLM-BB-005;CLM-BB-007;CLM-BB-010;CLM-BB-011;CLM-BB-013

Status. planned

Hypothesis. BayesBreak fails in predictable, detectable ways under declared boundary cases.

Competing explanation. Diagnostics remain optimistic under heavy tails, zero inflation, dense changes, unequal information, or adversarial priors.

Design. Heavy tails, zero inflation, no change, dense change, short segments, unequal subject information, grid refinement, approximation failure, and prior conflict.

Metrics. false discoveries, missed changes, posterior entropy/calibration, certificate failure, runtime, warning/error behavior.

Repetition and uncertainty. Predeclared repetitions and regime-wise intervals; report reversed and null outcomes.

Dependencies. CODE-BB-001;CODE-BB-005;CODE-BB-010;CODE-BB-012

Output. failure map and decision thresholds.

C.7 Claim status map

Table C.3. Claim status used by the implementation and experiment records.

ID	Claim	Status
CLM-BB-001	For any finite array of segment marginal likelihoods and a partition prior that factorizes over admissible segments or transitions, the same contiguous-partition recursions compute posterior sums and posterior maxima.	established by derivation; exact-regime implementation checks support the finite algorithms
CLM-BB-002	For supported regular exponential-family segment models with proper conjugate priors and finite normalizing constants, cumulative sufficient statistics determine exact segment marginal likelihoods and posterior moments.	exactly derived under the stated regularity, propriety, integrability, and moment conditions
CLM-BB-003	The sum-product recursion yields fixed-count marginal likelihoods, the posterior distribution of the segment count, changepoint marginals, and partition-averaged signal moments for the declared factorized model.	established by complete derivation and finite exhaustive-enumeration or invariant checks
CLM-BB-004	A separate max-sum recursion with deterministic backtracking returns the joint MAP partition; marginal changepoint modes need not do so.	established with a complete proof immediately after the statement
CLM-BB-005	Irregular coordinates can enter through distinct segment-cohesion and interior-boundary-hazard factors while preserving the factorization required by dynamic programming.	established at the model and recursion level; package implementation of the separated factors remains planned
CLM-BB-006	For aligned sequences that are conditionally independent given a common partition, sequence-specific segment marginal likelihoods multiply before the same partition recursion is applied.	established algebraically
CLM-BB-007	On a finite candidate set, the posterior for a common partition concentrates on the distinguished partition under a positive average expected log-score gap and suitable law-of-large-numbers conditions.	established under the stated finite-candidate assumptions

ID	Claim	Status
CLM-BB-008	The finite latent-group criterion admits a Jensen minorization with normalized auxiliary weights, and each template update is solved by a responsibility-weighted max-sum recursion.	the optimization statements are established for the stated finite criterion; one implementation path returns a stale final objective
CLM-BB-009	The current finite latent-group criterion is a normalized finite-mixture sampling density identifiable up to label permutation.	not established and not asserted by the corrected manuscript
CLM-BB-010	A uniform error bound on reachable segment log marginal likelihoods gives explicit bounds on partition marginal likelihoods and posterior odds.	established conditionally on the uniform segmentwise error assumption
CLM-BB-011	Specific Laplace, quadrature, variational, Poly-Gamma, and expectation-propagation routines satisfy uniform rates over all reachable segments.	proof obligation; no routine-wide rate is used as an established claim
CLM-BB-012	Posterior prediction must use the fitted observation family and an explicit coordinate-support or extrapolation rule.	statistical requirement established; package coverage is incomplete for Beta observations and out-of-range coordinates
CLM-BB-013	The populated synthetic recovery, calibration, latent-group, approximation, model-selection, and timing values are real executed results for their recorded designs.	empirically supported by archived outputs; scope remains design-specific
CLM-BB-014	The four populated real-data fits and populated comparator outputs are real executed computations, with interpretation determined by their coordinate, reference, model, and metric definitions.	all populated outputs are real executions; two historical calculations are excluded from their stated comparator or predictive conclusions

C.8 Executed-result interpretation map

Table C.4. Result provenance and permitted scientific interpretation.

ID	Class	Interpretation status	Release action
RES-BB-SYN-001	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-SYN-002	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-SYN-003	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-SYN-004	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-001	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.

ID	Class	Interpretation status	Release action
RES-BB-RD-002	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-003	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-004	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-005	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-006	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-007	real	valid for the stated interpretation, with limitations	Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-CMP-001	real diagnostic	real diagnostic	Use only as agreement/sensitivity diagnostic and relabel reference explicitly.
RES-BB-CMP-002	real computation; invalid comparison definition	excluded from comparator conclusions	Retain the historical value and documented axis mismatch; omit it from scientific comparison until a valid raw-matrix rerun receives a new result ID.
RES-BB-QA-001	real Phase 1 QA execution	implementation verification result	Use in implementation and reproducibility discussion, not as a performance result.
RES-BB-RD-007Q	real computation; invalid predictive definition	excluded from posterior-predictive conclusions	Retain the exact archived value and documented failure mechanism; replace its scientific use only after the Beta predictive calculation and coordinate-support rule are repaired and rerun.
RES-BB-QA-002	real Phase 1 QA execution	implementation verification result	Report 157 passed, 5 skipped, and 17 unresolved exactly; retain the historical 174/5 run only as an earlier verification record.
RES-BB-QA-003	real Phase 6 QA execution	implementation verification result	Report this as the current bounded verification and retain RES-BB-QA-002 as the historical Phase 1 verification state.

C.9 Test matrix

Table C.5. Required test classes.

Class	Scope	Acceptance
unit	Family calculations, prior factors, coordinate assignment, schema parsing, and metric edge cases.	Declared outputs and exceptions match hand-computed fixtures.
property	Sum-product and max-sum recursions over randomized finite cases, zero-support blocks, priors, and ties.	Marginal likelihoods, posterior normalizers, and MAP partitions match exhaustive enumeration.
numerical	Log-domain aggregation, nonconjugate reference comparisons, finite checks, and tolerance tightening.	No overflow-induced ranking change; reference discrepancies and failure states are recorded.
statistical	Calibration, repeated recovery, misspecification, fairness-controlled comparison, and posterior prediction.	Pre-specified repetitions, uncertainty summaries, reference definitions, and falsifying outcomes are reported.
regression	Archived result read-only behavior, corrected result parent links, figure/table semantics, and dataset cards.	No populated archived value changes; invalid historical calculations cannot re-enter excluded conclusions.
reproducibility	Pinned environment, seeds, hashes, commands, clean checkout, and artifact regeneration.	A clean run reproduces declared artifacts or reports an explicit incomplete state with owner and reason.

C.10 Logging, provenance, and resources

C.10.1 Required run records

- Record `result_id` and `parent_result_id` before writing a corrected result.
- Record data, configuration, code, and environment hashes for every new run.
- Store the coordinate axis, observation family, prior, split, extrapolation policy, metric version, and tuning budget with each relevant result.
- Report executed, valid for the stated interpretation, excluded from a named analysis, and planned as separate fields.
- Reject absolute author-local paths in release records.

C.10.2 Resource planning

- Profile the one test unresolved under the Phase 6 per-test cap before setting time or memory budgets.
- Account explicitly for the $O(n^2)$ segment-score table and $O(k_{\max} n^2)$ worst-case partition recursions.
- Measure sequence count, candidate count, observation family, nonconjugate method, repetitions, and comparator tuning as separate resource drivers.
- Produce pilot-based CPU, memory, storage, and wall-time estimates before the expanded experiment grid; the current archive does not support a defensible fixed estimate.

C.11 Explicit failure states

FAIL-BB-001

Condition. Nonfinite or dimensionally inconsistent segment scores.

Required behavior. Stop before partition inference and emit the offending family, segment coordinates, and configuration hash.

FAIL-BB-002

Condition. Comparator prediction and reference axes differ.

Required behavior. Reject metric computation; preserve the attempted run only as a diagnostic record.

FAIL-BB-003

Condition. Posterior prediction requested for an unsupported family or coordinate outside support.

Required behavior. Raise an explicit error unless a named, recorded extrapolation rule is selected.

FAIL-BB-004

Condition. Latent-group objective is nonmonotone beyond tolerance or returned objective differs from the final trace value.

Required behavior. Mark the restart invalid and do not select it as the optimum.

FAIL-BB-005

Condition. A required test exceeds its profiled bound.

Required behavior. Report unresolved with owner, environment, seed, and profile; do not count it as a pass.

FAIL-BB-006

Condition. A corrected result lacks a parent link or required hashes.

Required behavior. Reject release of the result and retain the archived parent unchanged.

C.12 Definition of done

- The exact title and generalized hierarchical Bayesian segmentation narrative remain unchanged across the book, main paper, executive summary, code handoff, and presentation handoffs.
- Exact finite-case dynamic-programming properties match exhaustive enumeration under seeded tests.
- Posterior prediction is family-specific and out-of-range coordinates follow an explicit recorded policy.

- Comparator axes, reference type, matching rule, and tuning budget are validated before metric computation.
- Every corrected result has a new ID, a parent link, and data, configuration, code, and environment hashes; archived populated values remain unchanged.
- Book appendix, standalone coding handoff, repository registries, figures, tables, and metadata pass synchronization checks.
- Every unresolved proof, test, computation, annotation, or release decision is stated directly rather than represented as complete.

Detailed experiment protocols

CHAPTER PATH

Claim-mapped protocols that distinguish executed evidence, planned recertification, uncertainty, fairness, failure modes, and falsifying outcomes.

INTERPRETATION RULE

A protocol may contain both immutable executed outputs and planned expansion. Existing populated values remain real computations; planned cells are never populated synthetically. Positive, mixed, null, reversed, and falsifying outcomes must all be retained in the run registry.

D.1 Cross-protocol controls

All simulations use deterministic seed registries and preserve ground truth separately from fitted objects. Repetitions and intervals are predeclared. Comparator budgets, axes, penalties, and search ranges are recorded before scoring. Data transforms are fitted on training data only. Runtime studies separate block construction, DP, backtracking, prediction, and I/O. Every output is linked to data/config/code hashes and a result ID.

D.2 EPR-BB-001: Randomized exact-DP and semiring invariants

Evidence status. executed plus expansion planned

Claims. CLM-BB-001;CLM-BB-003;CLM-BB-004

Hypothesis. Sum-product and max-sum implementations match exhaustive enumeration on all sampled admissible finite cases.

Competing explanation. Agreement is limited to hand-picked priors or benign support.

Design and provenance. Property-based generation over n , $k_{\max}$, local scores, zero-support blocks, priors, and ties; compare evidence, $P(k)$, marginals, and joint MAP.

Metrics. absolute/relative error, normalization error, exact partition equality, failure count with seed.

Repetitions, uncertainty, and statistics. At least 1,000 deterministic seeded cases per configuration family; report zero-failure upper bound.

Dependencies and stopping. CODE-BB-002;CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. tests and a machine-readable invariant report.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.3 EPR-BB-002: Conjugate-family recovery and null/misspecification grid

Evidence status. executed real values plus expanded protocol planned

Claims. CLM-BB-002;CLM-BB-013

Hypothesis. The common partition recursions produce calibrated segmentation behavior across supported families when their data model is correct.

Competing explanation. Apparent portability is driven by a narrow SNR or tuned prior.

Design and provenance. Factorial grid over family, n, segment count, spacing, SNR, weight/exposure patterns, null data, and selected misspecification.

Metrics. boundary precision/recall/F1 under external truth, k error, posterior coverage, log evidence, runtime.

Repetitions, uncertainty, and statistics. Predeclared repetitions with bootstrap or t-intervals by cell and family-level multiplicity control for headline contrasts.

Dependencies and stopping. CODE-BB-001;CODE-BB-002;CODE-BB-010. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. family performance table, calibration plots, and null-control table.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.4 EPR-BB-003: Boundary posterior calibration

Evidence status. executed ECE/Brier retained; recertification planned

Claims. CLM-BB-003;CLM-BB-013

Hypothesis. Bayes curve probabilities are calibrated for the declared model after marginalizing over k.

Competing explanation. Low aggregate ECE is caused by severe class imbalance or bin choices.

Design and provenance. Repeated simulated datasets with fixed-k and k-marginalized posterior variants; preregistered equal-width and equal-mass bins.

Metrics. Brier, log score, ECE, reliability slope/intercept, prevalence, calibration curves.

Repetitions, uncertainty, and statistics. Dataset-level intervals and paired comparison of fixed-k versus marginalized variants.

Dependencies and stopping. CODE-BB-002;CODE-BB-010. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. calibration figure with intervals and protocol sidecar.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.5 EPR-BB-004: Segment-count and SNR behavior

Evidence status. executed plus interpretation expansion planned

Claims. CLM-BB-003;CLM-BB-013

Hypothesis. Posterior uncertainty over k changes predictably with signal strength and prior concentration.

Competing explanation. Observed k behavior is a prior artifact or oversegmentation bias.

Design and provenance. Cross SNR, n, true k, minimum segment length, and prior settings; include no-change and dense-change controls.

Metrics. $P(k_{\text{true}})$, posterior entropy, k bias, boundary loss, false discoveries.

Repetitions, uncertainty, and statistics. Repeated-seed intervals and decomposition by prior regime.

Dependencies and stopping. CODE-BB-001;CODE-BB-002. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. k-posterior panels and falsification table.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.6 EPR-BB-005: Finite latent-group template stress test

Evidence status. archived 96 percent value real; rerun required after semantic/code repair

Claims. CLM-BB-008;CLM-BB-009;CLM-BB-013

Hypothesis. Coordinate updates monotonically improve the stated score and recover shared templates in identifiable finite simulations.

Competing explanation. Recovery is restart luck, label coding, or collapse under overspecified G.

Design and provenance. Vary S, $G_{\text{true}}/G_{\text{fit}}$, n, noise, template separation, imbalance, initialization, and collapsed/duplicate templates.

Metrics. objective monotonicity, final objective, label-invariant ARI, template distance, collapse rate, restart selection stability.

Repetitions, uncertainty, and statistics. At least 50 datasets per cell and multiple restarts; paired pre/post repair diagnostic.

Dependencies and stopping. CODE-BB-003;CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. stress-test table, objective traces, and regression report.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.7 EPR-BB-006: Shared-boundary weak-signal pooling

Evidence status. executed supplementary outputs plus fair comparison planned

Claims. CLM-BB-006;CLM-BB-007

Hypothesis. Exact shared-boundary pooling improves boundary recovery when multiple subjects share changes and individual signals are weak.

Competing explanation. Gains are caused by forced-sharing bias or tuned k .

Design and provenance. Vary subject count, information heterogeneity, shared fraction, subject-specific deviations, noise, and prior; compare shared, independent, and suitable multi-sequence methods.

Metrics. external-truth boundary loss, k error, calibration, predictive score under a common joint target, runtime.

Repetitions, uncertainty, and statistics. Repeated datasets with paired contrasts and heterogeneity-stratified intervals.

Dependencies and stopping. CODE-BB-004;CODE-BB-009;CODE-BB-010. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. pooling benefit/failure surface and model-assumption table.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.8 EPR-BB-007: Nonconjugate local certificate and global sensitivity

Evidence status. real benchmark values retained; certification expansion planned

Claims. CLM-BB-010;CLM-BB-011

Hypothesis. Measured reachable-block log-score tolerances predict the observed perturbation of evidence and posterior summaries.

Competing explanation. Average block error hides rare reachable blocks that dominate the posterior.

Design and provenance. Construct high-accuracy references, census or sample all reachable blocks, separate truncation from numerical error, and rerun DP at nested tolerances.

Metrics. max/quantile block error, evidence ratio, posterior-odds distortion, TV distance, boundary/k changes, time/memory.

Repetitions, uncertainty, and statistics. Multiple datasets and tolerance ladders; deterministic reference checks where possible.

Dependencies and stopping. CODE-BB-005;CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. segment-error records and posterior-bound table based on segmentwise error measurements.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.9 EPR-BB-008: Runtime and memory decomposition

Evidence status. real local slopes retained; extended scaling planned

Claims. CLM-BB-003;CLM-BB-013

Hypothesis. Measured cost decomposes into block construction and $O(k_{\max} n^2)$ DP transitions with predictable memory scaling.

Competing explanation. Near-linear local slopes reflect cache/warmup or a narrow n range, not algorithmic structure.

Design and provenance. Extend n and $k_{\max}$ until resource limits; separate cold/warm cache, block construction, DP, backtracking, and I/O; use fixed hardware and repeated runs.

Metrics. wall time, CPU time, peak RSS, transition count, slope by range, residuals against $n^2 k$ model.

Repetitions, uncertainty, and statistics. Median and dispersion across repetitions, machines reported separately.

Dependencies and stopping. CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. scaling curves, resource table, and profiler artifacts.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.10 EPR-BB-009: Well-log prior sensitivity case study

Evidence status. two executed real fits retained

Claims. CLM-BB-005;CLM-BB-014

Hypothesis. Cohesion and boundary-hazard choices change segmentation in interpretable ways without being mislabeled as external accuracy.

Competing explanation. Differences are arbitrary prior sensitivity with no physical interpretation.

Design and provenance. Reproduce archived index-uniform and length-aware fits, then vary repaired c/h choices on the same 507-point stride-8 series.

Metrics. k_{hat} , log evidence within a common model family, boundary stability, posterior uncertainty, runtime.

Repetitions, uncertainty, and statistics. Sensitivity grid rather than sampling uncertainty; external annotations only if independently verified.

Dependencies and stopping. CODE-BB-001;CODE-BB-011;CODE-BB-015. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. case-study figure, prior sensitivity table, provenance card.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.11 EPR-BB-010: Array-CGH shared-boundary case study

Evidence status. 43 by 2215 executed fits retained; incompatible comparator excluded from comparison

Claims. CLM-BB-006;CLM-BB-014

Hypothesis. The shared model provides a coherent common-boundary summary across the multisequence matrix.

Competing explanation. Forced sharing obscures subject-specific changes or comparator conclusions arise from axis mismatch.

Design and provenance. Reproduce shared and independent fits; validate axis before every comparator; use raw matrix for multisequence baselines and common joint targets for model comparison.

Metrics. boundary summaries, subject fit diagnostics, predictive/common-target scores, external annotations when verified, runtime.

Repetitions, uncertainty, and statistics. Subject bootstrap or simulation-based uncertainty where scientifically justified; no pseudo-replication across probes.

Dependencies and stopping. CODE-BB-004;CODE-BB-009;CODE-BB-011;CODE-BB-015. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. valid multiple-sequence comparison table and a link to the excluded archived calculation and failure analysis.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing

explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.12 EPR-BB-011: SPX regime case study

Evidence status. Gaussian and Bernoulli fits are executed real results

Claims. CLM-BB-014

Hypothesis. The two declared observation models expose different posterior regime summaries over 2015-2023.

Competing explanation. Event alignment is post hoc and family evidence values are incommensurate.

Design and provenance. Reproduce stride-4 $n=566$ fits, define response transformations, and predeclare any out-of-sample or contextual-event analysis.

Metrics. k_{hat} , within-model evidence, posterior uncertainty, predictive score under declared target, qualitative event overlay.

Repetitions, uncertainty, and statistics. Block-aware time split or rolling evaluation for predictive claims; event overlay remains descriptive.

Dependencies and stopping. CODE-BB-007;CODE-BB-011;CODE-BB-015. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. synchronized case-study figure/caption/data card.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.13 EPR-BB-012: MethylKit chr21 segmentation and predictive repair

Evidence status. segmentation result real; archived held-out value excluded from predictive conclusions; corrected run planned

Claims. CLM-BB-002;CLM-BB-012;CLM-BB-014

Hypothesis. A family-correct Beta-observation predictive distribution yields a valid held-out score on a scientifically justified in-support split.

Competing explanation. The apparent score was an artifact of Gaussian fallback and endpoint clipping.

Design and provenance. Use exact executed methylKit test1.myCpG chr21 source, hash weights/split, implement Beta predictive, keep test coordinates in declared support or use named extrapolation, repeat block-aware splits.

Metrics. total and mean log predictive score with denominator m , calibration where defined, k_{hat} , boundary stability, runtime.

Repetitions, uncertainty, and statistics. Repeated splits with intervals; second region/cell type remains planned until sourced.

Dependencies and stopping. CODE-BB-006;CODE-BB-007;CODE-BB-010;CODE-BB-011. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. new result record linked to the excluded archived value, with a corrected figure and table.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.14 EPR-BB-013: Comparator agreement, tuning fairness, and external accuracy

Evidence status. non-CGH archived agreement values real; fair comparison planned

Claims. CLM-BB-014

Hypothesis. Conclusions remain stable when matched-k agreement, independently tuned prediction, and external-truth accuracy are reported as separate tasks.

Competing explanation. Reported superiority is induced by BayesBreak-referenced labels or asymmetric tuning.

Design and provenance. Three strata: matched-k diagnostic, equal-budget prediction tuning, and external annotation/simulation truth; canonical one-to-one matcher.

Metrics. agreement F1/Jaccard/matched MAE, predictive proper score, external precision/recall/F1, runtime and tuning cost.

Repetitions, uncertainty, and statistics. Repeated splits/datasets and paired intervals; NA when no matched boundaries.

Dependencies and stopping. CODE-BB-008;CODE-BB-009;CODE-BB-010. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. task-separated comparator table and tuning ledger.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.15 EPR-BB-014: Family-specific posterior-prediction tests

Evidence status. partial tests; expansion required

Claims. CLM-BB-012

Hypothesis. Every family produces finite, normalized or correctly generalized predictive scores on supported inputs and rejects unsupported extrapolation by default.

Competing explanation. Base-class fallbacks hide family-specific errors.

Design and provenance. Unit and integration cases for each family, MAP versus model-averaged prediction, left/right out-of-range coordinates, zero weights, edge parameters, and invalid inputs.

Metrics. analytic/reference agreement, finite checks, normalization where applicable, expected exceptions, reproducibility.

Repetitions, uncertainty, and statistics. Deterministic tests plus randomized property cases.

Dependencies and stopping. CODE-BB-006;CODE-BB-007;CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. posterior-prediction verification report.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

D.16 Misspecification, boundary cases, and negative controls

Protocol ID. EPR-BB-015

Evidence status. planned

Claims. CLM-BB-005;CLM-BB-007;CLM-BB-010;CLM-BB-011;CLM-BB-013

Hypothesis. BayesBreak fails in predictable, detectable ways under declared boundary cases.

Competing explanation. Diagnostics remain optimistic under heavy tails, zero inflation, dense changes, unequal information, or adversarial priors.

Design and provenance. Heavy tails, zero inflation, no change, dense change, short segments, unequal subject information, grid refinement, approximation failure, and prior conflict.

Metrics. false discoveries, missed changes, posterior entropy/calibration, failure to establish the segmentwise error bound, runtime, warning/error behavior.

Repetitions, uncertainty, and statistics. Predeclared repetitions and regime-wise intervals; report reversed and null outcomes.

Dependencies and stopping. CODE-BB-001;CODE-BB-005;CODE-BB-010;CODE-BB-012. Stop only after the predeclared cells complete or are reported as failed/unresolved.

Outputs. failure map and decision thresholds.

Outcome interpretation. A positive result supports only the mapped claim at the tested regime. A mixed or null result narrows scope. A reversed result favors the competing explanation. A failed control or axis/schema violation invalidates the affected comparison rather than supplying negative evidence.

Reproducibility and traceability map

CHAPTER PATH

A navigable map from sources to claims, proofs or experiments, implementation modules, figures, tables, and result records.

E.1 Traceability requirements

Each scientific statement is linked to at least one of four sources of support: an in-place proof or exact derivation; an executed result that is valid for the stated interpretation; a verified external source; or an explicit statement that the proof or experiment remains planned. Whether a computation was executed is separate from whether it supports the stated scientific interpretation. Hashes identify source state; they do not establish correctness by themselves.

E.2 Claim map

E.2.1 CLM-BB-001: Generalized hierarchical Bayesian segmentation through observation-model-specific segment integration and common recursions over ordered partitions.

Status and confidence. established by derivation, implementation, and exact-regime tests; high.

Disposition. retain as the central thesis together with irregular designs, multiple-sequence hierarchies, and grouped or latent-group models.

Source anchor. docs/report/sections/2.problem.tex; 4.method.tex; src/bayesbreak/base.py; dp.py.

Obligation. State the exponential-family conditions, hierarchical extensions, and exact relationship to product-partition and multiple-changepoint literature; do not claim priority for established recursions.

Destinations. Workstreams REV-BB-002;REV-BB-005; book CH-BB-01;CH-BB-02;CH-BB-12;CH-BB-16; paper PPR-BB-01;PPR-BB-02;PPR-BB-03;PPR-BB-11; experiments EPR-BB-001.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.2 CLM-BB-002: Closed-form weighted conjugate exponential-family block evidences and moments from cumulative sufficient statistics.

Status and confidence. family-specific supported formulas are exactly derived; generic scalar-weight display is too narrow for all exposure/trial specializations; proof presentation incomplete; high after notation and presentation repair.

Disposition. retain and prove/reclassify in place.

Source anchor. docs/report/sections/4.method.tex:54-125, 1088-1267.

Obligation. Use an exposure-aware statistic $S(y, w)$ with $-w A(\theta)$; add immediate algebraic proofs or derived corollaries; state moment-existence conditions..

Destinations. Workstreams REV-BB-003; book CH-BB-03;CH-BB-04;CH-BB-13; paper PPR-BB-01;PPR-BB-04;PPR-BB-08;PPR-BB-09; experiments EPR-BB-002;EPR-BB-012.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.3 CLM-BB-003: Sum-product DP yields evidence, $P(k \rightarrow y)$, fixed-count boundary marginals, and Bayes curves for the declared factorized model.

Status and confidence. established by complete derivation and brute-force/invariant tests; high.

Disposition. retain; position against Fearnhead and Hutter.

Source anchor. docs/report/sections/4.method.tex:150-486; tests/test_dp*.py.

Obligation. State exact regime and prior normalization precisely; narrow novelty language..

Destinations. Workstreams REV-BB-005; book CH-BB-02;CH-BB-05;CH-BB-13; paper PPR-BB-01;PPR-BB-02;PPR-BB-03;PPR-BB-05;PPR-BB-08;PPR-BB-09; experiments EPR-BB-001;EPR-BB-003;EPR-BB-004;EPR-BB-008.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.4 CLM-BB-004: Separate max-sum/backtracking recursion yields the joint MAP segmentation rather than marginal boundary modes.

Status and confidence. established, but the named theorem defers/duplicates its proof; high.

Disposition. retain; move one complete proof immediately after the theorem.

Source anchor. docs/report/sections/5.algorithms.tex:149-164; 8.appendix.tex:439-462.

Obligation. Remove duplicate proposition and satisfy in-place proof policy..

Destinations. Workstreams REV-BB-005; book CH-BB-06; paper PPR-BB-05; experiments EPR-BB-001.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.5 CLM-BB-005: Design-aware segment-length cohesion handles irregular coordinates and gives large physical gaps greater boundary prior mass.

Status and confidence. partially supported; the Poisson boundary-gap derivation and implemented segment-length cohesion are different priors; medium for general design-aware prior, low for current gap interpretation.

Disposition. repair without changing the DP core.

Source anchor. docs/report/sections/4.method.tex:580-697.

Obligation. Separate segment cohesion $c(i,j)$ from boundary hazard $h(j)$, or remove the Poisson-process rationale from $g(\text{length})$.

Destinations. Workstreams REV-BB-004; book CH-BB-02;CH-BB-03;CH-BB-07;CH-BB-14; paper PPR-BB-01;PPR-BB-02;PPR-BB-03;PPR-BB-04;PPR-BB-06;PPR-BB-08;PPR-BB-10; experiments EPR-BB-009;EPR-BB-015.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.6 CLM-BB-006: Shared-boundary pooling across conditionally independent subjects is exact after multiplying per-subject block evidences.

Status and confidence. established algebraically; high.

Disposition. retain.

Source anchor. docs/report/sections/4.method.tex:699-785; src/bayesbreak/replicates.py.

Obligation. Keep evidence-comparison semantics model-consistent and add numerical-stability checks..

Destinations. Workstreams REV-BB-006; book CH-BB-02;CH-BB-08;CH-BB-13;CH-BB-14; paper PPR-BB-01;PPR-BB-02;PPR-BB-03;PPR-BB-06;PPR-BB-08;PPR-BB-09;PPR-BB-10; experiments EPR-BB-006;EPR-BB-010.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.7 CLM-BB-007: The stated shared-boundary identifiability result follows when one subject distinguishes every alternative boundary vector.

Status and confidence. unsupported as stated; low.

Disposition. replace with a finite-candidate posterior-consistency theorem.

Source anchor. docs/report/sections/4.method.tex:787-827.

Obligation. Require a positive average expected log-score gap and LLN conditions; remove the claim that exponential-family identifiability makes the condition automatic..

Destinations. Workstreams REV-BB-006; book CH-BB-08;CH-BB-15; paper PPR-BB-01;PPR-BB-06;PPR-BB-11; experiments EPR-BB-006;EPR-BB-015.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.8 CLM-BB-008: Finite latent-template EM monotonically optimizes the documented template-score objective.

Status and confidence. partially established as monotone alternating optimization of the stated finite latent-group criterion; one convergence path returns a stale restart objective; medium.

Disposition. retain as finite latent-group template estimation; use the stated criterion and test every convergence path.

Source anchor. docs/report/sections/4.method.tex:843-1016; src/bayesbreak/mixture.py:397-478.

Obligation. Remove normalized likelihood/generative language, rename sequence likelihood API, return objective[-1], and test all convergence/restart-selection exits..

Destinations. Workstreams REV-BB-007; book CH-BB-02;CH-BB-09;CH-BB-13; paper PPR-BB-01;PPR-BB-02;PPR-BB-03;PPR-BB-07;PPR-BB-08;PPR-BB-09; experiments EPR-BB-005.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.9 CLM-BB-009: The latent-template construction is a normalized finite-mixture density identifiable up to label permutation.

Status and confidence. unsupported under the current S_g definition; low.

Disposition. remove from established claims; retain normalized mixture as reserve research.

Source anchor. docs/report/sections/4.method.tex:883-919; 8.appendix.tex:220-252.

Obligation. Do not apply finite-mixture density identifiability to current scores; a normalized branch needs a new model, proof, and rerun of affected results..

Destinations. Workstreams REV-BB-007; book CH-BB-09;CH-BB-15; paper PPR-BB-01;PPR-BB-07;PPR-BB-11; experiments EPR-BB-005.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.10 CLM-BB-010: Uniform segment log marginal-likelihood error implies controlled partition marginal likelihoods and posterior odds.

Status and confidence. established conditionally on the uniform error assumption; high.

Disposition. retain as the main approximation theorem.

Source anchor. docs/report/sections/4.method.tex:1753-1851.

Obligation. Sharpen/cap the TV corollary and distinguish odds control, probability control, and ranking margins..

Destinations. Workstreams REV-BB-008; book CH-BB-10;CH-BB-13;CH-BB-15; paper PPR-BB-01;PPR-BB-07;PPR-BB-08;PPR-BB-09;PPR-BB-11; experiments EPR-BB-007;EPR-BB-015.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.11 CLM-BB-011: Laplace, quadrature, JJ, PG mean-field, and EP satisfy the stated routine-specific uniform rates over reachable blocks.

Status and confidence. unsupported or overstrong as stated; low.

Disposition. remove the method-wide rate theorem; replace it with method-specific numerical error analysis.

Source anchor. docs/report/sections/4.method.tex:1450-1884; 8.appendix.tex:464-505.

Obligation. Strengthen Laplace hypotheses, separate quadrature tail and discretization error, and avoid unproved JJ/PG/EP evidence rates..

Destinations. Workstreams REV-BB-008; book CH-BB-10; paper PPR-BB-07; experiments EPR-BB-007;EPR-BB-015.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.12 CLM-BB-012

Claim. Posterior-predictive calculations and exported segmentation and Bayes-curve quantities are partially implemented.

Status and confidence. partially implemented; exact-family coverage is incomplete, BetaObs uses a Gaussian fallback, and coordinate assignment silently clips out of range; medium for interfaces; low for current methyl held-out interpretation.

Disposition. retain the statistical quantities after family-specific posterior prediction and coordinate-support corrections.

Source anchor. docs/report/sections/4.method.tex:1960-2140; src/bayesbreak/prediction.py.

Obligation. Define observation-scale mean/predictive targets, implement Beta predictive density, reject or explicitly govern extrapolation, and rerun predictive evidence..

Destinations. Workstreams REV-BB-009; book CH-BB-11;CH-BB-14;CH-BB-15; paper PPR-BB-01;PPR-BB-07;PPR-BB-08;PPR-BB-10;PPR-BB-11; experiments EPR-BB-012;EPR-BB-014.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.13 CLM-BB-013: Synthetic recovery, calibration, latent-group, approximation, model-selection, and runtime outputs are real executed results.

Status and confidence. empirically supported by archived outputs; generality is scope-limited; high as provenance, medium as broad evidence.

Disposition. retain all populated values read-only.

Source anchor. docs/report/figures; tables; scripts; sidecars.

Obligation. Add uncertainty, repetitions, falsification criteria, and avoid converting local scaling slopes into asymptotic claims..

Destinations. Workstreams REV-BB-010; book CH-BB-12;CH-BB-13;CH-BB-15;CH-BB-16; paper PPR-BB-01;PPR-BB-08;PPR-BB-09;PPR-BB-11; experiments EPR-BB-002;EPR-BB-003;EPR-BB-004;EPR-BB-005;EPR-BB-008;EPR-BB-015.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.2.14 CLM-BB-014: Four real-data case-study fits and populated comparator outputs are real executed results.

Status and confidence. all populated outputs are real executions; scientific interpretation varies and two outputs are excluded from comparator or predictive conclusions; high confidence in execution provenance, limited support for comparative efficacy.

Disposition. retain all populated values; relabel and exclude invalid comparisons.

Source anchor. docs/report/figures/realdata_metrics.json; baselines_metrics.json; fitcache.

Obligation. Preserve the values; exclude the incompatible-axis array-CGH comparator and the invalid Beta/endpoint-assigned methylation calculation from their stated conclusions; correct sources, captions, and model labels.

Destinations. Workstreams REV-BB-010;REV-BB-011; book CH-BB-14;CH-BB-16; paper PPR-BB-01;PPR-BB-08;PPR-BB-10;PPR-BB-11; experiments EPR-BB-009;EPR-BB-010;EPR-BB-011;EPR-BB-012;EPR-BB-013.

Language rule. Use no stronger language than is justified by the mapped proof, derivation, or empirical result.

E.3 Real-result and interpretation map

Table E.1. Immutable real-result records and interpretation decisions.

Result ID	Asset	Interpretation status			Interpretation / release action
RES-BB-SYN-001	Boundary calibration figure/table	valid	for the stated interpretation, with limitations		Archived synthetic protocol; uncertainty/repetition reporting must be expanded. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-SYN-002	Latent-group simulation	valid	for the stated interpretation, with limitations		Behavior under current template-score objective; not proof of normalized mixture identifiability. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-SYN-003	Runtime scaling	valid	for the stated interpretation, with limitations		Local empirical slopes, not algorithmic complexity or evidence of asymptotically linear runtime. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.

Result ID	Asset		Interpretation status		Interpretation / release action
RES-BB-SYN-004	Conjugate-family and nonconjugate tables/-figures		valid	for the interpretation, with limitations	Routine-specific theoretical rates are not established by the benchmark. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-001	Well-log uniform fit	index-	valid	for the interpretation, with limitations	No verified external-boundary ECE; keep absent cell planned. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-002	Well-log length-aware fit		valid	for the interpretation, with limitations	Prior is real as implemented, but its Poisson-gap interpretation is not supported. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-003	Array-CGH boundary fit	shared-	valid	for the interpretation, with limitations	Not directly comparable as a Bayes factor with the independent evidence sum. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-004	Array-CGH independent fits	independ-	valid	for the interpretation, with limitations	Different factorization; no single k_{hat} ; invalid single-trace comparator artifact excluded. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-005	SPX Gaussian fit		valid	for the interpretation, with limitations	Macro-event alignment is contextual/qualitative, not ground-truth accuracy. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-006	SPX Bernoulli threshold fit		valid	for the interpretation, with limitations	Different observation model; evidence not directly compared across responses without a common data model. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.
RES-BB-RD-007	Methylation region-A segmentation fit		valid	for the interpretation, with limitations	Executed methylKit test1.myCpG chr21 case study; second region/cell type and external atlas metrics remain planned; weight hash missing. Retain exact value; calibrate interpretation; any corrected rerun gets a new result ID.

Result ID	Asset	Interpretation status	Interpretation / release action
RES-BB-CMP-001	Well-log/SPX/methylation comparator sidecar	real diagnostic	Agreement with BayesBreak, not external accuracy; asymmetric tuning. Use only as agreement/sensitivity diagnostic and relabel reference explicitly.
RES-BB-CMP-002	CGH comparator sidecar rows	excluded from the stated analysis	Exclude from comparator conclusions and preserve as an archived diagnostic record. Omit from scientific comparison until a validated raw-matrix recomputation receives a new result identifier.
RES-BB-QA-001	Mixture stale-objective diagnostic	QA evidence	Implementation diagnostic, not a manuscript performance result. Use in implementation/reproducibility discussion, not as performance evidence.
RES-BB-RD-007Q	Methylation archived held-out score	excluded from the stated analysis	BetaObs uses Gaussian fallback and all post-training coordinates are silently clipped to the endpoint MAP block; not a valid Beta held-out predictive likelihood. Preserve the exact archived value and failure mechanism in the result history; use a replacement predictive result only after family-correct posterior prediction, an explicit coordinate-support rule, and a new computation with a new result identifier.
RES-BB-QA-002	Historical Phase 1 test verification	QA evidence	The archived state is 179 collected, 157 passed, 5 skipped, and 17 unresolved; it remains immutable provenance.
RES-BB-QA-003	Independent Phase 6 bounded test verification	QA evidence	The current rerun is 179 collected, 173 passed, 5 skipped, 1 timeout under a 20-second per-test cap, and 0 failed; no package code was changed.

E.4 Canonical diagram map

DIA-BB-001: **Segment marginal likelihoods and inference over ordered partitions.** Source `shared/figures/tikz/workflows/segment_to_partition.tex`. Question: How do observation-model segment calculations enter sum-product posterior inference and max-sum MAP optimization? Uses: book;main paper;technical handoff.

DIA-BB-002: **Exact, approximate, executed, planned, and excluded results.** Source `shared/figures/tikz/theory/result_status_map.tex`. Question: Which quantities are exact, numerically approximated, executed with a valid interpretation, planned, or excluded from a stated analysis? Uses: book;executive paper;presentation shared handoff.

DIA-BB-003: **Segment cohesion versus boundary hazard.** Source `shared/figures/tikz/models/cohesion_boundary_hazard.tex`. Question: How do segment-length beliefs and local boundary intensity enter the ordered partition prior? Uses: book;main paper.

DIA-BB-004: **Sum-product posterior versus max-sum joint MAP.** Source `shared/figures/tikz/workflows/semiring_inference_routes.tex`. Question: Why are marginal boundary modes not the joint MAP partition? Uses: book;main paper;technical handoff.

DIA-BB-005: **Shared-boundary multi-sequence plate.** Source `shared/figures/tikz/models/shared_boundary_plate.tex`. Question: What is shared and what remains subject-specific in exact pooled segmentation? Uses: book;main paper.

DIA-BB-006: **Finite latent-group template criterion.** Source `shared/figures/tikz/models/latent_group_templates.tex`. Question: What criterion is optimized by the latent-group extension, and why is it not a normalized finite-mixture likelihood? Uses: book;main paper optional;technical handoff.

DIA-BB-007: **Propagation of segment marginal-likelihood error.** Source `shared/figures/tikz/theory/segment_error_propagation.tex`. Question: How does a uniform error bound over admissible segments control partition marginal likelihoods, posterior odds, and modal changepoints? Uses: book;main paper.

DIA-BB-008: **Theorem dependency and status map.** Source `shared/figures/tikz/theory/theorem_dependency_map.tex`. Question: Which formal results are established, repaired, conditional, targets, or removed? Uses: book;technical handoff.

DIA-BB-009: **Claim, experiment, result, and interpretation map.** Source `shared/figures/tikz/experiments/claim_experiment_result_map.tex`. Question: How does each claim map to a statistical experiment, an executed output, and a permitted interpretation? Uses: book;main paper methodology;executive paper.

DIA-BB-010: **Posterior prediction, support, conditioning, and metrics.** Source `shared/figures/tikz/workflows/posterior_prediction_workflow.tex`. Question: Which posterior predictive distribution is evaluated, at which coordinates, under which treatment of segmentation uncertainty, and with which metric? Uses: book;main paper.

DIA-BB-011: **Repository, provenance, and artifact dependency map.** Source `shared/figures/tikz/infrastructure/repository_provenance_map.tex`. Question: How do data, package modules, configs, tests, result sidecars, figures, and manuscripts remain synchronized? Uses: book;coding handoff;presentation shared handoff.

DIA-BB-012: **Implementation and validation stages.** Source `shared/figures/tikz/roadmaps/implementation_validation_stages.tex`. Question: In what order should implementation corrections, tests, controlled recomputations, and release checks be completed? Uses: book;executive paper;executive handoff only.

E.5 Validation status

The technical book contains all 17 chapters, all five appendices, 37 bibliography annotations, the formal-result status table, and editable TikZ sources. The journal paper has been rebuilt under the author-approved title and narrative. The executive paper, complete pinned test run, corrected research computations, and final release verification remain later work.

E.6 Canonical-source synchronization

The handoff JSON hash appears in Appendix C and the standalone Markdown. Annotation source hashes appear in the annotation manifest. Claim, result, experiment, code-task, diagram, and acceptance-criterion registries are copied to shared machine-readable sources. A release validator must recompute these hashes and compare stable-ID sets before packaging.

E.7 Source hierarchy used for this book

The source order is: integrity and the user's directive; approved Phase 2 decisions; editable manuscript and real assets; compiled baseline; archived sidecars and implementation; Phase 1 verified literature; and explicitly labeled inference. Existing populated result values are never modified by prose repair. Where source semantics conflict, the execution artifact is retained and the interpretation is narrowed or excluded from the stated analysis.

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